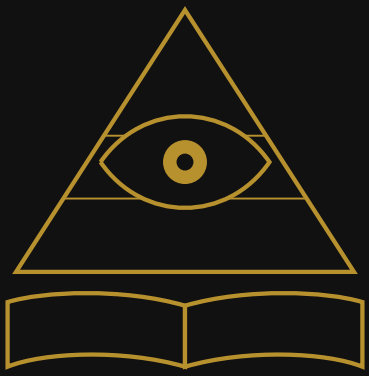


MEECHIE'S WORLD INC • CONSCIENTIA

THE PYRAMIDS OF EGYPT

How They Were Built, What They Were For, and the Myths Around Them

A MEECHIE'S WORLD INC PUBLICATION

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About accuracy. This book is a work of general nonfiction about ancient Egypt and the scholarship around it. Dates are approximate; Egyptologists disagree about many of them, and where the evidence is debated this book says so. Figures such as heights, weights and block counts are standard published estimates and vary by source. Archaeology moves quickly: findings reported as recently as 2025 are included, and later work may refine or overturn them. This book was prepared with the assistance of artificial intelligence and should be read as an introduction, not as a substitute for the primary scholarship listed in the further-reading appendix.

About the emblem. The Conscientia emblem, an all-seeing eye in a pyramid over an open book, is the mark of Meechie's World Inc.

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Preface: A Question Four and a Half Thousand Years Old

Why a pile of stone on the edge of the desert still starts arguments, and how this book tries to settle them.

Stand at the foot of the Great Pyramid of Giza and the first thing you notice is not its size. It is the silence. The traffic of Cairo is a few miles away, the vendors are shouting somewhere behind you, and yet the wall of stone in front of you seems to absorb all of it. Each block comes up to your chest. Then you look up, and the blocks keep coming, course after course, narrowing toward a point that is farther away than it has any right to be. For nearly four thousand years this was the tallest human-made structure on Earth. It was already ancient when Rome was founded. Cleopatra lived closer in time to the Moon landing than to the building of the Great Pyramid.

It is no surprise, then, that people have always asked the same three questions. *How was it built? What was it for? And who, really, was behind it?* This book takes all three seriously.

The first question has a better answer than most people realize. Over the past forty years, archaeologists have found the quarries, the tools, the harbor, the bakeries, the workers' tombs, and even a logbook written by a man who hauled stone to Giza while the Great Pyramid was being built. We no longer have to

guess at the broad outline. We can describe, with real evidence, how the stone was cut, how it was moved, how the monument was aligned to the points of the compass, and how a nation fed and organized tens of thousands of people for the task.

The second question also has a strong answer, though it comes with honest gaps. The pyramids were part of a royal burial system: a tomb, a temple, a causeway, a harbor, a town of priests and workers, and a belief about the king's journey after death. That answer is supported by texts carved into pyramid walls, by the layout of the complexes, and by the wider pattern of Egyptian religion. But it has a notable hole in it. Nobody has ever found an intact royal burial in a Fourth Dynasty pyramid. This book will not hide that fact. It will explain what we can and cannot say because of it.

The third question is where the arguments get loud. For more than a century, writers have claimed that the pyramids were power plants, or observatories of a lost super-civilization, or landing beacons for visitors from other worlds, or repositories of secret mathematics. Some of these ideas are fun. Some contain a grain of something real, such as an observation about a shaft, an alignment, or a strange-looking carving. Some, sadly, rest on the assumption that ancient Egyptians could not have done the work themselves. This book gives every one of these ideas a fair hearing, states what its supporters actually claim, and then sets the claim beside the evidence.

How this book works

The book is divided into five parts. **Part One** sets the scene: the land, the people, and the long road from simple graves to the first pyramid. **Part Two** walks through the age of the great pyramids, one reign at a time. **Part Three** is the technical heart of the book, covering surveying, quarrying, transport, lifting, labor, and energy. **Part Four** is about belief, ritual, burial, and the long afterlife of the monuments themselves, including tomb robbers and early explorers. **Part Five** takes on the modern mysteries and myths, and then turns to what science is finding now.

Throughout, you will find shaded boxes labeled *Key Point*, *Myth and Fact*, or *Worked Example*. They are there for readers who like to skim, and for readers who want to check the numbers. Where a figure is an estimate, the text says so. Where scholars disagree, the text says that too.

A word about certainty

A good history book should tell you not only what we know but how we know it. In this book you will meet four kinds of statements, and it helps to keep them straight.

- **Well established.** Supported by multiple independent lines of evidence, such as the dates of the Fourth Dynasty or the identity of Khufu as the builder of the Great Pyramid.
- **Probable.** The best current reading of limited evidence, such as the claim that wetted sand was used to ease sledge transport, or that the work crews were organized in rotating gangs.

- **Debated.** Where serious specialists disagree, such as the exact ramp system used to raise stone to the upper levels.
- **Unsupported.** Claims for which no credible evidence has been offered, such as the existence of acid-and-zinc machinery inside the monument.

Sorting claims into these four boxes is the single most useful habit you can take from this book. It applies far beyond Egypt.

A note on names and dates

Egyptian kings are known by several names. The builder of the Great Pyramid appears in Egyptian as *Khnum-Khufu* and is usually called Khufu. The Greek writers who visited Egypt thousands of years later called him Cheops, and you will still see that spelling in older books. His successors Khafre and Menkaure are Chephren and Mycerinus in the Greek tradition. This book uses the Egyptian forms and mentions the Greek names when it helps.

Dates are given as years BCE and are approximate. Egyptologists build their chronology from king lists, astronomical records, and radiocarbon dating, and different schemes can differ by a few decades in the Old Kingdom. When you see a date such as "c. 2589 BCE," read the "c." as a reminder that it is the middle of a range.

Why it matters

The pyramids matter because they are proof of something that is easy to forget. Human beings, with copper tools, rope, wood, mud brick, grain, and an extraordinary talent for organization, once built something that has outlasted every empire since. They did it without engines, without iron, and without the wheel as a heavy-transport device. The monuments are an argument for human ability, and they do not need an outside explanation to be remarkable.

If you finish this book and still wonder about the pyramids, good. The best of the open questions are still open, and some of them are being answered right now, by researchers using cosmic-ray detectors, radar, thermal cameras, satellite imagery, and endoscopes thinner than a finger. Chapter 22 describes where that work stands as of this writing.

Turn the page. The story starts, as all Egyptian stories do, with a river.

PART 1

The Land and the Idea

Before the pyramids, there had to be a country

CHAPTER 1

The Gift of the Nile

How a river, a desert, and a very good harvest made the pyramids possible.

The Greek historian Herodotus, who visited Egypt around 450 BCE, called it "the gift of the river." The phrase has been repeated so often that it risks sounding like a cliché, but it is exactly right. Egypt is a long green ribbon in the middle of one of the driest regions on Earth. Away from the river, the land is sand and stone. Within a few miles of the river, it is some of the most fertile farmland the ancient world ever knew. Everything the Egyptians did, including building pyramids, grew out of that contrast.

A country shaped like a lotus

The Nile flows north for about 6,650 kilometers, one of the longest rivers on Earth. Egypt occupies only its last stretch. From the first cataract at Aswan, where granite outcrops break the river into rapids, the Nile runs about 900 kilometers to the Mediterranean. In the south, it runs through a narrow valley, seldom more than a few kilometers wide, with cliffs and desert on both sides. North of modern Cairo, it fans out into the Delta, a broad, green, marshy triangle that Greeks named for its resemblance to the letter delta.

The Egyptians themselves saw their country as two lands: Upper Egypt, the narrow southern valley, and Lower Egypt, the northern Delta. Even after the two were unified under a single king, the king's titles and crowns kept reminding everyone of the old division. The white crown belonged to the south, the red crown to the north, and the double crown combined them. The Egyptian name for the country, *Kemet*, meant "the Black Land," after the dark silt left by the flood. The desert was *Deshret*, "the Red Land." The line between them could be crossed in a single step, and Egyptians knew it.

> **KEY POINT**
> The pyramids stand on the western edge of the valley, on desert plateau just above the floodplain. The location mattered: it was close to the river for transport, high enough to stay dry, and on the side where the sun sets, which Egyptians linked to the realm of the dead.

The flood that ran the calendar

Every summer, rain far to the south, in the Ethiopian highlands, swelled the Nile. Between about July and October, the river spilled over its banks and covered the fields with water and a fresh layer of fine silt. When the water went down, farmers planted. Between roughly November and March they grew crops. From March to July they harvested.

Egyptians organized their year around these three seasons. *Akhet* was the inundation. *Peret* was the growing season. *Shemu* was the harvest. The start of the year was linked to the reappear-

ance of Sirius, which they called Sopdet, in the eastern dawn sky just before sunrise, an event that happened around the time the flood began. This is one of many places where astronomy, agriculture, and religion were mixed together in ancient Egypt.

The flood also gave the state a calendar of labor. When the fields were under water, farmers could not work them. This meant there was a large, available workforce during the very months when the river could carry heavy loads close to the building sites. This overlap is not an accident that Egyptians failed to notice. Evidence suggests the state used the flood season to bring in stone, as we will see in Chapter 11.

Surplus: the secret ingredient

The Egyptian farmer's life was hard, but the system was productive. Barley and emmer wheat were the staple crops, and they made the two basic foods of ancient Egypt: bread and beer. The Nile also provided fish and waterfowl. Marshes yielded papyrus for writing and rope. Cattle, sheep, goats and pigs grazed on the floodplain.

In a good year, the harvest produced more food than farmers needed. That surplus is the secret ingredient in the story of the pyramids. A society that grows only what it needs cannot spare anyone to haul stone or chisel hieroglyphs. A society with a dependable surplus, and a way to collect and store it, can feed specialists: scribes, craftsmen, priests, soldiers, and builders.

Egypt had a way of collecting the surplus, and it was tax. Taxes were paid in grain, cattle, linen, and labor, and recorded by scribes on papyrus. Granaries stored the grain. This adminis-

trative machinery, one of the oldest on the planet, is as important to the pyramids as any quarry or ramp. A pyramid is, among other things, a monument to bookkeeping.

From farming villages to a kingdom

Humans lived along the Nile for hundreds of thousands of years, but the story of the Egyptian state begins in the **Predynastic Period**, roughly 5000 to 3100 BCE. During this time, farming villages along the river grew into larger towns. In the south, major centers developed at places such as Naqada, Hierakonpolis, and Abydos. In the north, communities such as Merimde and Maadi flourished.

The southern cultures, known collectively as Naqada, left behind rich graves. Over centuries, burials became more unequal. Some people were buried with only a pot or two; others with carved stone vessels, jewelry, and imported goods such as lapis lazuli from Afghanistan. This is a sign that rank and wealth were growing. At Hierakonpolis, archaeologists have found what may be an early temple, a brewery, and large decorated tombs. One of them, known as Tomb 100, has painted walls that show boats, hunting, and what looks like a ruler smiting captives.

Around 3100 BCE, according to later Egyptian tradition, a king named **Menes** united the two lands. Archaeologists debate who Menes really was. He may have been the king known from monuments as **Narmer**, shown on a famous ceremonial palette wearing the white crown on one side and the red crown on the other. Or he may have been Hor-Aha, Narmer's successor. The exact name matters less than what happened. By about 3100

BCE, a single government ruled the whole length of the valley and the Delta. It had a capital at Memphis, near where Cairo stands today, and a king who was understood to be more than human.

The king as a god

To understand the pyramids, it is necessary to understand how Egyptians saw their ruler. The king, whom we call the pharaoh, was not just a political leader. In the Old Kingdom he was considered the living embodiment of the god **Horus**, the sky-falcon, and the son of the sun god **Ra**. After death he became **Osiris**, lord of the dead, while his successor became the new Horus.

This was not a symbolic idea in the way we might use the word today. The Egyptians believed the king kept order in the universe. The Egyptian word for that order was **Maat**: truth, justice, balance, the right way of things. If the king did his job, the Nile flooded, the sun rose, the crops grew, and the cosmos kept running. If the king failed, chaos threatened.

It followed that a dead king could not be treated like any other dead person. He had to be preserved, protected, and sent on to the next world with proper rituals, because the whole country depended on his safe arrival. The pyramid was the answer to that need, at huge scale.

The earliest royal tombs

The first kings of united Egypt were buried at Abydos, in the south. Their tombs were underground chambers lined with mud brick and wood, covered by a low mound of sand. Beside them were rows of small graves for servants, who were likely buried at the same time. Nearby stood large rectangular enclosures built of mud brick, which are thought to have been the kings' funerary monuments. The best preserved, the so-called "fort" of King Khasekhemwy, still has walls over ten meters high.

Meanwhile, the officials of the new state were buried near the capital at **Saqqara**, in tombs that experts call **mastabas**. The name comes from the Arabic word for "bench," because from the outside, the tomb looks like a low bench: a flat-topped, rectangular structure of mud brick or stone, with sloping sides. Under the mastaba, a shaft led down to a burial chamber. Above it, a chapel held offerings.

A mastaba could be modest or enormous. Some of the earliest at Saqqara were over fifty meters long. Their outer walls were decorated with a pattern called "palace façade," which imitated the paneled walls of a royal residence. They show us that Egyptians were already thinking about a tomb as a permanent house for the dead.

What the Early Dynastic Period gave the pyramids

By about 2700 BCE, the Egyptians had invented several things that the pyramid builders would later depend on.

- **A writing system.** Hieroglyphs appeared around 3200 BCE and were soon joined by hieratic, a cursive script for papyrus. Writing made record-keeping and long-distance administration possible.
- **A standard system of measurement.** The **cubit**, based on the length of the forearm, became the main unit of length. The royal cubit measured about 52.5 centimeters, and was divided into 7 palms of 4 fingers each. We will see it again in the pyramid's dimensions.
- **Skilled stoneworking.** Egyptians had already been carving hard stone vessels for centuries, including granite, basalt, and diorite. This knowledge was a direct ancestor of pyramid masonry.
- **A professional bureaucracy.** Officials with titles such as vizier, treasurer, and overseer of works managed the state's resources and workforce.
- **A theology of kingship.** The king's divine role, his relationship with Ra and Osiris, and his need for an eternal home.

Nothing on this list required outside help. Each of them can be traced, step by step, in the archaeological record of Egypt itself. By the time of the Third Dynasty, the country had everything it needed to attempt something unprecedented.

The man who changed the plan

Around 2670 BCE, a king named **Djoser** came to the throne. His reign would prove to be a turning point, and the person usually credited with the change was not the king himself but his chief official, a man named **Imhotep**.

The next chapter will tell what Imhotep did. For now, it is enough to say that by the time Djoser took the throne, Egypt was a mature state: unified, literate, productive, and confident. It had the labor, the stone, the surplus, and the belief. All that was missing was the idea.

KEY TAKEAWAYS

Egypt's pyramids depended on five things: a river that provided transport and annual fertility; a surplus of grain taxed and stored by the state; a king believed to be divine and essential to cosmic order; a literate bureaucracy that could manage tens of thousands of workers; and a long tradition of stoneworking. Remove any one and the pyramids do not happen.

CHAPTER 2

The Step Pyramid: Imhotep's Stone Revolution

How one king and one remarkable official turned a mudbrick tradition into a mountain of stone.

Every history of the pyramids has a moment where the story lurches forward. In Egypt that moment came at Saqqara, around 2670 BCE, in the reign of King Djoser. Before it, royal tombs were made of mud brick, timber, and small amounts of stone. After it, kings built in stone, and the idea of a monumental tomb, a structure meant to last for eternity, had a form to match its ambition. The result is the Step Pyramid, the oldest of Egypt's pyramids and one of the earliest large buildings of cut stone anywhere in the world.

King Djoser and the Third Dynasty

The Third Dynasty began around 2686 BCE and is generally taken to mark the start of the Old Kingdom. The dynasty's first strong ruler is often listed as Nebka, though some Egyptologists place **Djoser** (whose Horus name was Netjerikhet, "divine of body") first. Later Egyptian king lists, such as the Turin Canon, credit Djoser with a reign of about nineteen years. Within that span he commissioned military campaigns in Sinai, where turquoise and copper were mined, and he sponsored a building project that dwarfed anything the country had seen.

We know surprisingly little about Djoser as a person. A seated limestone statue of him, found in a small closed chamber beside the pyramid, shows a king in a long robe and a striped wig, with large, intent eyes that once held inlays of rock crystal. The statue sits in a sealed box with two small holes at eye level. The holes were meant so that the king's spirit could look out at the offerings and the northern stars. A replica stands there today; the original is in the Egyptian Museum in Cairo.

The Man Who Designed It

Next to the pyramid, a fragment of a statue base was found in 1926 bearing the name of King Djoser and a list of titles. One of them, belonging to a man named **Imhotep**, reads in effect: chancellor of the king of Lower Egypt, first after the king of Upper Egypt, administrator of the great palace, hereditary noble, high priest of Heliopolis, and chief of sculptors. The title list strongly suggests that Imhotep was the king's chief minister and the head of the royal workshops.

Later Egyptian tradition remembered Imhotep as a sage, a scribe, an architect, and a healer. Over time, his reputation grew until he was worshipped as a god of medicine. The Greeks, who encountered his cult in the Late Period, identified him with their own healing god, Asclepius. A priestly account written more than two thousand years after Djoser, by the Egyptian historian Manetho, credits a man he calls Imouthes with inventing building in hewn stone, though the surviving fragments of Manetho are second-hand and need careful handling.

It is tempting to say Imhotep "invented" the pyramid, but the evidence is more interesting than that. We do not have the plans, and the monument evolved while it was being built. What we can say is that Imhotep was close to the center of a team that made a series of confident, creative decisions, and the later Egyptians credited him with the result. Whether he alone deserves the credit, the project was a decisive turning point.

From Mastaba to Pyramid

The Step Pyramid started as something familiar. The first phase was a **mastaba**, the flat-topped, rectangular tomb that officials and kings had been building for centuries, but with one big difference: it was made of stone. Archaeologists, using the term M1 for this first version, find that it was about 63 meters square and 8 meters high, built of small blocks of local limestone.

Then the plan changed. The mastaba was extended to the east, in a second phase called M2. After that, the builders added a series of rectangular layers on top, each smaller than the one below. The first pyramid phase, called P1, had four steps. The final form, P2, had six. The result was a stepped tower rising about 62 meters above the desert, with a base of approximately 109 by 125 meters. At that time it was the tallest structure humans had ever built.

> **KEY POINT**
> The Step Pyramid is not a single design executed once. It is the record of a series of changes made as the builders learned what worked. Reading it from the inside out is like reading a diary of architectural ideas.

The stepped profile has been interpreted in several ways. One possibility is that the steps were a symbolic stairway to the sky. Later religious texts speak of a stairway or ladder that the king uses to ascend to the heavens, and a stepped pyramid is an obvious image of that. Another is that stacked mastabas gave the

builders a way to build tall using small, manageable blocks and a stable layered construction. Both ideas can be true at once.

Small Stones, Big Ambition

One of the most striking details of the Step Pyramid is that it is built of **small stones**. The blocks are around 30 by 15 by 25 centimeters in the core, about the size and weight of a large mud-brick, which is roughly what the builders were used to handling. A single mason could lift one. The builders laid these blocks in sloping courses that lean inward toward the center, so that each layer rests against the one inside it. This produces a stable, self-supporting structure, a little like a stack of shallow bowls.

These inward-leaning layers were a practical way to build tall without having to lift heavy loads. They were also a sign that the builders were still thinking in terms of mud-brick construction. The blocks were stone, but the mind was brick. It was only over the following century that Egyptian builders would learn to handle much larger blocks.

The outer face was different. The pyramid was clad in a smooth casing of fine white limestone brought from the quarries at **Tura**, across the Nile. The same stone would later be used on the Great Pyramid. Little of the casing survives, but fragments show that it was dressed with care.

Below the Pyramid

If the pyramid is the visible part of Djoser's monument, the real core is underground. A square shaft, about seven meters on each side, plunges roughly 28 meters beneath the center of the structure. At the bottom is the burial chamber, made of granite blocks from Aswan, nearly 800 kilometers up the Nile. The roof of the chamber was sealed with a granite plug weighing around three and a half tonnes. The workers lowered the plug through a hole in the ceiling after the burial and then filled the shaft above.

Radiating from this central shaft is a network of tunnels and chambers, about five and a half kilometers in all. Some of the galleries are lined with blue-green faience tiles in a pattern that imitates reed matting, which would have given the underground rooms the appearance of a palace. In one of the chambers, three panels show Djoser in relief, running a ritual race and wearing the crowns of Upper and Lower Egypt. The ceiling of the burial chamber was painted with stars.

In the galleries, excavators found an extraordinary storehouse: tens of thousands of stone vessels, estimates run to about forty thousand, many of them carved from banded alabaster, schist, breccia, and other hard stones. Some bear the names of kings of the earlier dynasties. Why did Djoser stock his tomb with the relics of his predecessors? It may have been a pious act, a way of gathering the heritage of the kingdom around him in the afterlife. It may also have been a way of claiming legitimacy.

Like almost every royal tomb in Egypt, this one was plundered in antiquity. The excavators found only scraps of human remains in the burial chamber, and it is uncertain whether they belonged to the king.

The Enclosure: A City of the Dead

The pyramid does not stand alone. It sits in a huge walled enclosure, about 544 meters long and 277 meters wide, surrounded by a limestone wall about 10.5 meters high. The wall is decorated with a pattern of recessed panels and bastions, a design called **palace façade**, which imitates the architecture of royal palaces built in mud brick. There are fourteen gateways in the wall, but only one of them is real. The other thirteen are stone imitations, and they are false doors for the spirits.

Inside the wall, the architecture is a kind of stone theater designed for rituals. A long colonnaded entrance hall leads in from the south-east. Its ceiling imitates logs, and its limestone columns are carved to look like bundled reeds, a way of translating plant materials into stone. A large court was probably used for the king's **Heb-Sed**, a jubilee festival in which an aging king demonstrated his vigor by running a ceremonial course. Around the court stand dummy chapels, solid stone façades with no interior, which represent the shrines of the gods of Egypt's provinces. By "attending" the festival forever, in stone, the king would renew his rule in the afterlife.

Other features include the South Tomb, a second burial chamber and a mysterious structure whose purpose is still debated, and the "House of the North" and "House of the South," rep-

resenting the two halves of the kingdom. A statue chamber, called the **serdab**, contains the seated statue mentioned above. Altogether the complex is a model of the entire state, preserved in stone.

> **KEY POINT**
> At Saqqara, Egyptians did something new: they translated an architecture of perishable materials, such as reeds, wood, brick, and matting, into permanent stone. Nearly every column, doorway, and wall decoration in the enclosure imitates a plant or timber original.

The Cost and the Method

How did the builders manage a project on this scale? The answer is organization. The enclosure alone required quarrying, shaping, and hauling hundreds of thousands of blocks. The stone itself came largely from local quarries, including one beside the complex itself, and from the Tura quarries across the river. The granite came from Aswan, the alabaster from Hatnub, and the timber from Lebanon, which was beginning to be traded.

The builders left clues about their methods. The small block size allowed quick work with short ramps and simple lifting. The use of a local quarry reduced hauling. The sheer repetition of the form made the work easy to subdivide among crews. It was an enormous undertaking, but it was a controlled and logical one. The Egyptians were learning how to build in stone at scale, and the Step Pyramid served as a school.

Restoring the Complex

By the time modern archaeologists arrived, much of the enclosure lay in ruins. Excavation began in earnest in the 1920s under the British archaeologist Cecil Firth, and after 1931 it was taken up by the French architect **Jean-Philippe Lauer**, who devoted most of his long life to the site, working there for some seventy years and dying in 2001 at the age of 98. He reassembled the fallen columns, rebuilt sections of the wall, and reconstructed the entrance colonnade piece by piece, a feat of architectural jigsaw. Restoration work has continued in the twenty-first century. The pyramid itself was reinforced in recent years after concerns about its stability, and the monument was reopened to visitors in 2020.

Successors and Experiments

Djoser's immediate successors tried to build their own stepped pyramids, but they did not finish them. **Sekhemkhet**, who ruled a few years after Djoser, began a larger complex just to the southwest of Djoser's. It was abandoned when only the lowest layer was complete. Excavators in the 1950s found a sealed alabaster sarcophagus in its underground chambers, and when it was opened before an audience, it was empty. Farther north, at Zawyet el-Aryan, the so-called Layer Pyramid, usually attributed to a king named **Khaba**, also stands in an unfinished state.

Several small stepped pyramids, probably built in the late Third Dynasty, stand at provincial sites across Egypt, at Seila, Elephantine, Edfu, and elsewhere. Their purpose is uncertain.

They may have been markers of royal authority in the provinces, a way for the king to make his presence visible throughout the land.

Finally, at Meidum, **Huni**, the last king of the dynasty, or **Sneferu**, the first king of the next, began a monument that would become the first true pyramid. That story is the subject of Chapter 4.

What the Step Pyramid Teaches Us

The Step Pyramid is more than a curiosity or a stepping stone to something greater. It is a work of art and architecture in its own right, and it reveals some basic facts about how the pyramid age began.

- **The pyramid grew out of Egyptian tradition.** The mastaba, the mound, the enclosure, and the stone vessel all had earlier roots. Nothing about the Step Pyramid required a source outside Egypt.
- **The builders learned by doing.** The changing plans, small blocks, and inward-leaning courses show a technology in development.
- **The project was about the state as much as the king.** The complex is a model of Egypt, and the effort required to build it was a way of organizing the country.
- **Imhotep's reputation was earned.** Whatever the details of his role, the Egyptians remembered the architect of this complex as someone special, and they were right.

KEY TAKEAWAYS

The Step Pyramid at Saqqara (c. 2670 BCE) was the first large stone monument in Egypt. It grew from a mastaba into a six-step pyramid about 62 meters high, with an underground labyrinth, a granite burial chamber, and a vast ritual enclosure. It was built of small blocks, laid in inward-leaning courses, and its architects translated reed and timber forms into stone. Its design is credited to Imhotep, later worshipped as a god.

CHAPTER 3

Why Build a Pyramid? Sun, Stars, and the Dead King

The beliefs behind the shape, and the practical reasons that kept the project going.

A pyramid is an odd choice for a tomb. It is wasteful of stone. It is laborious to build. It is visible from miles away, which is the opposite of what a person wants if the aim is to hide a burial from robbers. And yet for roughly a thousand years Egyptian kings chose it again and again. To understand why, we have to set aside our own assumptions about what a tomb is for and think about what an Egyptian king needed.

The two destinies of the king

The earliest detailed evidence for royal beliefs comes from the **Pyramid Texts**, a collection of spells carved on the walls of the burial chambers and corridors of pyramids from the end of the Fifth Dynasty onward. We will look at them closely in Chapter 17. For now, what matters is that they describe the dead king's journey in two ways that seem to overlap.

In the first, the king joins the sun god **Ra**. He ascends to the sky, boards the sun's boat, and travels with Ra across the heavens by day and through the underworld by night. Each dawn he

is reborn along with the sun. In the second, the king becomes **Osiris**, the god of the dead and of fertility, who rules the underworld, and he is also linked with the stars that never set, the "imperishable ones," circling the northern sky.

Egyptians did not feel the need to pick one. Different spells emphasize different destinies. The pyramid made sense in both of them.

The pyramid and the sun

The most widely accepted explanation for the pyramid's shape connects it with the cult of the sun at **Heliopolis**, a city just northeast of modern Cairo. Heliopolis, which the Egyptians called Iunu, was home to the most powerful priesthood in Old Kingdom Egypt. Its theology said that in the beginning there was only a dark, endless sheet of water called Nun. From it rose a mound of earth, the **primeval mound**, and on it appeared the creator god Atum, who then produced the other gods. In Heliopolis the mound was represented by a sacred stone, the **benben**.

The benben is thought to have been a pointed or conical stone. By the time of the pyramids, the top of a pyramid and, later, the top of an obelisk were called a *benbenet*. The tip of a pyramid is a stone version of the first spot of land to catch the first rays of sunrise.

Look at a pyramid on a hazy day, when the sun breaks through the clouds and sends shafts of light down to earth. The rays fan out and strike the ground in a triangular pattern very like the pyramid's sides. Egyptian texts describe the sun's rays as

a ramp or stairway on which the king climbs to heaven. It is likely that the Egyptians saw the pyramid's smooth, sloping sides as a petrified version of those rays. The idea is poetic, but it fits the language of the Pyramid Texts, and it is accepted by many scholars as part of the explanation.

> **KEY POINT**
> Many Egyptologists read the true pyramid as a solar symbol. Its name for Khufu's monument, *Akhet-Khufu*, means "the Horizon of Khufu." The *akhet* was the place where the sun rises and sets, a point of transition between worlds. A pyramid was a place of transition for a king.

The timing supports this reading. The first true pyramids were built during the Fourth Dynasty, when the sun cult was growing in influence. Royal names began to include the name of Ra, as in Djedefre and Khafre, and the king took the title "Son of Ra." During the Fifth Dynasty, kings also built dedicated **sun temples** with massive obelisk-like monuments, though only two of them have been found. The pyramid was an expression of a royal theology that centered on the sun.

The pyramid and the stars

There is a second thread in the story, and it concerns the night sky. Egyptians watched the stars carefully. The stars that circled the north celestial pole without ever setting were called the **imperishable stars** or "the ones who know no weariness." To the Egyptians they were a model of what it meant to live forever. In the Pyramid Texts, the king is said to join them.

This helps explain the orientation of pyramids. Their sides are aligned very accurately with the four cardinal directions, and the entrances to most Fourth Dynasty pyramids face north, toward the region of the sky where the circumpolar stars turn. The inner passages generally rise or descend along a north-south axis. In the Great Pyramid, the passages known as the "air shafts" from the King's Chamber point at the sky, one roughly toward the southern sky and one toward the northern, and these are often discussed in connection with the stars. We will return to them in Chapter 15.

The sun and the stars are not in competition. In Egyptian thought, the cosmos was a layered whole in which the king might travel by day with the sun, by night through the underworld, and take his place among the circumpolar stars. A tomb that connected earth and sky, and oriented to true north, expressed all of these ideas at once.

Osiris and the underworld

The other dimension is the underworld, called the **Duat**. Osiris, in Egyptian myth, was a king murdered by his brother Set, revived by his wife Isis, and made ruler of the dead. His son Horus defeated Set and took the throne of the living. The king, in life, was Horus. After death, he became Osiris, and his heir became the new Horus. This cycle gave the transfer of power from one king to the next a mythic frame.

The Pyramid Texts mention Osiris extensively, especially in the later pyramids. Many spells treat the king's body, limb by limb, as sacred, and they work to reassemble it as the body of

Osiris was reassembled. A tomb was thus also a place where the king's body was transformed from a corpse into a god.

A machine for resurrection

The pyramid itself was only one part of a larger system. The complex included a **valley temple** at the edge of the floodplain, a long **causeway** leading up to the plateau, a **mortuary temple** against the east face of the pyramid, and a small **cult pyramid**, sometimes called a satellite pyramid. In the temple, priests presented offerings of food, drink, linen, and incense every day. The king's spirit was thought to receive them, and the offerings were supposed to continue forever.

Seen this way, the pyramid is less like a gravestone and more like the central piece of a machine, one meant to ensure the king's resurrection. The burial chamber protected the body. The temple fed the spirit. The inscriptions and rituals provided the magic. The shape connected the king to the sun. The orientation tied him to the stars. The location, on the western edge of the valley, associated the pyramid with the realm of the dead. Everything worked together.

The practical side: power, prestige, and the state

Religion explains the form of the pyramid, but it does not by itself explain the scale. Why were the Great Pyramids so enormous? Several answers have been proposed, and they are not exclusive.

Royal authority. The pyramids were statements of power. Khufu's pyramid, rising above the plateau, told every person who saw it that the king was divine and that the state could mobilize on an unmatched scale. Constructing it also strengthened that state.

National integration. In his book *The Riddle of the Pyramids* (1974), the physicist Kurt Mendelssohn argued that the real purpose of the pyramids was social. Building one required the coordination of the whole country, and the act of building taught people to work together, creating a bureaucratic state out of scattered communities. The idea has been criticized for assuming too much about the primitive state of Egypt before the pyramids, but a milder version of it is widely accepted. The pyramids were both a product of a strong state and a tool for strengthening it.

Redistribution. Workers were paid in food, clothing, and status. A pyramid project could be seen as a giant public works program, redistributing the surplus from the countryside to the labor force and the craftsmen.

Competition and emulation. Kings wanted to outdo their predecessors, and the pyramid became a symbol of the dynasty. The size peaked with Khufu and then dropped, which suggests that the cost was felt.

Security. A solid mass of stone is hard to rob. Egyptians understood that robbers were a threat, and a pyramid, with its sealed entrances, hidden passages, and blocking stones, was a heavy-duty answer. As later history shows, it was not entirely successful.

What the pyramid was not

Over the centuries, people have proposed other purposes. It helps to list the most common ones and say why they do not hold up.

- **A granary.** In late antiquity and the Middle Ages, Christian and Muslim writers sometimes thought the pyramids were the granaries that Joseph built in the Bible's Book of Genesis. A thirteenth-century mosaic in St. Mark's Basilica in Venice shows pyramids with doors, labeled as granaries. But the pyramids have no space for storing grain, and no ancient Egyptian text suggests it.
- **An observatory.** The pyramids are carefully aligned to the stars and the sun, but they are solid stone masses. Nothing about the structure allows observation from within or on top, and the Egyptians had no need for it. Their astronomy was practiced from the ground and was tied to religion and the calendar.
- **A water pump, a power plant, or a landing beacon.** These come from twentieth-century speculation, and we will examine them in Part Five. None has any basis in the physical evidence.
- **Not a tomb at all.** Some people argue that the Great Pyramid never held a body because none was found. This is a fair observation, but it is not an argument for another purpose. We will come to the evidence in the next chapters, but all pyramids except the Great Pyramid have burial evidence, and the Great Pyramid's design is the same.

MYTH AND FACT

Myth: "No mummy was ever found in the Great Pyramid, so it can't have been a tomb." **Fact:** The pyramid was robbed in antiquity, and by the time medieval explorers entered, it was already empty. An absence of body is just what we would expect from a plundered tomb. The structure's chambers, sarcophagus, passages, plugs, and temple complex all follow the pattern of a royal burial monument.

Doubts in the ancient world

It is interesting that doubt about the pyramids' purpose is quite old. The Greek historian Diodorus Siculus, who visited Egypt around 60 BCE, reported that the priests told him the kings who built the two largest pyramids, Khufu and Khafre, were not buried in them. He said they had feared the people's hatred, and had ordered their bodies to be hidden elsewhere. This statement is not supported by evidence, and it is a good example of how stories about the pyramids can arise when time passes. By the first century BCE, the pyramids were already some two and a half thousand years old, and many of the details had been forgotten. We should treat Diodorus's account as hearsay, not as evidence of anything about Khufu's actual burial.

A tomb is not the end of the story

One more point deserves attention. The Egyptians did not stop caring for a king when he was buried. Priests were assigned to each pyramid, and endowments of land and produce were set aside to support the cult. Some pyramid towns, such as the one beside Sneferu's Bent Pyramid, were occupied for centuries. The cult of Khufu and Khafre continued into the Middle Kingdom and, in revived form, even into the Late Period, some two thousand years after their reigns. A pyramid was a long-term institution.

KEY TAKEAWAYS

The pyramid shape probably reflects the sun cult of Heliopolis (the primeval mound and the sun's rays), the star-centered beliefs of the Pyramid Texts, and the king's identification with Osiris. The monument was part of a larger funerary complex staffed by priests. Its scale reflects royal power and the ability of the Egyptian state to organize labor. Alternative explanations, such as granaries, observatories, or machines, have no good evidence behind them.

PART 2

The Age of the Great Pyramids

Four generations, three kings, and the biggest stone buildings on Earth

CHAPTER 4

Sneferu's Experiments: Collapse, Bend, and Success

The Fourth Dynasty's founder built three pyramids, learned from each, and got it right on the third try.

The Great Pyramid of Khufu is the famous one, but it did not come from nowhere. Its immediate parent was a king named **Sneferu**, who ruled for roughly twenty-four years starting around 2613 BCE. Sneferu is one of the great builders of world history, and yet most people have never heard of him. During his reign Egyptian architects tried, failed, adapted, and finally succeeded at making a smooth-sided pyramid. Without Sneferu's three monuments there would have been no Giza.

The scale of the work

Sneferu is credited with building, or completing, at least three large pyramids: one at **Meidum**, and two at **Dahshur**, known as the **Bent Pyramid** and the **Red Pyramid**. Some Egyptologists add a small stepped pyramid at **Seila** in the Faiyum. The combined volume of stone in his large pyramids is often estimated at more than three and a half million cubic meters, which is greater than the 2.6 million cubic meters in Khufu's Great Pyramid. In other words, Sneferu moved more stone than his son did.

That fact alone should change how we picture the Fourth Dynasty. The Great Pyramid was not a freak project. It was the result of a generation of intensive building, of administrators who had already learned how to quarry, haul, and set vast quantities of stone, and of a labor system that had been running for decades. When Khufu started his pyramid, the people who knew how to do it were already in place.

Meidum: the first true pyramid, and a failure

The oldest of Sneferu's pyramids, at **Meidum**, about 100 kilometers south of Cairo, is a curious sight today. It looks like a three-stage tower rising out of a hill of rubble. The rubble is the pyramid's own outer casing, which collapsed.

The monument seems to have begun as a step pyramid in the style of Djoser's, possibly under Sneferu's predecessor Huni. Over time, it was enlarged in stages. Egyptologists label these phases E1, E2, and E3. In its first form it had seven steps, in its second, eight. In the final phase, the steps were filled in with packing stone and covered with smooth limestone casing, producing the first true pyramid, a structure with smooth, sloping sides.

The inside of Meidum has some features that look forward to the Great Pyramid. The entrance is on the north face and leads down a long, sloping passage into a corridor, then up a short shaft to a burial chamber with a corbelled roof, a vault made of stones laid in overlapping courses so that each projects slightly beyond the one below. The chamber sits at the level of the pyramid's base, not below it.

The collapse of Meidum is a famous puzzle. In the 1970s, the engineer-physicist Kurt Mendelssohn argued that the pyramid had suffered a catastrophic failure during construction, that the outer layers had slid off, and that the disaster had panicked the builders and sent Sneferu to a new site. This story was popular for a long time, but the evidence for it is weak. There is no firm proof that Meidum collapsed during Sneferu's reign. Excavations have shown that the pyramid was still in use, with a mortuary temple and a cult, long after the supposed disaster. A more modern view is that the outer casing came down much later, perhaps over centuries, as a result of structural weaknesses combined with stone-robbing and weathering. The timing of the collapse is still debated.

KEY POINT

Meidum teaches two lessons. It shows the Egyptians learned how to convert a step pyramid into a smooth one. And it shows the danger of a pyramid built by adding layers to an earlier core: the new layers were not bonded to the old, and the outer casing sat on a base of loose sand, which made it vulnerable to slipping.

The Bent Pyramid: a change of mind

About 50 kilometers north, at **Dahshur**, Sneferu started again. The result is the **Bent Pyramid**, one of the most distinctive structures in Egypt. The lower portion rises steeply, at an angle of about 54 degrees 27 minutes. About halfway up, at a height of roughly 47 meters, the slope abruptly changes to about 43 degrees 22 minutes. The upper half is considerably flatter. The result is a shape like a pyramid with a hip, and it makes the monument impossible to confuse with any other.

The pyramid is 188 meters on each side and about 105 meters high. It is far better preserved than most. The outer casing of fine Tura limestone still covers much of the surface, and from a distance it gives a good idea of how the Great Pyramid must have looked when it was new.

Why the bend? Several explanations have been offered.

- **Structural problems.** The pyramid was built on a layer of marl and clay that could not support the weight of a steep, tall structure, and builders noticed signs of distress, such as cracks and settling, in the inner chambers. To reduce weight, they flattened the angle. Evidence for this includes cracks in the interior and a change in construction method: the lower part was built with inward-leaning courses, while the upper part was laid in horizontal courses.
- **Time pressure.** The king may have died or grown ill, and the builders, needing to complete the monument quickly, reduced the volume of stone needed to finish it.

- **Deliberate design.** A few scholars have suggested the shape was intentional, though this is not widely accepted.

The first two explanations are not mutually exclusive, and many Egyptologists favor a mixture: the builders discovered a problem, corrected it, and finished the structure in a form that would stand. The Bent Pyramid is still standing after four and a half thousand years, which suggests the adjustment worked.

Two entrances, two chambers

The interior of the Bent Pyramid is unusual. It has two separate entrances, one on the north face, as was usual, and one on the west face, high up on the structure. The northern passage descends to a lower chamber with a high corbelled vault, about 17 meters tall. The western passage leads to an upper chamber, also corbelled. A tunnel links the two systems.

This unusual layout probably reflects changes during construction. Experts think the western entrance and upper chamber may have been added when the builders found problems in the lower chamber. Wooden beams of Lebanese cedar, still visible in the corbelled chamber, were used as bracing, possibly to prevent the walls from collapsing as the pyramid settled. Sliding stone slabs, or portcullises, were installed in the western passage to seal the way after the burial. The system of blocking devices was an early version of the techniques that would be developed further in the Great Pyramid.

A small satellite pyramid stands next to the main one, and a temple on the east side held a pair of royal stelae. To the east, a

valley temple is decorated with relief carvings that show processions of estates personified as women bringing offerings, a good example of how the king's divine economy was represented.

The Red Pyramid: the first success

Sneferu's third and final pyramid is the **Red Pyramid**, also at Dahshur, about two kilometers north of the Bent. It takes its name from the reddish color of its local limestone core, which became visible after the casing was stripped. It is roughly 220 meters wide and 105 meters high, which makes it the third-largest pyramid in Egypt, after the Great Pyramid and the Pyramid of Khafre.

The Red Pyramid was built with an angle of about 43 degrees 22 minutes from the beginning, the same shallow angle as the top half of the Bent Pyramid. The lower slope made the structure far more stable, and the pyramid has survived in good condition. The courses are laid in horizontal layers, the best method for supporting weight, and the foundations were laid on firm bedrock.

Inside, the Red Pyramid has three chambers in a line, each with a corbelled roof about 12 meters high. The stone courses of the vaults are beautifully fitted. A single entrance on the north face leads down a long descending corridor. The pyramid is open to visitors today, and it is one of the best places to experience how a pyramid's interior was designed. The chamber's smell, because of the ammonia from centuries of bat droppings, can be overwhelming, but the architecture is remarkable.

There was apparently a burial: during excavations in the early 1950s, workers found a few human remains, including pieces of a mummified body, in the burial chamber. Whether they belonged to Sneferu cannot be proven.

The Red Pyramid is the first successful true pyramid in history. It is the answer to a problem the Egyptians had been working on for decades: how to build a tall, smooth-sided, stable pyramid. The solution was simple, and it was based on experience. Keep the angle low, build in horizontal courses, lay it on bedrock, and use good stone.

> **KEY POINT**
>
> **Three pyramids, three lessons.** Meidum taught the builders how to turn a stepped structure into a smooth one, and how not to bond the new layers. The Bent Pyramid taught the importance of foundations and of controlling weight. The Red Pyramid put all the lessons together.

The learning curve

The sequence of Sneferu's pyramids is one of the best pieces of evidence for the Egyptian origin of pyramid technology. A reader might be forgiven for thinking that the pyramids were designed by a super-intelligent mind that knew exactly what to do. Instead, we find a series of experiments. The builders tried a design and found its limits. They adjusted. They added passages and blocking devices. They changed the slope. They went back to basics and succeeded.

This is what real technological development looks like. Skilled people learn from their errors, record what they learn, and apply it on the next project. Notice also that many of the mistakes are what a competent but inexperienced team might make: building on soft ground, building too steeply, and bonding new layers badly. The Great Pyramid, in comparison, was built on the best bedrock on the plateau, at a moderate angle, with carefully fitted horizontal courses. The lessons had been learned.

MYTH AND FACT

Myth: "The pyramids appeared suddenly, fully formed, with no precursors, so they must have had outside help." **Fact:** The sequence of pyramids from Djoser to Sneferu to Khufu spans about 150 years and includes visible mistakes, redesigns, and unfinished projects. It is a classic example of incremental engineering.

Sneferu the person

Our picture of Sneferu is helped by a surprising source: the stories told about him a thousand years later. In the Middle Kingdom, Egyptian scribes wrote tales about him. The *Westcar Papyrus*, a collection of tales set in the court of Khufu, includes a story in which Sneferu's chief lector priest tells the bored king to row on the palace lake accompanied by twenty beautiful women in fishnet dresses. When one of the women drops a turquoise pendant in the water, the priest folds back one half of the lake to retrieve it. The story is fiction, written long after Sneferu's death, but it shows that he was remembered as an easygoing and generous ruler.

A prophecy text from the Middle Kingdom, the *Prophecy of Neferti*, likewise looks back on Sneferu's reign as a golden age. His reputation contrasts with that of his son Khufu, who appears in the Westcar tales as an imperious, curious king, and in Herodotus as a tyrant who closed the temples and forced his people to labor. It is worth remembering that Khufu's bad reputation is a late tradition. It does not appear in sources from his own time, and Egyptians continued to worship him as a god for centuries.

Where Sneferu was buried

One mystery remains. Sneferu built three large pyramids, but he can be buried in only one of them. Which one? Historians have debated this for decades. The Red Pyramid, as the final and most successful structure, is the most likely candidate, and the fragments of a body found there support that view. But the Bent Pyramid also has a complete funerary complex, and the Meidum pyramid has a burial chamber too. A plausible explanation is that each pyramid served a function: one may have been a cenotaph, a symbolic tomb. Or the king may have switched his plans as the projects progressed. We do not know.

The legacy

By the end of Sneferu's reign, Egyptian architects had learned the key lessons of pyramid construction. They knew how to quarry and move millions of cubic meters of stone, how to set the foundation, how to select an angle, and how to build corbelled vaults to support a chamber under immense weight. They also had a trained workforce, a logistical system, and a royal ideology that put the pyramid at the center of the monarchy. The stage was set for his son.

KEY TAKEAWAYS

Sneferu (c. 2613–2589 BCE) built three large pyramids and moved more stone than Khufu. Meidum was converted from a step pyramid, with a debated collapse. The Bent Pyramid changed angle partway, most likely in response to structural distress. The Red Pyramid, at a stable 43 degrees, was the first successful true pyramid. The sequence shows trial, error, and learning, which is strong evidence for the human, Egyptian development of pyramid technology.

CHAPTER 5

Khufu and the Great Pyramid

The largest and most precisely built of the pyramids: what it is, what is inside it, and what we know about the man who built it.

Between about 2589 and 2566 BCE, a king named **Khufu** ruled Egypt and built a monument that would remain the tallest human-made structure on the planet for nearly four thousand years. Its Egyptian name was *Akhet-Khufu*, "the Horizon of Khufu." The Greeks called him Cheops, and called his monument the Great Pyramid, a name that has stuck. It is the oldest of the Seven Wonders of the Ancient World and the only one that still stands.

The man

For the most famous builder in history, Khufu is strangely hard to see. His only securely identified statue is a small ivory figure, about 7.5 centimeters high, found by the British archaeologist Flinders Petrie at Abydos in 1903. It shows a king in the red crown, seated on a throne, holding a flail. That is nearly all we have of his portrait. The rest of his image comes from the monument itself.

He was the son of Sneferu and Queen Hetepheres I. His reign is credited with about 23 years in the Turin Canon, though other evidence suggests it may have lasted into the mid-twenties;

Herodotus claimed 50 years, which is certainly too long. His family tree matters, because many members were buried in tombs beside the pyramid. His half-brothers, sons, and officials had tombs laid out in neat rows around the great monument, as if the whole court had moved in. His mother's tomb was found in 1925, a shaft near the pyramid containing her funerary furniture, including a gilded bed, a carrying chair, and a box that held her canopic jars. The sarcophagus was empty, which has led to long debate over where her body went.

> **KEY POINT**
> We know Khufu mainly through his monuments and the monuments of his family and officials. This is the pattern for most of Egyptian history: the people we know best are the ones who built tombs.

Choosing the site

Khufu abandoned the sites that his father had used at Dahshur and moved north to the **Giza plateau**, on the west bank opposite the old capital at Memphis. It was a smart choice. Giza was close to the river, so that stone and supplies could arrive by water. It sat on a broad limestone plateau with solid bedrock close to the surface. It had a quarry on the spot, immediately south of the pyramid site, which supplied much of the core masonry. And it was high enough to stay dry during the inundation but close enough to the floodplain for a harbor.

The builders chose a spot at the northeastern edge of the plateau, where the bedrock rises into a natural knoll. They left part of this rock in place as the core of the pyramid, which cut the volume of stone they had to bring in. Using a natural rise saved effort, and the pyramid still stands on one of the finest foundations in Egypt.

The numbers

The Great Pyramid's dimensions have been measured repeatedly, beginning with Petrie in 1880 and 1882. Here are the main numbers, using modern surveys. They are given alongside the Egyptian unit, the **royal cubit**, which is about 52.4 centimeters.

Feature	Metric and imperial	Cubits
Original height	about 146.6 m (481 ft)	280
Current height	about 138.5 m (454 ft)	—
Base side (average)	about 230.3 m (756 ft)	440
Slope of faces	about 51° 51′	rise 14 : run 11
Base area	about 53,000 m² (13 acres)	—
Volume	about 2.6 million m³	—
Number of blocks	about 2.3 million (estimates 2.3–2.6 million)	—
Estimated mass	about 5.9 million tonnes	—

Feature	Metric and imperial	Cubits
Typical block weight	about 2.5 tonnes (range: under 1 to over 15)	—

The loss of height is because the capstone and the top courses are missing and the casing stones have been stripped off. The slope of 14 vertical to 11 horizontal, known as a *seked* of 5½ palms in Egyptian terms, is expressed in the same unit that later Egyptian mathematical texts such as the Rhind Papyrus use for the slopes of pyramids. This is evidence that the builders had a simple, practical system for specifying slopes. They did not need advanced mathematics, only a consistent rule.

A well-known curiosity is that the perimeter of the base divided by twice the height comes close to the value of pi. Using 440 cubits for each side and 280 for the height, the figure is 4 × 440 / 280 = 6.2857, which is 2 × 3.1428. This is within a fraction of a percent of 2π. Some writers have taken this as proof that the Egyptians knew pi to several decimal places. The simpler explanation is that a seked of 5½ palms produces a ratio close to pi, a consequence of the slope chosen for other reasons, and that the Egyptians never wrote about it as a deliberate relationship. We return to this in Chapter 21.

Precision

The Great Pyramid is not just huge. It is also astonishingly accurate. The base is almost a perfect square: the four sides differ in length by no more than about 4 centimeters over a length of 230 meters. The corners are nearly right angles. The entire base is level to within a couple of centimeters across more than 53,000 square meters. And the sides are aligned with true north, south, east, and west with an error of only about three arcminutes, or one-twentieth of a degree.

For a monument built with copper tools, cord, wood, and water, this is a remarkable achievement. Chapter 9 explains how it might have been done. For the moment, note that the precision shows the effort and skill of the builders. They were not accidental, and not sloppy. They were committed to getting it right.

The outside

When finished, the Great Pyramid was covered with an outer casing of fine white limestone from the **Tura** quarries across the river, with joints so tight that Herodotus and later visitors reported that the whole structure looked like one continuous surface. The casing blocks were cut and dressed on-site, and the final smoothing was probably done from the top down after the core was complete. The sides sloped slightly inward at their centers, a feature that can be seen from the air when the sun is low, producing a subtle eight-sided shape. Egyptologists are not sure whether this was a deliberate refinement or a settling effect.

Most of the casing is gone. Medieval builders in Cairo stripped it for buildings and, in some accounts, an earthquake in 1303 loosened many stones, which were then reused. A few casing stones survive at the base on the north side, and the stretch of the casing still stands at the top of Khafre's pyramid next door, giving us a sense of what the original looked like.

At the apex, the pyramid was crowned by a capstone called a **pyramidion**. The original is lost. Some researchers suggest that it was covered with a thin layer of gold or electrum, a natural gold-silver alloy, which would have caught the sun, but there is no direct evidence for this. Later Egyptian pyramidia, for example from the pyramids of the Twelfth Dynasty, were made of black granite and inscribed with sun symbols.

Entering the pyramid

The original entrance is on the north face, about 17 meters above the ground, in a gabled limestone frame. A narrow, sloping corridor, the **Descending Passage**, runs about 105 meters down through the masonry and into the bedrock, ending in a rough, unfinished chamber about 30 meters below the base. This **Subterranean Chamber** was never completed. Why the builders left it unfinished is unknown, but the most common view is that the king's plans changed during construction, and the burial was moved higher up in the pyramid.

A short distance inside the Descending Passage, a second corridor, the **Ascending Passage**, branches off and climbs upward. In its lower part, the entrance to this passage was blocked by a series of three large granite plugs, designed to be slid into

place after the funeral, sealing the upper chambers. The plugs are still there. When the caliph's workmen entered the pyramid in the ninth century, they could not break the granite, so they dug a tunnel around the plugs through the softer limestone to reach the passage beyond. That tunnel, the **Robbers' Tunnel**, is the one visitors use today.

The Ascending Passage opens into a spectacular space, the **Grand Gallery**.

The Grand Gallery

The Grand Gallery is perhaps the finest piece of architecture in the Old Kingdom. It is a high, narrow corridor, about 47 meters long and 8.6 meters tall, rising at the same slope as the passage below, about 26 degrees. Its walls are made of polished limestone, and they rise in seven stepped overhangs, each course projecting inward by about 7.5 centimeters beyond the one below. At the top, the ceiling is only about a meter wide. The result is a corbelled vault: a sort of pointed stone arch.

Along each side at the floor are shallow ramps, with rectangular holes cut into them at regular intervals. Their purpose is debated. Some propose that they held wooden beams that helped to store the granite plugs before they were released. Others see it as part of the structure's ritual symbolism. Whatever the purpose, the gallery is a masterpiece of stone engineering, and it was built to carry heavy loads. Above it, the pyramid's weight presses down, and the corbelled walls transfer that weight sideways and downward in a controlled way.

At the upper end of the Grand Gallery, a high step leads to a low, narrow antechamber and then to the **King's Chamber**.

The King's Chamber

The King's Chamber is a plain rectangular room, about 10.5 meters long, 5.2 meters wide, and 5.8 meters high. It is built entirely of red granite from Aswan, with walls lined with polished blocks and a flat ceiling made of nine huge granite slabs. Above these are five further low chambers, called **relieving chambers**, separated by more granite slabs. The topmost has a gabled roof of limestone. Together the granite beams and the stacked layers help to spread the weight of the masonry overhead to the sides, protecting the chamber from crushing.

The estimated weights of the granite slabs above the chamber range from about 25 to about 80 tonnes each. The total weight of the nine ceiling slabs alone is perhaps 400 tonnes. Moving and lifting them more than 40 meters above the base is one of the great engineering feats of the project, and we will come back to it in Chapter 12.

The only object in the room is a granite **sarcophagus**, lidless and cracked at one corner. It is slightly too big to have been brought in through the entrance passage, so it must have been placed there during construction, before the chamber was roofed over. It is carved from a single block of red granite and sits aligned with the north-south axis of the pyramid. The lid, if there was one, is lost. In 1837 the explorer Howard Vyse, searching for hidden chambers, used gunpowder to blast his way into the relieving chambers above the King's Chamber. In the

topmost of those chambers, he found red-painted graffiti left by the work gangs, including the cartouche of Khufu. These inscriptions are among the best direct evidence that the pyramid was built in Khufu's time.

The Queen's Chamber and the shafts

Back at the lower end of the Grand Gallery, a level passage leads horizontally to a smaller room that has been known for centuries as the **Queen's Chamber**. The name is a medieval misnomer. It does not seem to have been intended for a queen, and the queens of Khufu were buried in separate small pyramids beside the main one. The room has a high, gabled roof, and in the east wall, a tall niche that is corbelled. The niche may have held a statue, but no one knows.

From the north and south walls of both the King's Chamber and the Queen's Chamber, narrow shafts about 20 centimeters across lead outward. The ones in the King's Chamber emerge at the outside of the pyramid. The ones in the Queen's Chamber stop short of the surface. They were discovered in 1872 by the engineer Waynman Dixon, who noticed a crack in the wall and followed it. The shafts have since been explored with robots, in 1993 and 2002, and with a miniature camera, in 2011. In the southern shaft of the Queen's Chamber, the first robot found a small limestone door with two copper handles. In 2002 a robot drilled through it and found another door behind it. In 2011 a robot with a flexible camera, called Djedi, saw painted marks on the back of the door, in red ocher. No one knows what they mean.

The shafts are often described as "air shafts," but the Egyptians had no need to ventilate a sealed tomb. A more likely explanation is that they have a religious function, perhaps allowing the king's spirit to leave the pyramid and ascend to the sky, and they may also have been aligned with certain stars. We will come back to this in Chapter 15.

> **KEY POINT**
> The Great Pyramid has three chambers at three levels: the unfinished Subterranean Chamber in the bedrock, the so-called Queen's Chamber in the middle, and the King's Chamber high up. This arrangement suggests that the plan changed at least twice during construction as the builders moved the burial chamber upward.

The complex around the pyramid

Like all pyramids, the Great Pyramid was part of a complex. On the east side stood a **mortuary temple**, of which only the black basalt paving survives. From its entrance a long **causeway**, partly roofed and decorated with relief carvings, ran down to a **valley temple** near the harbor. The causeway has been largely destroyed, but its foundations have been traced for several hundred meters.

Next to the pyramid are three smaller pyramids for queens, labeled G1a, G1b, and G1c, and a small cult pyramid just southeast of the main one. Five pits cut in the bedrock around the pyramid once held **boats**. Two of them, on the south side, were sealed with limestone slabs. In 1954, the Egyptian archaeologist

Kamal el-Mallakh opened one of them and found a complete ship, 43.6 meters long, made of Lebanese cedar, which had been taken apart in 1,224 pieces and carefully packed. It was rebuilt over years and displayed in a museum beside the pyramid. The first ship was moved to the Grand Egyptian Museum in 2021. A second pit, opened in the 1980s, held another disassembled ship, which an Egyptian-Japanese team has been studying and restoring.

Around the pyramid, in neat grids, are the tombs of Khufu's family and his court officials, arranged in what archaeologists call the *Eastern Field* and the *Western Field*. Most are mastabas. They contain some of the finest early reliefs, and several held the strange limestone "reserve heads," portrait heads without bodies that were placed in the burial shafts.

The overseer: Hemiunu

Among the tombs in the Western Field is that of **Hemiunu**, Khufu's nephew and his vizier. Among his titles was "Overseer of all the king's works," which makes him a good candidate for the chief architect or project manager of the Great Pyramid. His tomb, labeled G4000, is one of the largest in the cemetery. In its serdab, archaeologists found a life-sized limestone statue of him, now in Hildesheim, Germany. It shows him as a heavy, seated man with a rather stern expression and the folds of fat that indicated a person who ate well, which marked social status in ancient Egypt.

It is not safe to say that Hemiunu designed the pyramid single-handedly, and no plan or drawing survives. But the overlap

of his titles and his status within the court makes him a plausible candidate for the person who directed its construction.

How long did it take?

If Khufu reigned for about 23 to 27 years, and work on the pyramid presumably began early in his reign, the construction could have taken about 20 years. This is the figure usually given, and it matches an old claim by Herodotus. The evidence from the logbook found at Wadi al-Jarf, which we will meet in Chapter 11, dates work on the casing stones to around Khufu's 26th or 27th year, which suggests that the pyramid was nearing completion at that time.

Assuming the 20-year figure, simple arithmetic tells us how fast the work must have been. With about 2.3 million blocks to place, and working perhaps 300 days a year, the crew had to place about 380 blocks a day. If they worked ten-hour days, that is roughly 38 blocks an hour, or one block every one and a half to two minutes, day after day, for two decades. That is a stunning pace, but it was possible with a large, well-organized workforce working on many fronts at once, since the quarrying, hauling, dressing, and laying all happened in parallel.

WORKED EXAMPLE

Blocks per day. 2,300,000 blocks ÷ (20 years × 300 days) = about 383 blocks per day. At 10 working hours, 383 ÷ 10 = about 38 blocks per hour, or one every 1.6 minutes. Early courses at the wide base would go much faster because most of the blocks lie there; the final courses near the top would go much slower because fewer blocks were needed.

Khufu and the memory of the Great Pyramid

Khufu's reputation has been a matter of controversy for 2,500 years. Herodotus, writing around 450 BCE, said that Cheops was a tyrant who closed the temples and forced all of Egypt to work for him, and that he even forced his own daughter into prostitution to raise money. The story is a legend, and not a reliable one. Contemporary evidence suggests that Khufu was a powerful king who commanded a sophisticated state and whose workers, as we will see in Chapter 13, were fed, housed, and given medical care.

Khufu's name lived on. He was venerated for centuries, and a cult of the king was maintained at Giza, especially in the Late Period, when the Egyptians looked to the Old Kingdom as a golden age. By the time of the Greeks and Romans, tourists were visiting the pyramid and carving their names on it. The Great Pyramid has been a site of pilgrimage for more than four thousand years.

KEY TAKEAWAYS

The Great Pyramid was built for Khufu (c. 2589–2566 BCE) at Giza as the centerpiece of a vast complex. Its original height was about 146.6 m and its base side about 230.3 m; it contains roughly 2.3 million blocks. Its interior includes the Descending Passage, an unfinished Subterranean Chamber, the Grand Gallery, the so-called Queen's Chamber, and the granite King's Chamber with five relieving chambers above it. Work-gang graffiti, nearby tombs, and the Wadi al-Jarf logbook tie it securely to Khufu.

CHAPTER 6

The Giza Plateau: Khafre, Menkaure, and the Sphinx

Three pyramids, one great statue, and a necropolis that was a city of the dead.

Today, when people say "the pyramids," they usually mean the three at Giza. They stand in a diagonal row along the edge of the plateau, with the Great Pyramid of Khufu to the northeast, Khafre's to the middle, and Menkaure's, much smaller, to the southwest. Around them are smaller pyramids, temples, tombs, and the Sphinx. To understand the whole, we need to walk through the plateau as the Egyptians planned it, and to look at how it changed over about eighty years.

After Khufu: a short, strange interlude

When Khufu died, his throne passed to his son **Djedefre**, who ruled for about eight years. Djedefre did something unexpected. He moved the royal necropolis north to **Abu Rawash**, a few kilometers from Giza, and started his own pyramid there on a high, isolated hill. It was planned as a large monument, but it was left unfinished, and in later centuries it was quarried so heavily that only a stump remains.

Djedefre is a significant figure for another reason. He is the first king known to have used the title "Son of Ra" as part of his royal name. A fine quartzite head of Djedefre, in the Louvre, is

among the best royal portraits of the Old Kingdom. Some Egyptologists have wondered whether the Great Sphinx might even have been made in his image, though most favor his successor. We will come to that.

There may have been a family dispute. The tombs of some of Khufu's heirs suggest a complicated succession, and Djedefre's reign is short. But the evidence is too thin for confident conclusions. What matters here is that, after Djedefre, the royal necropolis moved back to Giza.

Khafre's pyramid

Khafre was another son of Khufu. He ruled for about 24 or 25 years, from around 2558 BCE, and built the second-largest pyramid at Giza. In its original form the Pyramid of Khafre measured about 215 meters on each side and stood about 143.5 meters tall. It was only three meters shorter than Khufu's. It was also steeper, at an angle of about 53 degrees 10 minutes, and it was built on a higher part of the plateau, so that from certain angles it looks taller than the Great Pyramid.

That is a good example of how perspective can deceive. From the Cairo side, to the northeast, Khafre's pyramid sits higher and nearer the viewer, so it seems to loom over its larger neighbor. A good vantage point to compare them is from the air, or from the plateau's far edge.

At the top of Khafre's pyramid, a cap of the original Tura limestone casing is still intact, and it gives the best sense of what the original smooth faces looked like. At the bottom, the lowest courses of the casing were made of red granite from

Aswan. The interior is simpler than the Great Pyramid's. There are two entrances, both on the north face, one about 11 meters above the base and one at ground level, and the passages lead to a single burial chamber cut into the bedrock, which contains a granite sarcophagus sunk into the floor. The lid is broken. Giovanni Belzoni, the Italian circus strongman turned explorer, opened this pyramid in 1818 and wrote his name in the burial chamber, and his inscription can still be seen.

Khafre's valley temple and statues

Khafre's complex is the best preserved at Giza. His **valley temple**, at the foot of the causeway, is a masterpiece of Old Kingdom architecture. It is built of massive limestone blocks cased in red granite, with a floor of white alabaster. Inside is a T-shaped hall with square granite pillars, each a single block, and a total absence of decoration. The effect is solemn and stark. Some of the limestone blocks in its core weigh well over 100 tonnes, and the granite casing was cut with such accuracy that the joints are almost invisible.

In 1860, the French archaeologist Auguste Mariette found a well in the floor of the temple filled with broken statues. Among them was a life-sized seated statue of Khafre carved from a dark, very hard stone traditionally called diorite, now one of the stars of the Egyptian Museum in Cairo. The king sits on a throne decorated with intertwined lotus and papyrus, which symbolize the unification of Egypt. Behind his head the falcon-god Horus spreads his wings protectively. Diorite is extremely hard, and the sculpture's detail and polish demonstrate the skill of Old

Kingdom sculptors. The temple originally held about twenty-three statues of the king.

The Sphinx

East of Khafre's valley temple, at the edge of the plateau, sits the **Great Sphinx**. It is about 73 meters long and 20 meters high, with a face about 4 meters wide. It is the largest monolithic statue in the world, and one of the oldest.

The Sphinx was carved from a knoll of limestone that the builders had left standing when they quarried stone for the neighboring pyramids. The knoll was uneven in quality. The harder layers at the head and the softer layers at the body explain why the head has survived well while the body has eroded badly. For centuries, the Sphinx was covered by drifting sand up to its shoulders, and it was periodically cleared. The body, which is a lion's, was later cased in stone blocks during various repairs, including by the Romans.

The face is generally believed to be that of **Khafre**. The evidence for this includes the Sphinx's position in his complex, the style of the face, and the fact that its temple is aligned with his valley temple. Some scholars have suggested Khufu or Djedefre as alternatives, but there is no strong evidence for that. A few writers have argued that the Sphinx is much older than the Fourth Dynasty. We will address that claim in Chapter 21.

The Sphinx's nose is missing, and many people believe that Napoleon's soldiers shot it off. This is a myth. Drawings made before Napoleon's expedition, including an illustration from 1737 by the Danish explorer Frederic Norden, already show the

Sphinx without a nose. The medieval Egyptian historian al-Maqrizi, writing in the fifteenth century, reported that a Sufi zealot named Muhammad Sa'im al-Dahr had damaged the face in 1378 because local peasants were making offerings to it. That is the likelier explanation.

> **MYTH AND FACT**
>
> **Myth:** "Napoleon's army shot the nose off the Sphinx." **Fact:** Drawings made decades before 1798 show the nose already missing, and a fifteenth-century Egyptian historian blames a fourteenth-century religious zealot. The Sphinx's beard, incidentally, was a later addition, and pieces of it are in museums in London and Cairo.

A king who was also a lion

The Sphinx is a combination of a lion's body and a king's head, which was a long-standing symbol in Egypt. The lion stood for royal power, strength, and the vigilance of the sun, and the Sphinx guards the entrance to the royal necropolis. In later centuries, the Egyptians called it *Hor-em-akhet*, or Horus in the Horizon, which the Greeks rendered as Harmachis. Around 1400 BCE, a young prince named Thutmose, who would become Thutmose IV, fell asleep in the Sphinx's shadow and dreamed that the god promised him the kingship if he cleared away the sand. When he became king, he did, and recorded the story on a granite slab, the **Dream Stele**, which stands between the paws today.

The word "sphinx" itself is Greek. It was the name of a female monster in the Oedipus legend, a creature with a woman's head and lion's body that asked a riddle. The Greeks applied the name to the Egyptian statue, and the riddle story has become attached to it in popular memory, though it has no Egyptian origin.

Menkaure's pyramid

The third pyramid at Giza belongs to **Menkaure**, who ruled for somewhere between 18 and 28 years, depending on the source, and was probably the son of Khafre. His pyramid is much smaller than the others: about 104.6 meters on each side and about 65 meters high. Its volume is only about one-tenth that of the Great Pyramid. The decline in size is consistent with a pattern across the Fourth Dynasty, in which the pyramids reached a peak under Khufu and then got smaller, as the cost of the monument weighed on the economy, or as royal priorities changed.

What Menkaure's pyramid lacks in size, it makes up in materials. The lower sixteen courses were cased in red granite from Aswan, while the upper courses were cased in white Tura limestone. The contrast would have been beautiful. The granite casing was never finished. Many of the blocks still have their rough outer surfaces, with the final polishing left undone. That suggests Menkaure died before the project was complete, and his successor finished the pyramid and the temple in cheaper material, mudbrick. This tells us something about the speed of royal succession and the willingness of the next king to complete a predecessor's monument at less cost.

On the north face of the pyramid, there is a long vertical gash. It is the result of an attempt to dismantle the pyramid in 1196 CE, ordered by al-Aziz Uthman, the son of Saladin. His workers labored for months and found the stone too hard to remove. The project was abandoned after they had managed to strip only a part of the casing from one face. It is a vivid reminder of how much effort it took even to destroy a pyramid.

Inside, Menkaure's pyramid has a descending passage leading to a series of chambers, including a paneled antechamber and a burial chamber with a vaulted roof. In 1837, Howard Vyse found a wooden coffin lid and some human bones in the pyramid. The lid is now in the British Museum, and later study showed that the coffin and the remains belonged to a much later period, probably an intrusive burial. The original basalt sarcophagus was lost at sea off the coast of Spain in 1838 and never reached England.

Menkaure's triads

The best-known works of art from the pyramid age come from Menkaure's valley temple. When the American archaeologist George Reisner excavated it in 1908–1910, he found a series of beautiful slate statues known as **triads**. In each, the king stands between the goddess Hathor and a personification of one of Egypt's provinces. There is also a famous double statue, in the Museum of Fine Arts in Boston, of Menkaure and his queen, whose arm is around his waist. The statues are among the great works of Egyptian art, noted for their smooth, rounded forms and ideal proportions.

The queens' pyramids and the cemetery

The three Giza pyramids are each accompanied by smaller satellite pyramids. Khufu has three for queens and one for a cult, Khafre has one, and Menkaure has three for queens and one for a cult. Between and around them stretch the tombs of the nobles: large stone mastabas with decorated chapels, arranged in orderly streets. Officials held offices such as "overseer of the king's works," "royal scribe," or "inspector of priests," and they were buried near the king in hope of sharing in his eternity.

One of the most remarkable tombs belongs not to a king but to a queen. **Khentkaus I**, who lived at the end of the Fourth Dynasty, has a massive stepped tomb, cut from a natural rock outcrop, with a small adjoining town for her priests. Her titles include "Mother of two kings of Upper and Lower Egypt." She is a striking figure in the history of the period, though the details of her relationships remain a subject of discussion.

Shepseskaf: an end to the pyramid line

Menkaure's successor **Shepseskaf** did something unusual. He abandoned the pyramid form and built an enormous mastaba shaped like a sarcophagus at South Saqqara. It is known today as the **Mastabat al-Fara'un**, Pharaoh's Bench. Some historians read the change as a religious or political shift, a break from the sun-centered ideology of the pyramid. Others point out that his tomb was built with similar techniques and may have been a rare variant or a response to the limited resources of the time. In either case, the Fifth Dynasty restored the pyramid, and the next chapter takes up that story.

Who lived at Giza?

The Giza plateau was a workplace, as well as a necropolis. To the south of the plateau, beyond a huge stone wall called the Wall of the Crow, archaeologists have found the remains of a settlement known as **Heit el-Ghurab**, the "Lost City of the Pyramid Builders." It contained bakeries, breweries, barracks, workshops, and storage areas, together with the cemeteries of the workers. We will visit it in Chapter 13. The existence of this settlement, which dates from the time of Khafre and Menkaure, is among the most convincing proofs that the pyramids were built by organized Egyptian laborers.

KEY TAKEAWAYS

Giza has three main pyramids: Khufu's (the largest), Khafre's (almost as tall, with surviving casing at its peak), and Menkaure's (much smaller, with granite lower courses). The Sphinx was carved from quarry bedrock, most likely in Khafre's reign. The smaller size of Menkaure's pyramid and the unfinished casing suggest that the great scale of Khufu's project could not be sustained. A thousand years later, kings were still restoring the plateau and worshipping its monuments.

CHAPTER 7

The Later Old Kingdom: Smaller Pyramids, Sacred Texts, and Collapse

From the sun-temple kings of the Fifth Dynasty to the end of the Old Kingdom.

After the Fourth Dynasty, the great pyramids were over. Egypt did not stop building pyramids, and in some ways the next two dynasties were its most productive in religious and artistic terms. But the monuments became smaller, the construction cheaper, and the emphasis shifted. What the later pyramids lost in size they made up in words. For the first time, the walls of the burial chambers were covered with writing.

A change of dynasty

The Fifth Dynasty began around 2494 BCE with **Userkaf**. Later Egyptian tradition, preserved in the *Westcar Papyrus*, told that the first three kings of the dynasty were triplets, sons of a priest of Ra and a woman named Rededjet, fathered by the sun god himself. The story is a legend, but it reflects something real: the priesthood of Ra at Heliopolis had become a rival center of power, and the new dynasty owed it a great deal.

Userkaf marked the change in two ways. He built his pyramid not at Giza but at Saqqara, close to Djoser's old complex.

And he built something new: a **sun temple** at Abusir, about 15 kilometers to the north. Six kings of the dynasty are known to have built sun temples, though only two have been found: Userkaf's and Niuserre's. These were open-air courtyards dominated by a squat stone obelisk on a base, with an altar facing the sky. They were dedicated entirely to Ra. The pyramid was no longer the only monument of a king's devotion to the sun.

The Abusir pyramids

The kings who followed Userkaf, beginning with **Sahure**, shifted the royal necropolis to **Abusir**, where they built a line of pyramids along the desert edge. They were much smaller than the Giza giants: around 80 meters per side and 50 meters high, with a volume of only a few percent of Khufu's.

They were also built differently. The Fourth Dynasty kings had used large blocks of good limestone throughout the core. The Fifth Dynasty builders built an outer shell of fine limestone around a core of small, rough stone, sand, and rubble, held in place by a framework of internal retaining walls. This was cheaper and faster but far less durable. Over the centuries, once the casing was stripped off, the rubble cores spilled out and the pyramids collapsed into heaps. That is why many Fifth Dynasty pyramids look like low, shapeless mounds today.

The decline in structural quality is a measure of how the state's priorities changed, but it should not be read as a failure of skill. The Fifth Dynasty builders concentrated their efforts on the temples, which were richly decorated. The mortuary temple of Sahure is one of the finest examples of Old Kingdom art. Its

walls carried reliefs of ships, foreign peoples, and offering bearers, and its columns were carved as palm trees. The artists were as good as ever.

The Abusir papyri

Abusir also gave us something almost priceless: papers. In the 1890s and again in the twentieth century, excavators found the remains of a temple archive in the funerary temples of kings Neferirkare and Neferefre. These **Abusir papyri**, written in hieratic script, are the oldest substantial group of administrative papyri from Egypt. They include duty rosters, inventories of temple property, delivery records, and accounts of offerings.

They reveal how a mortuary cult worked day to day. Priests were organized into five groups, each serving in rotation for a month at a time. Each group counted the temple's equipment, checked its seals, and made the offerings. Part-time priests, who held other jobs in the community, worked their rotation and then returned to civilian life. This is exactly the kind of system the state used for labor on the pyramids: rotation, recording, and accountability. Egypt ran on paperwork.

Unas and the Pyramid Texts

The last king of the Fifth Dynasty, **Unas** (c. 2375–2345 BCE), built a small pyramid at Saqqara, just beside Djoser's complex. It is only about 43 meters high, and from the outside it is a modest ruin. But when archaeologists entered it in 1881 they found something extraordinary. The walls of the burial chamber and the antechamber were covered with columns of hieroglyphs, painted blue-green, more than 200 spells in all. These are the **Pyramid Texts**, the oldest religious writings in Egypt and among the oldest known religious texts in the world.

The Pyramid Texts continued in the pyramids of the Sixth Dynasty kings Teti, Pepi I, Merenre, and Pepi II, and in some queens' pyramids. They are not the same in every pyramid. Each king's selection of spells differs, and over time the corpus grew to about 800 separate utterances. The spells include rituals for offerings, protective charms against snakes and enemies, hymns to the sun, and descriptions of the king's ascent. We will look at them in more detail in Chapter 17.

For scholars, the Pyramid Texts are crucial. They give us the language in which the Egyptians themselves explained what the pyramid was for. They also provide evidence of an extensive royal theology already in place by 2350 BCE. The Great Pyramid had no such texts, which is one of the reasons its purpose has been so debated. By the time of Unas, the builders were willing to say it out loud.

The Sixth Dynasty: Teti to Pepi II

The Sixth Dynasty (c. 2345–2181 BCE) began with **Teti**, who built his pyramid at Saqqara, north of Djoser's. Its surroundings became a prestige necropolis for his officials, including his vizier **Mereruka**, whose mastaba has 33 rooms and some of the most detailed relief scenes of daily life in the Old Kingdom: fishing, boat-building, hunting in the marshes, and the preparation of food.

Pepi I followed and built a pyramid at South Saqqara. He and his son **Merenre** presided over an expanding Egyptian presence in Nubia and the Sinai. A famous biographical inscription by an official named **Weni**, who served under Teti, Pepi I, and Merenre, describes in detail how he organized the quarrying and transport of stone for royal monuments, including a granite sarcophagus and its lid, and how he led a military campaign against Asiatic nomads. His text is one of the best surviving descriptions of how an Old Kingdom official managed stone, labor, and ships. It makes clear that it was a matter of state business, planned and carried out by trained men.

Pepi II succeeded to the throne as a child and, according to Egyptian tradition, reigned for more than 90 years. Modern scholars think the figure is exaggerated and that the real figure was about 64 years, though even that would be a record. His pyramid at South Saqqara is about 52 meters high and has a complex of fine reliefs and texts, and three small queens' pyramids.

Smaller pyramids, bigger state

If the pyramids got smaller after Khufu, how are we to understand the shift? Historians have proposed several explanations.

- **Cost and economy.** The Great Pyramid had strained the country's resources. Reducing the scale of subsequent pyramids freed resources for other purposes, such as temples, trade expeditions, and sun temples.
- **Shift in ideology.** The cult of Ra took center stage. The pyramid became one part of a larger religious system that also included the sun temple.
- **Changing power structure.** Over time, provincial governors and high officials gained wealth and independence, and the king could no longer command the whole country in the way Khufu had.
- **Learning.** By using local rubble and small blocks, the builders cut the cost and the effort. They may have decided that grand scale was unnecessary because the spells carved on the walls were enough.

> **KEY POINT**
> The Pyramid Texts changed the way pyramids communicate. Fourth Dynasty pyramids are silent. Fifth and Sixth Dynasty pyramids have their purpose written on the walls. This is the single best written source for what the pyramid builders believed.

The end of the Old Kingdom

The Old Kingdom ended around 2181 BCE, and the causes have been debated for a century. A number of factors appear to have contributed.

First, the central government gradually lost power. Over generations, the offices of provincial governors became hereditary, and local rulers built impressive tombs in their own regions. The king had to pay for their loyalty with land and tax exemptions, and these concessions drained the treasury.

Second, there is evidence of climatic stress. Studies of ancient climate show that the late third millennium BCE was marked by a period of reduced rainfall across the Near East, sometimes called the 4.2 kiloyear event. Nile floods may have been low for several years in a row. Some Egyptian inscriptions of the period, including the autobiographies of nomarchs, mention famine and the efforts of local leaders to feed their people. The famous text known as the *Admonitions of Ipuwer* describes a world turned upside down, though it was written much later and may be a literary work, not a historical record.

Third, the length of Pepi II's reign may have contributed to a succession crisis. With a very old king and many heirs who died before him, the throne may have passed to a series of short-lived rulers, as the Egyptian king lists suggest.

Historians generally avoid a single explanation. The Old Kingdom's decline was probably a combination of political decentralization, economic strain, climate variability, and dynastic instability. Whatever the causes, the result was the **First Inter-**

mediate Period (c. 2181–2055 BCE), a time when Egypt had no strong central government and regional rulers competed for power.

What the late pyramids teach us

The later Old Kingdom shows that the Egyptians were not trapped in a single approach. They adapted. They experimented with different construction methods, changed the form of the monument, added texts, and reallocated resources according to their needs. They also continued to maintain the cults of their ancestors. Priests were still giving offerings to Khufu and Sneferu long after the kings who built them were dead. And they were collecting data, writing it down, and passing on knowledge.

The Great Pyramid thus appears not as an isolated marvel but as the highest point of a long process, a moment when the Egyptian state had the resources, the knowledge, and the motivation to do something enormous. Other kings did not have those exact conditions, and they did what they could with what they had.

KEY TAKEAWAYS

After Khufu, pyramids shrank and were built with rubble cores. The Fifth Dynasty added sun temples and moved the cemetery to Abusir. Unas (c. 2375–2345 BCE) is the first king to inscribe the Pyramid Texts, which tell us directly what the pyramids were for. The Old Kingdom ended around 2181 BCE through a mixture of political decentralization, climate stress, and succession problems.

CHAPTER 8

Pyramids After the Old Kingdom: Revival, Decline, and the Nubian Kings

The Middle Kingdom's mudbrick pyramids, the last royal pyramid of Egypt, and the Sudanese pyramids that outnumber them all.

The Old Kingdom did not take the pyramid with it when it fell. After a century and a half of division, Egypt was reunited, and its new rulers turned to the pyramid again. They built differently, with new engineering ideas and new ways to defeat tomb robbers. Then, after about 1550 BCE, Egypt's kings stopped building them, but the form lived on. Hundreds of miles to the south, in Nubia, the pyramid acquired a second life that lasted nearly a thousand years longer than Egypt's own.

Reunification and a new capital

The First Intermediate Period (c. 2181–2055 BCE) was a time of regional rule. Power was held by local governors, and in the south, a line of kings based at Thebes eventually defeated their northern rivals. Around 2055 BCE, **Mentuhotep II** reunited the country. He built a remarkable temple-tomb at **Deir el-Bahari**, across the river from Thebes, with terraces and colonnades, possibly topped by a small pyramid-like structure. The design would later inspire Queen Hatshepsut's temple nearby.

Around 1985 BCE, a vizier named **Amenemhat** took the throne and founded the Twelfth Dynasty. He moved the capital to a new city, **Itjtawy**, "Seizer of the Two Lands," near the modern village of Lisht, south of Memphis. Next to it he built a pyramid. The Twelfth Dynasty is generally regarded as the high point of Egypt's Middle Kingdom, a period of strong government, public works, and literature.

A pyramid built from older pyramids

The pyramid of **Amenemhat I** at Lisht reveals something unexpected about the Middle Kingdom builders: their attitude toward the past. When excavators took the pyramid apart, they found that its core included hundreds of decorated blocks taken from the Old Kingdom pyramid complexes at Giza, Saqqara, and Abusir. Some bore the names of Khufu and Khafre. Fragments of relief from the causeway of Unas were also found.

Why would a king take stone from the monuments of earlier rulers? The simplest answer is convenience: the Old Kingdom

stone was already quarried and dressed, and the old pyramid sites were by then 600 years old and, in many cases, in disrepair. Another possibility is symbolic: by incorporating the stone of the great kings into his pyramid, Amenemhat linked himself to the glory of the past. A third possibility is that it was an act of piety toward earlier kings, since the blocks were placed, not discarded. These explanations are not mutually exclusive.

The new way to build: mud brick and hidden entrances

Middle Kingdom builders returned to the pyramid, but they changed the construction method. Instead of solid limestone throughout, many of the pyramids were built with a core of **mud brick**, supported by a framework of stone walls and covered in limestone casing. This was cheaper and faster, but, like the Fifth Dynasty's rubble cores, it left the pyramid vulnerable once the casing was stripped. Today, most Middle Kingdom pyramids are brown heaps of eroded brick.

The kings also devoted new attention to the problem of robbers. A tomb that looks like a mountain is an advertisement, and by the Middle Kingdom, everyone in Egypt knew what pyramids had held and how often they had been robbed. The architects responded with elaborate security.

- **Hidden entrances.** Instead of facing north on the pyramid's face, as at Giza, entrances were moved to the south side, to the east, or even beneath the ground at some distance from the pyramid.

- **False passages and dead ends.** Corridors that led nowhere were meant to confuse intruders.
- **Massive blocking stones.** Huge slabs of quartzite or granite, sometimes weighing tens of tonnes, were lowered into place after the burial.
- **Trap chambers.** Some designs included concealed shafts and hollows.

At **Lahun**, the pyramid of **Senusret II** (c. 1880 BCE) was built around a natural spur of rock, which gave the structure a stable, solid core, and then completed with mud brick and a limestone casing. The entrance was hidden. It was not on the pyramid at all but down a shaft in the courtyard to the south, where a hidden stair led into a labyrinth of passages and chambers. At the heart of the structure, the burial chamber contained a pink granite sarcophagus. The robbers of antiquity found it anyway. When Flinders Petrie entered in 1889, he found the chamber looted but the sarcophagus still in place. Years later, in 1914, his colleague Guy Brunton found the tomb of a royal woman, the princess Sithathoriunet, nearby, still holding spectacular jewelry that the thieves had overlooked.

Near the pyramid, Petrie excavated a planned town, which he called **Kahun**, home to the priests and workers who served the pyramid. It was laid out in a grid, with a district of large houses and another of smaller ones, and the remains included papyri with accounts, letters, medical texts, and veterinary advice, as well as toys, tools, and household goods. It offers a rare glimpse of the daily life of the community around a pyramid.

The last great pyramid builder

Amenemhat III (c. 1860–1814 BCE) was one of the longestreigning and most prosperous kings of the Middle Kingdom, and he built two pyramids. The first, at **Dahshur**, is known as the **Black Pyramid** because its mud brick core has weathered to a dark color. It was built with ambition but ran into trouble. The weight of the pyramid caused the underground chambers to crack and the corridors to subside, and some of the passages bear signs of emergency repair. The king abandoned it as a burial place, though he may have kept it as a symbolic monument, and built a second pyramid at **Hawara**, in the Faiyum oasis.

The Hawara pyramid contained the most sophisticated security of any Egyptian pyramid. Its burial chamber was carved from a single enormous block of quartzite, weighing an estimated 110 tonnes, and was roofed with giant slabs lowered into place and sealed. The passages included sliding stone doors, false corridors, and a dead-end that returned to the entrance. Even so, thieves broke in during antiquity. Petrie opened the pyramid in 1889 after a lengthy effort and found the chambers plundered and flooded with groundwater. The burial chamber held two sarcophagi, both emptied by thieves.

The labyrinth

Adjacent to the Hawara pyramid, there once stood a gigantic mortuary temple that struck visitors with awe. Herodotus visited it around 450 BCE and described it as even more impressive than the pyramids: 3,000 rooms, half above ground and half below, arranged around twelve covered courts. Later Greek and Roman writers called it the **Labyrinth**, evidently comparing it to the mythical labyrinth of Crete.

Almost nothing remains of it. Quarrying in antiquity and later periods reduced the structure to a vast field of broken limestone chips. Petrie, and later scholars, traced the outline of its foundation, which covered an area of roughly 300 by 250 meters, or about the size of the entire Step Pyramid enclosure, in the 1880s. In recent years, a team using ground-penetrating radar has claimed to detect large buried structures beneath the site, but those findings need further investigation. As things stand, the Labyrinth is known mostly through the words of ancient writers.

The end of the royal pyramid in Egypt

The Twelfth Dynasty ended around 1800 BCE, and a succession of short-lived kings followed in the Thirteenth, who continued to build smaller pyramids. Then Egypt fragmented again, and a foreign dynasty of rulers from western Asia, the **Hyksos**, took control of the north. A line of native Egyptian kings at Thebes resisted them. Around 1550 BCE, **Ahmose I** drove out the Hyksos and reunified Egypt, beginning the New Kingdom.

Ahmose I built a pyramid at Abydos, with a temple and a cenotaph chapel, and it was the last royal pyramid in Egypt. After him, the kings of the New Kingdom chose a different kind of tomb. They had their burial chambers cut deep into the cliffs of a remote valley across the Nile from Thebes, the **Valley of the Kings**, and had their offering temples built separately along the river. The reasons are debated. The most common explanation is security: a hidden tomb is harder to find, and a hundred years of experience had shown that pyramids did not keep out robbers. A second is religious: the cult of Amun had grown in importance, and there was a new theology of the afterlife that emphasized the underground journey of the sun through the Duat.

The valley itself is dominated by a natural pyramid-shaped peak called **al-Qurn**, "the Horn," which many Egyptologists believe was a conscious choice: the kings were buried beneath a mountain that looked like a pyramid. The pyramid shape never completely vanished. It survived in small pyramids, often with steep sides, that capped the tombs of workmen at **Deir el-Medina**, the village that housed the artisans who built the royal tombs. The capstone of an obelisk was called a pyramidion, too, and the shape stayed a powerful symbol for the sun.

KEY POINT

The royal pyramid in Egypt lasted about 1,100 years, from Djoser (c. 2670 BCE) to Ahmose I (c. 1550 BCE). During that time, the form changed, from solid limestone to mud brick, and the security became more elaborate, but the idea of a monumental, sun-linked tomb stayed the same.

The pyramids of Nubia

South of Egypt, in the land called Kush, in what is now Sudan, another civilization had grown up along the Nile. For centuries it was alternately an Egyptian colony, a trading partner, and a rival. Around 750 BCE, a Kushite king named **Piye** marched north and conquered Egypt, founding the Twenty-Fifth Dynasty. The Kushite kings ruled Egypt for about a century, until the Assyrians expelled them in 664 BCE.

The Kushites loved Egyptian culture. They worshipped Amun, wrote in hieroglyphs, and built temples in the Egyptian style. And they built pyramids. At **El-Kurru** and **Nuri**, near the Fourth Cataract, Kushite kings, including **Taharqa**, were buried beneath small, steep-sided pyramids with an underground burial chamber. After the Kushite capital moved south to **Meroë**, in the sixth century BCE, the kings and queens of the Meroitic kingdom continued the custom for another thousand years, until about 350 CE.

The result is a remarkable landscape. Archaeologists have counted more than 200 pyramids in Sudan, which is more than twice as many as survive in Egypt, though they are far smaller.

Typical Nubian pyramids are 6 to 30 meters high, built of sandstone blocks in stepped layers, and have a much steeper angle, perhaps 70 degrees. Each was accompanied by a small chapel on the east side, decorated with reliefs, where offerings were made. The Meroitic cemeteries, with their rows of pointed silhouettes, are among the most beautiful sights in Africa.

> **MYTH AND FACT**
>
> **Myth:** "Egypt has the most pyramids in the world." **Fact:** Sudan has more, probably over 200, though they are much smaller than Egypt's. Egypt has between about 118 and 138, depending on how one counts the smaller and ruined ones.

The Nubian pyramids suffered a famous act of destruction. In 1834, an Italian physician and treasure-hunter, **Giuseppe Ferlini**, demolished the top of a pyramid at Meroë, thought to be that of Queen Amanishakheto, and found a hoard of gold jewelry. He then proceeded to damage dozens of other pyramids, but he found nothing else. His jewelry was sold to museums in Berlin and Munich, where it remains. The episode is a harsh reminder of how archaeology looked in the nineteenth century.

Pyramids elsewhere in the world

Pyramids are not unique to Egypt and Nubia. The shape appears across the world, because a pyramid is one of the simplest ways to build a tall, stable structure from stacked material. A broad base and tapering sides need little more than gravity and stones.

- **Mesopotamia.** The **ziggurats** of ancient Iraq were stepped temple towers built of mud brick, starting around 2100 BCE. They were platforms for temples, not tombs.
- **Mesoamerica.** From about 300 BCE onward, builders at Teotihuacan, Tikal, Chichen Itza, and other sites erected stepped pyramids topped by temples. Most were platforms for ritual, but some, such as the Temple of the Inscriptions at Palenque, contain royal tombs. The tomb of King K'inich Janaab' Pakal, who died in 683 CE, was found inside it in 1952.
- **China.** Large pyramid-shaped mounds, built of rammed earth, cover the burials of the early emperors. The tumulus of Qin Shi Huang, the first emperor, near Xi'an, dates to about 210 BCE.
- **Elsewhere.** There are also stepped pyramids in the Canary Islands, Sudan, and Indonesia.

These monuments differ in purpose, construction, and symbolism. There is no evidence that they are related to each other through cultural contact. The better explanation is independent invention. A mound is a natural way to mark a grave or a holy place, and a stepped mound is a natural way to make it higher.

KEY TAKEAWAYS

After the Old Kingdom, Middle Kingdom kings built pyramids with mud brick cores and elaborate defences against robbers, including hidden entrances and giant sealing stones. The last royal pyramid in Egypt was built by Ahmose I (c. 1550 BCE), after which kings were buried in the Valley of the Kings. The Kushite kings of Nubia then built more than 200 steep pyramids in Sudan over nearly a thousand years. Pyramids elsewhere in the world arose independently.

PART 3

How They Were Built

The tools, the methods, the people, and the numbers

CHAPTER 9

Surveying and Planning: Squares, Levels, and True North

How the builders marked out a base the size of ten football fields and made it level, square, and aligned to the stars.

Before a single block was cut, someone had to decide where the pyramid would go and how it would sit on the ground. This stage is easy to overlook, because it leaves little visible trace, but it is one of the most impressive parts of the achievement. The Great Pyramid's base is nearly a perfect square, almost perfectly level, and almost perfectly aligned with the cardinal directions. Getting a monument of this size right at the start was essential, because every error in the base would be magnified as the pyramid rose.

The tools of the surveyor

Egyptian surveyors had a small kit of simple instruments, but they used them skillfully. Many of the tools are known from tomb paintings, models, and actual finds.

- **The cubit rod.** The basic unit of length was the **royal cubit**, about 52.3 to 52.5 centimeters, divided into seven palms of four fingers each. Wooden and stone measuring rods, marked off in palms and fingers, have survived from the

New Kingdom, and similar ones were surely used in the Old Kingdom. Measurements were also made with knotted cords.

- **The stretched cord.** A rope, rubbed with chalk or ocher and snapped against a surface, drew a straight line. A rope held between two stakes gave a straight edge. Ropes were so important to surveying that one of the king's ritual acts at the foundation of a temple was called "stretching the cord," and a Greek tradition called Egyptian surveyors "ropestretchers."
- **The plumb bob.** A weight hung on a string establishes a true vertical. Egyptian plumb bobs of stone and metal are known.
- **The set square and the A-frame level.** A right angle can be checked with a wooden set square. A level was made from an A-shaped frame with a plumb line hanging from the apex. When the line fell on a mark at the center of the cross-bar, the feet were at the same height.
- **The merkhet.** This was an instrument for observing the stars, consisting of a bar with a sighting slit and a plumb line. Examples survive from the Ptolemaic period, and the device may have been much older.
- **Sighting rods and water.** A level surface can be found with water, simply because water finds its own level.

No tool on this list is surprising. All were available, in recognizable form, throughout the Egyptian Bronze Age, and none requires anything more than wood, string, stone, and metal.

Choosing and preparing the ground

The Great Pyramid sits on a limestone plateau, and the builders' first step was to prepare a firm, level platform. They did not level the entire site. Instead, they left a natural rise of bedrock in the middle, a small hill that they incorporated into the pyramid's core. The hill rises about seven meters at the center and is still visible inside the lower part of the structure. This saved effort, but it also meant that the builders had to level the ground around the rise.

How did they level an area of roughly 53,000 square meters to within a couple of centimeters? The most popular explanation is a **water-leveling system**. In this approach, the builders cut a grid of shallow channels in the bedrock, flooded them with water, and used the water surface as a reference. They then cut away the rock between the channels down to a common level, and the remaining ridges gave them the height of the platform. This method needs only water and a hoe, and it is accurate. Direct evidence at Giza is limited, since the channels would have been cut away as the work proceeded, but the method is simple, accurate, and consistent with the practice of using water as a reference in later Egyptian construction.

A second approach was to use many sighting rods and string levels across the site, checking one against another. Egyptians used long wooden "boning rods" in a kind of sight-and-line

method, which could have worked at this scale, though it would take longer.

Squaring the base

A square base needs four right angles and four equal sides. The most straightforward way to construct a right angle with ropes is to use a triangle whose sides are in the ratio 3:4:5, a so-called Pythagorean triple. A popular claim says that Egyptians used a rope with twelve equally spaced knots, forming a 3-4-5 triangle. The idea is appealing, but we have no Egyptian document that describes it. Greek writers mention it centuries later, and modern authors repeated it. It is a plausible method, but not proven.

Another way to find a right angle is to use the **diagonals**. If the two diagonals of a four-sided figure are equal in length, and the opposite sides are equal, it is a rectangle. If all four sides are equal as well, it is a square. Checking that two ropes stretched across the corners are the same length is simple. Surveyors can then adjust the corners until the diagonals match. It is an easy method, requiring no calculation.

The results speak for themselves. The side lengths of the Great Pyramid vary by only a few centimeters, and the corners deviate from true right angles by only a few seconds of arc. Whatever method the builders used, they were extremely careful.

Finding north

The most famous precision of the Great Pyramid is its orientation. The sides face the cardinal directions with an error of only about three arcminutes, or one-twentieth of a degree. How did the builders find true north without a compass? Two main answers have been proposed.

The sun method. Set a vertical pole, called a **gnomon**, on a flat, level surface. Mark the end of its shadow in the morning, and draw a circle around the pole with a radius equal to the shadow's length. In the afternoon, the shadow will reach the circle again, and the line between the two points the shadow touched will run east-west. A line at right angles to it is north-south. This method was suggested by the Egyptologist I. E. S. Edwards in 1961 and developed by others. It works best around the equinoxes, when the sun's path is nearly a straight line, and it can produce an accuracy of about a tenth of a degree. In 2016, the engineer Glen Dash argued, on the basis of field tests at Giza, that a refined version of the method could account for the small error seen in the Great Pyramid.

The star method. In 2000, the Egyptologist Kate Spence proposed in the journal *Nature* that the builders used two stars in the northern sky: Kochab in the Little Bear and Mizar in the Big Dipper. At certain times, when the two stars were aligned vertically, they would form a plumb line pointing to the north celestial pole. Because of the slow wobble of the Earth's axis, called precession, the alignment would shift over the centuries, and Spence showed that the small errors in the orientation of Old

Kingdom pyramids seem to correlate with their dates in a way that matches the movement of these stars. Her hypothesis would let us date the pyramids to within a decade or two. Other researchers have challenged it, pointing to uncertainties in the dating, the practical difficulty of making observations, and the possibility that the correlation is coincidental.

KEY POINT

Neither the sun method nor the star method has been proven. Both are workable with simple tools, and both are consistent with the accuracy seen. It is possible that Egyptian surveyors used more than one approach, and that they tested the result with a second method.

The accuracy in context

Three arcminutes is a very small angle. A good way to appreciate it is to convert it into distance. If you were to stand at one corner of the Great Pyramid and sight along a side, 230 meters long, an error of three arcminutes would shift the far end sideways by about 20 centimeters.

WORKED EXAMPLE

How big is a three-arcminute error? One degree equals 60 arcminutes, so 3 arcminutes is 0.05 degrees. The tangent of 0.05° is about 0.00087. Multiply by the 230-meter side: 230 × 0.00087 ≈ 0.20 meters. In other words, over the length of one side, the pyramid's alignment error amounts to about the length of a hand.

That is an extraordinary achievement for a pre-industrial culture. It is also not magic. A careful team with a gnomon, a cord, and enough time could achieve it. What the result shows is a deep investment in craftsmanship, a repeated cycle of measurement and correction, and probably a team that included several experienced surveyors.

The mystery of the drift

The orientation of other pyramids is less perfect. The pyramids of Meidum and Dahshur are a little further from true north. The pyramids of Khafre and Menkaure show larger errors than Khufu's. The Great Pyramid seems to be the most accurate of all. This pattern raises an interesting question: why was the orientation of Khufu's pyramid better than those of both his predecessors and his successors? Put another way, did the care taken over the orientation rise and then fall within a few decades?

A plausible answer is that Khufu's builders had access to an unusually capable surveyor, or invested extra care because the project was so important. Another is that by the later Fourth Dynasty, the orientation was no longer treated as a priority. This is speculation, and we do not know.

Planning on paper

No architectural plan of a pyramid has survived. Egyptians did not leave behind a blueprint of the Great Pyramid, and we cannot be sure they drew one. But we do have indications that Egyptian builders used drawings.

- **Sketches on limestone flakes.** At the Step Pyramid, archaeologists found a flake of limestone with a curve drawn on it and numerical measurements, thought to be a plan for the curved roof of a chamber. Similar sketches were used by later builders.
- **Grid-based drawings.** Egyptian artists used a square grid to proportion figures and, probably, architectural elements. Squared grids appear on unfinished tomb walls.
- **Guidelines on the stone.** At Meidum and Giza, lines were drawn on the surface of casing blocks to mark the intended slope. In the Great Pyramid, red ocher marks of various kinds can still be seen.
- **Models.** Small models, like those used in later times for temples, may have been used to explain designs to the work crews.

A more surprising source of information is that the builders appear to have changed their plans several times, as we have seen in the successive chambers of the Great Pyramid. This implies that the design was not fixed in advance. It was developed, adjusted, and tested as the work went on. That is another sign of a human process.

The foundation ritual

Surveying was also ritual. In Egyptian temples, from the earliest period, the foundation of a new building began with a ceremony called **stretching the cord**. A king, often accompanied by the goddess of writing and measurement, **Seshat**, drove stakes into the ground and stretched a cord between them to mark the corners. Scenes of the ceremony survive in temples. We do not have direct evidence of such a ceremony at the Great Pyramid, but it is likely that it was performed, given how consistently the rite appears in later periods.

This reminds us that for the Egyptians, practical and religious acts were not separate. Aligning the pyramid with the heavens was both a surveying problem and a way of locating the king in the cosmos.

Why the planning mattered

The precision of the base had real structural consequences. A pyramid built on an uneven foundation can tilt or crack, as the Bent Pyramid shows. A pyramid built with slightly different slopes on its four sides will not meet at a point. A pyramid with a base that is not square will have faces that twist. The Great Pyramid's faces meet in a point to within a few centimeters, and the structure has stood for 4,500 years with no major structural failure. The care taken with the base paid off.

KEY TAKEAWAYS

The builders used simple tools: cubit rods, cords, plumb bobs, A-frame levels, set squares, and water. The base was leveled by cutting channels filled with water and a central knoll was left as core. North was probably found by the sun's shadow (Edwards, Dash) or two circumpolar stars (Spence); both are plausible and neither is proven. The result, a square base with a 3-arcminute orientation error, is explained by careful craftsmanship, not exotic knowledge.

CHAPTER 10

Quarrying: Limestone, Granite, and Copper Chisels

Where the stone came from, how it was cut, and what tools could do it.

The Great Pyramid contains about 2.3 million blocks. Every one of them had to be cut from living rock, shaped, and moved. If there is one part of the project that tests belief, this is it. Cutting stone is slow, noisy, dangerous work, and the Egyptians did it with tools that, by modern standards, look pitifully inadequate. How did they do it? The answer comes from the quarries themselves, which are still there, with the tool marks, the abandoned blocks, and the debris left by the workers.

The stones of the pyramid

The pyramid builders used several types of stone, each for a different purpose. The materials were sourced from different places, some close to the building site and some very far away.

Stone	Source	Main use
Local limestone (coarse, yellowish)	Giza plateau, quarried immediately south and east of the pyramid	The bulk of the core masonry

Stone	Source	Main use
Fine white limestone	Tura (Turah) and Maasara, across the Nile, about 13–15 km upstream	Outer casing, passage linings, and some interior finishing
Red granite	Aswan, about 800 km upstream	King's Chamber walls, ceiling beams, sarcophagus, plugs, and lower casing at Menkaure's pyramid
Basalt	Widan el-Faras, in the Faiyum, about 70 km	Pavement of the Mortuary Temple of Khufu
Gypsum (burnt and ground)	Local	Mortar, to bed and level the blocks
Alabaster (calcite)	Hatnub, in Middle Egypt	Floors and decorative elements in temples

The first two are the most important. The Great Pyramid is a mass of local limestone with an outer skin of Tura limestone. Of the total, something like 95 percent or more is local stone. The granite, much of the white limestone, and the other special stones were a small share of the total by volume, but they required the most complex logistics.

The Giza quarry

Archaeologists have mapped the quarries on the plateau and found that most of the stone for the Great Pyramid came from just south and southwest of the site. A vast quarry, partly shaped like a bowl, lies only a few hundred meters from the pyramid. It is a good example of efficient planning. By taking stone from nearby, the builders cut their transport distance to a minimum. The quarry also supplied the bedrock rise that the builders left as the pyramid's core.

At the quarry, we can still see how the work was done. The blocks were cut from the rock face in horizontal layers, working from the top down. Quarrymen first cut narrow trenches, about the width of a man's arm, around the sides of a block, marking out a shape. They then undercut the block from below, probably by tunneling beneath it with small chisels. When the base was cut through, the block was levered free. Some partially detached blocks, still attached to the bedrock, are visible in the Giza quarries today, frozen in the middle of the process.

The stone itself is **nummulitic limestone**, formed in the Eocene, about 40 to 50 million years ago, from the shells of tiny marine creatures. It is full of fossil nummulites, small coin-shaped shells, which the Greeks called "lentils." The local limestone is relatively soft, a hardness of about 3 on the Mohs scale, which is about the same as copper. That makes it practical to cut with copper tools.

The tool kit

The materials available to Old Kingdom workers were few: stone, wood, copper, and fiber. Iron was unknown, and bronze, an alloy of copper and tin, would not be common in Egypt until some centuries later. The main tools were these.

- **Copper chisels and picks.** Copper is soft, but it can be hardened by hammering and by adding small amounts of arsenic or other metals. Chisels of copper, with a wooden mallet, work well on limestone, and they were used for trimming and dressing. Many copper tools have been found in Old Kingdom contexts, including at Giza.
- **Stone hammers and pounders.** The hardest tool the Egyptians had in large numbers was a ball of **dolerite**, a very hard igneous rock, about the size of a bowling ball. Held in the hand and swung against granite, it crushes the surface into powder. The method is slow but works. Thousands of dolerite pounders have been found at the quarries of Aswan.
- **Wooden wedges and levers.** Wooden wedges, driven into slots and wetted, expand and can split rock along a line. Some evidence for this comes from quarries of later periods, and it was a common practice in the ancient world.
- **Copper saws with abrasive sand.** A copper blade is too soft to cut hard stone, but with a sharp quartz sand carried in the cut, it works as an abrasive. The saw moves the sand, and the sand does the cutting. Tool marks on granite sarcophagi and statues show the kerfs from this process.

- **Drills.** Tubular drills, probably made of copper or reed, turned with sand, were used to hollow out stone vessels and to cut out the interiors of sarcophagi. A drill core from Giza, now in the Petrie Museum, shows the spiral grooves left by such a tool.
- **Flint and chert tools.** Hard flint was used for fine scraping and to engrave harder stones.
- **Wooden mallets, plumb lines, and squares.** For marking and checking.

MYTH AND FACT

Myth: "You can't cut granite with copper tools, so the Egyptians must have used advanced technology." **Fact:** Copper alone cannot cut granite, and the Egyptians knew it. They used the abrasive-sand method for cutting and dolerite pounders for shaping. The marks are still on the stones, and the quarries contain the tools. Modern experiments have reproduced the process, though it is slow: a few centimeters per day.

Quarrying granite at Aswan

Granite was the most demanding material. The red granite used for the King's Chamber and its ceiling beams was quarried at **Aswan**, on the First Cataract. The quarries are still there. They include a spectacular and famous object: the **Unfinished Obelisk**, which was cut from the rock but cracked during the work and abandoned. If it had been completed, it would have measured about 42 meters and weighed about 1,200 tonnes. It was probably carved in the New Kingdom, long after the pyramids, but the marks around it reveal the method.

To remove a granite block, workers first identified a suitable section and marked the outline. They then pounded a trench around it with dolerite balls, working the stone down to the required depth. In the trench they could insert wooden wedges and use levers. To detach the block from below, they hammered out a series of slots in the base and drove in wedges, splitting the stone along a plane. The dust was cleared away by hand. Experiments have shown that a team of workers can pound a trench with dolerite at a rate of a few centimeters in a day, depending on the hardness of the rock.

Considering the scale of the Great Pyramid, the amount of granite is small: the walls and ceiling beams of the King's Chamber, the five relieving chambers above it, the sarcophagus, the plugs, and the lining of the antechamber. Even so, it comes to some thousands of tonnes, by most estimates. Each piece was cut and transported 800 kilometers. The effort of transporting

these loads on the river would be daunting, as we will see in Chapter 11.

Tura: the white stone

The fine white limestone that sheathed the Great Pyramid came from quarries at **Tura** and **Maasara**, on the east bank of the Nile, about 13 to 15 kilometers upstream of Giza. The stone there is of exceptional quality: fine-grained, pale, and so uniform that it can be cut and polished into smooth faces with narrow joints. The quarries are still partly visible. They consist of long underground galleries, worked inward along the best layers, with pillars of rock left to support the roof.

An inscription from the Fourth Dynasty records that stone was brought from Tura and loaded on boats for Giza. We know the details of one such shipment because of the logbook of the inspector Merer, which was found in 2013 and which we will read in the next chapter.

Marks, codes, and a paperwork system

The workers who cut and moved the stone left their marks on it. On many of the blocks in the Great Pyramid, especially in hidden places, there are red-painted marks of several kinds.

- **Gang names.** Teams of workers had names that expressed pride in their work, such as "The Friends of Khufu," found in the relieving chambers. At Menkaure's pyramid, a gang called "The Drunkards of Menkaure" is known.

- **Dates and counts.** Quarry marks sometimes record regnal years, months, and days. They show that the work was done on a schedule and that it was recorded.
- **Position marks.** Lines, arrows, and numbers show where a block was to be placed in the course.

Marks of this kind exist at the Great Pyramid, and they indicate that work crews were organized into named teams, that each team took responsibility for specific sections, and that supervisors tracked production. They also show that there was friendly rivalry between the teams, and that the workers took pride in being part of a royal enterprise.

Dressing and fitting

At the quarry, blocks were roughly shaped, which cut the weight and made them easier to move. At the pyramid, they were further dressed. For the interior masonry, a fairly rough fit was acceptable, and gaps were filled with gypsum mortar and chips of stone. The mortar did not act as glue. Rather, it filled the spaces between blocks, provided a smooth bed on which to slide each block into place, and spread the load.

The outer casing was another matter. The Tura limestone casing blocks were cut to fit each other with astonishing precision. The joints between casing stones were so thin, around half a millimeter, that a knife blade cannot be inserted. Each stone was cut to the specific slope of the pyramid face, and the surfaces were dressed with copper chisels, stone rubbers, and fine sand. The finishing was done after the blocks were in place,

working from the top down, so that the surface was smooth and continuous.

The fit of the casing stones, and the fact that they were dressed to a common plane while in place, is the reason that the pyramid looked, from a distance, like a single polished surface.

The size of the blocks

The common idea that the Great Pyramid is built entirely of enormous blocks is wrong. Most blocks are modest. Typical core blocks weigh about 1.5 to 2.5 tonnes, and many are smaller. The blocks in the lower courses are larger and thicker, up to about 1.5 meters tall, and the courses diminish in height as the pyramid rises, to about half a meter near the top. This arrangement makes sense for the builders: the heavy blocks are at the bottom, where they have the shortest route and the most stable footing, and the lighter blocks are at the top, where they have to be raised the highest.

There are exceptions. The granite beams over the King's Chamber, each of which weighs between about 25 and 80 tonnes, are the heaviest. The sarcophagus weighs about three tonnes. The plugs in the Ascending Passage weigh several tonnes each. And the Tura casing stones, which were thick wedges, weighed up to about 15 tonnes. But these were a minor portion of the total.

How many quarrymen?

How big was the quarry crew? Estimating this is difficult, but we can set bounds. If the Great Pyramid took 20 years and required 2.3 million blocks, the rate was about 115,000 blocks per year. A skilled team of quarrymen, working with copper tools on soft limestone, could probably produce a rough block in a day or two. So the quarry required perhaps a few thousand workers, on top of those transporting, lifting, and laying the stone. The figure fits well with the estimates of a total workforce of 20,000 to 30,000 that archaeologists derive from the size of the workers' settlement.

WORKED EXAMPLE

A rough quarry estimate. Suppose each quarry worker produced one rough block every three days, working 300 days a year, which is about 100 blocks a year. To supply 115,000 blocks a year, you need about 1,150 workers. Add a factor of 2 to 3 for the work of cutting the trench, clearing debris, repairing tools, and so on, and the figure becomes 2,500 to 3,500 quarry workers. This is a reasonable fraction of a total workforce of about 20,000.

The question of tools

It is fair to ask why more copper tools have not been found at Giza. After all, we would expect thousands. The answer is simple: copper was valuable and recycled. Worn or broken tools were melted down and recast. This is also the reason that few of the pyramid builders' tools survive. When a tool was too blunt to use, it went back to the smelter. The tools that do survive are mostly from unusual circumstances, such as a tool left in a sealed tomb or lost in a crevice.

What the quarries tell us

The quarries are the best answer to a question that is often asked: how can we be sure that the builders were Egyptians using the tools of their time? The answer is that the evidence is there. We have the quarry faces, the trenches, the half-cut blocks, the abandoned obelisk, the dolerite pounders, the copper tools, the grooves from saws, the quarry marks, the paintings of stone-working scenes in tombs, and the written records of expeditions. Taken together, they form a consistent story. There is no gap that needs to be filled by anything exotic.

KEY TAKEAWAYS

The Great Pyramid's core is local Giza limestone. The casing is white limestone from Tura. The granite came from Aswan, 800 km away, and was shaped by pounding with dolerite balls and cutting with sand-fed copper saws. Copper chisels, wooden wedges, and levers handled the limestone. The blocks were mostly modest, about 2.5 tonnes. Quarry marks, tools, and half-finished blocks show a well-organized human operation.

CHAPTER 11

Moving Stone: Sledges, Wet Sand, Boats, and a Logbook

How millions of blocks, and a few giant ones, reached the Giza plateau, and the diary of a man who helped move them.

Cutting stone is only half the problem. The other half is moving it. A block of limestone weighing 2.5 tonnes does not roll, slide, or float on its own. It has to be hauled, often over long distances, on ground that offers no hard roads. For the Great Pyramid's 2.3 million blocks, and for its few hundred heavy granite pieces, the builders needed a system of transport as carefully organized as the quarrying itself. We now know a good deal about how it worked, thanks to a handful of discoveries made within living memory.

The problem of the sledge

The Egyptians did not use the wheel to haul heavy loads. Wheeled vehicles, in the form of chariots, did not reach Egypt until about 1600 BCE, and heavy carts were not used for monument construction. Instead, builders used the **sledge**, a wooden platform with runners, pulled by ropes. Sledges survive from the Egyptian past. Several from the Middle Kingdom, built of cedar or sycamore with sturdy cross-pieces and curved runners, were found at Dahshur. Tomb paintings show them in use, loaded with statues, shrines, and stone blocks.

A sledge on ordinary desert sand works poorly. Dry sand piles up in front of the runners like a small wave, and the sledge has to climb over it. The resistance is high, and haulers tire quickly. A well-known scene suggests that Egyptians had found a remedy.

The painting that shows how

In a tomb at **Deir el-Bersha**, in Middle Egypt, belonging to a nobleman named **Djehutihotep** (about 1850 BCE), a painted relief shows the transport of a colossal statue of the owner. The statue sits on a sledge, and 172 men haul it on four ropes, in four rows. Above, the inscription says that the statue was cut from alabaster and stood about 13 cubits tall, or about 6.8 meters. The scene contains a small detail that has fascinated researchers: a man standing at the front of the sledge, on a platform, pouring liquid from a jar onto the ground ahead.

For years, scholars interpreted this as a ritual act, a libation. But in 2014, a team of physicists at the University of Amsterdam, led by Daniel Bonn, tested the idea. Their experiment, published in the journal *Physical Review Letters*, measured the force required to drag a loaded sledge across sand with different amounts of water. They found that on dry sand, the sledge piled up a heap in front of it and the force climbed. On slightly damp sand, the water created capillary bridges between the sand grains, giving the surface extra stiffness, and the heap did not form. At the right level of moisture, about two to five percent water by volume, the force required fell by about half.

KEY POINT

The Amsterdam experiment shows that wetting the sand in front of a sledge can reduce the pulling force by half. This was within the reach of any team of Egyptian workers with jars and water. It means a task that needed 40 haulers might need only 20.

The paper also noted that too much water is as bad as too little: the sand becomes soupy and the advantage is lost. A skilled worker, pouring the water at the front, would have to judge the amount. That the figure in the tomb is doing exactly this, in a scene about 4,000 years old, is remarkable. It does not prove that the same technique was used at Giza, since the tomb is 700 years later. But the technique is simple, the materials are cheap, and nothing suggests it was a recent invention.

Hauling on roads and rails

Egyptians did not simply drag stones across the desert. At Giza, they built **haul roads**, graded paths that led from the quarries to the pyramid. Some were paved or lined with stone and mud plaster. The best-preserved example is at the basalt quarry of Widan el-Faras, in the Faiyum, where a stone-paved road of about 11 kilometers leads to an ancient harbor on the shore of a lake in the Faiyum. This is the oldest known paved road in the world.

Another invention was the use of **wooden rails or sleepers**. A line of planks laid across the path under a sledge can reduce friction, and wooden sleepers have been found in a few places. Rollers are sometimes proposed, but they are not well documented for monument-building, and they perform poorly on soft sand.

WORKED EXAMPLE

How many people to pull a block? Take a 2.5-tonne block, which weighs about 24,500 newtons. On wet sand, a coefficient of friction of about 0.25 is reasonable, so the pulling force is 0.25 × 24,500 ≈ 6,100 N. If an adult can pull steadily with about 200 N for hours, you need around 30 people. On a ramp with a 1-in-8 slope (about 7 degrees), the extra force needed to lift the block is about 24,500 × 0.12 ≈ 3,000 N, bringing the total to about 9,000 N, or around 45 people. These numbers are rough, but they show why the Egyptians used gangs of 20 to 40 haulers.

The river road

The Nile was the main transport artery. To carry stone and supplies, the Egyptians built large boats, and tomb reliefs show cargo vessels carrying blocks, granite columns, and obelisks. The best evidence from the pyramid age comes from the causeway of **King Unas**, about 200 years after Khufu, where reliefs show boats carrying huge granite palm columns from Aswan to Saqqara. Weni, the official we met in Chapter 7, boasted that he had a fleet of cargo and towing boats built from acacia wood in a single year to fetch granite from Elephantine.

The seasons were crucial. During the **inundation**, from July to October, the Nile flooded the fields and spread across the plain. Boats could travel much closer to the edge of the desert, and at Giza, they could bring stone to a harbor near the plateau's foot. Granite blocks from Aswan, loaded onto boats in the summer, could drift north on the current and be unloaded at Giza. The ships were steered by oars and carried by the river's flow, and they could return upstream with the prevailing north wind filling their sails.

The river was not just a way to move cargo, but also a way to move people. This has implications for the labor force, as we will see: when the fields were flooded, farmers had no work to do on the land, and they could be sent to the project.

The harbor at Giza

The Giza plateau has no natural port, but the builders made one. Archaeologists have found evidence of a basin, canals, and harbor facilities near the base of the plateau, connected to the Nile by channels. Recent geological work has suggested that a now-vanished branch of the Nile, which scientists call the **Ahramat Branch**, once ran along the western edge of the floodplain, close to the pyramids. A study published in 2024 by Eman Ghoneim of the University of North Carolina Wilmington used satellite radar, sediment cores, and other data to trace this ancient channel for about 64 kilometers and showed that it passed very close to 31 pyramids, including Giza. Its existence helps to explain why so many pyramids are on the edge of the desert. A waterway at the base of the plateau would have made delivering stone much easier. We return to this study in Chapter 22.

The valley temple of Khufu, at the end of the causeway, would have stood beside this harbor, and the causeway itself ran from it to the pyramid, a distance of several hundred meters.

Merer's logbook

In 2013, a French-Egyptian team led by the Egyptologist **Pierre Tallet** of the Sorbonne, working at **Wadi al-Jarf**, an ancient port on the Red Sea coast, found something extraordinary. In a man-made gallery sealed by a stone, they found about 30 rolls of **papyrus**, many in fragments, dating to the reign of Khufu. They are the oldest inscribed papyri ever found. Most are administrative records of the harbor, but a group of them are the personal log of an inspector named **Merer**.

Merer was an official, a "sab," in charge of a team of about 40 men. Their job was to haul and ship stone for Khufu's pyramid. His logbook, written in hieratic script on papyrus, records day-by-day entries over several months. The entries are dated to year 26 or 27 of Khufu's reign, in the final years of the pyramid's construction.

The entries tell a clear story. Merer and his crew sailed to the quarries at **Tura**, on the east bank of the Nile, where they loaded boats with blocks of white limestone for the pyramid's casing. They then sailed down to Giza, a trip that took about a day each way, and unloaded the blocks at a harbor basin that the logbook calls *She Akhet-Khufu*, "the Lake of the Horizon of Khufu," which appears to have been connected by a canal to the Nile. A typical sequence reads, in paraphrase: *Day 1: the crew spent the day hauling stone at Tura, and spent the night at Tura. Day 2: they loaded the boats and sailed. Day 3: they unloaded at the Lake of Khufu. Day 4: they returned.* A round trip took about four days, and Merer's crew repeated this round trip many times.

KEY POINT

The log shows the Great Pyramid's casing stones moving by boat from Tura to a harbor at Giza, with a named official, a crew size, a schedule, and a destination. It is a direct written record of the construction process from someone who took part in it.

The logbook also names the official in charge of the harbor. He was **Ankhhaf**, a half-brother of the king who held a senior post at the pyramid's harbor and is named in connection with the arrival of the stone. The logbook gives a human face to the project: a regular crew, with regular hours, a boss, and a payroll.

Merer's papyri are not the only finds at Wadi al-Jarf. The site is a harbor, built about 4,600 years ago, with a stone jetty, anchors, ropes, and storage galleries holding dozens of large jars. The port was used as the starting point for expeditions to Sinai, where the Egyptians mined copper and turquoise. It tells us that Khufu's government was running naval and mining expeditions at the same time as the pyramid project. It also gives the physical evidence: an Egyptian harbor, run by Egyptian officials, on the shore of an Egyptian sea.

The ropes

Hauling on this scale required rope, and lots of it. Egyptian ropes were made of papyrus, halfa grass, palm fiber, and flax, twisted by hand in a long process, and they survive from many Egyptian contexts, including harbor stores and the boat pits at Giza, where the cedar planks of Khufu's ship were lashed together with rope. Rope-making was an industry in itself, and the expertise to make strong ropes was widely shared. In the tomb scenes of haulers, the ropes are drawn as thick as a man's wrist.

The granite problem

Granite from Aswan posed the toughest transport challenge. A single ceiling beam for the King's Chamber, weighing perhaps 50 tonnes, needed to be dragged from the quarry to the Nile, loaded on a boat, and floated for about 800 kilometers. At Giza, it was unloaded at the harbor and dragged from the harbor, along the causeway and up a ramp, to the pyramid.

The ships had to be big. Egyptians knew how to build them. The **Khufu ship**, the cedar vessel found next to the pyramid, is 43.6 meters long and built with great sophistication. It was built from planks sewn together with rope, with no nails, and has a flat hull with a shallow draft. Whether it was used for ceremonies on the river or only for the afterlife is debated, but its design shows the high level of Egyptian shipbuilding. Boats of that size, with a reinforced bottom and sturdy cross beams, could carry loads of many tonnes.

MYTH AND FACT

Myth: "The Egyptians could not have moved 50-tonne blocks without modern machinery." **Fact:** Egyptians moved even larger loads. The colossal statue in the tomb of Djehutihotep has been estimated at about 58 tonnes, and the painting shows it hauled by 172 men. Obelisks weighing more than 300 tonnes were later shipped by barge and hauled by large teams of men. The pyramid builders' loads were well within their capacity.

The logistics

The transport system was supported by a larger network. Cargo needed unloading, supplies needed to get to the haulers, and workers needed food and water. Workers, in turn, needed tools, rope, sledges, and replacement planks. The Egyptians managed this by a combination of centralized planning, local administration, and written record-keeping. It is the paper trail that makes the project work: Merer's logbook is a small part of it.

The scale of the logistics is almost as impressive as the lifting. We are not just talking about a single transport route but a regional network that took in Tura, Aswan, the Faiyum, and the Red Sea, as well as the Nile and the road network. Every day, blocks of stone, loads of grain, jars of beer, and bundles of rope were arriving at Giza. It was the largest construction logistics operation of its time, and it was run on papyrus and muscle.

KEY TAKEAWAYS

Heavy blocks were hauled on wooden sledges, with water poured on the sand to halve the friction (Djehutihotep's tomb painting; Amsterdam experiment, 2014). Stone came by river, with the flood bringing boats close to Giza. Merer's logbook (Wadi al-Jarf, found 2013) records a crew hauling Tura limestone to Giza in the 26th–27th year of Khufu. Egyptian ships and ropes were big and sophisticated enough to move the heaviest loads.

CHAPTER 12

Raising the Blocks: Ramps, Levers, and the Great Debate

The question that engineers and Egyptologists still argue about: how did the stone get to the top?

If you ask a group of people how the Great Pyramid was built, most will say the same thing: ramps. They are right, but the word hides a long and lively argument. Everyone agrees that ramps were used. What nobody has proven is exactly what kind, how many, where they ran, and what they were made of. In this chapter we look at the problem, the main proposals, and the evidence that has recently begun to settle some questions.

Where is the work?

The first thing to understand is that the work of raising stone is not evenly distributed. A pyramid is wide at the bottom and narrow at the top, so most of its volume lies near the base. This means that most of the blocks never had to be lifted very high.

For a pyramid, the fraction of volume lying below a given height is easy to calculate. If H is the total height and h the height of interest, the volume below h is $1 - (1 - h/H)^3$ of the total.

Height reached	Share of height	Share of volume below
37 m	25%	about 58%
73 m	50%	about 87.5%
110 m	75%	about 98%
146.6 m	100%	100%

More than half the stone is below 37 meters, and nearly 90 percent is below 73 meters. Only about 2 percent of the pyramid's blocks have to rise above the 110-meter mark. That changes the question. Moving blocks up to the first 40 or 50 meters is a large job, but moving them to the top is a small one. A system good enough for the lower half of the pyramid does most of the work.

The straight ramp

The simplest idea is a single ramp, built of rubble, brick, and mud, leaning against one face of the pyramid and growing higher as the pyramid grows. This is the picture many people have in mind, and it is supported by one thing: ramps are known to have been used in Egypt. Remains of construction ramps have been found at the pyramid of Senusret I at Lisht, at Meidum, and at many temple sites. Tomb scenes show ramps used to raise stones and bricks, and a New Kingdom scribal text, *Papyrus Anastasi I*, mentions the dimensions of a ramp for a colossal statue in the form of a mathematical problem.

The straight ramp has a problem, however. A ramp must keep a gentle slope or the haulers cannot pull the sledge up it. At a slope of 1 in 10, a ramp that reaches the top of the Great Pyramid would be nearly 1.5 kilometers long. Its volume would be a

large fraction of the pyramid itself, and it would need to be rebuilt, higher and longer, as the pyramid grew. Allowing for its width, it would need a large fraction of the pyramid's own volume, perhaps half or more. At Giza, there is no sign of the remains of such a monstrous ramp, which would have left a huge amount of rubble.

WORKED EXAMPLE

Size of a straight ramp. A ramp to a height of 146 meters at a slope of 1 in 10 would be 1,460 meters long. If it is 10 meters wide, its volume is about ½ × 1,460 × 146 × 10 ≈ 1.07 million cubic meters, around 40 percent of the pyramid's volume. A steeper slope of 1 in 5 would halve the length to 730 meters but roughly double the part of the hauling force that goes into lifting each block.

A straight ramp could work for the lower levels. After that, it becomes unwieldy. Some experts think the builders used a straight ramp for the first third or half of the pyramid and then switched to another method for the upper part.

The zigzag ramp

To get around the problem of length, some proposals have the ramp running back and forth up the face of the pyramid, in a zigzag. This design would use less material than a straight ramp, since the ramp is carried by the pyramid's own slope. Several scholars have proposed versions of this, in which the ramps run up the sides of the pyramid on temporary platforms.

The zigzag has its difficulties as well. At the turns, the haulers must pull the sledge around a sharp bend, which is awkward with large blocks. A ramp lying against the pyramid's face would cover the casing stones and make it difficult to level and dress them as the pyramid rose. And the evidence for it is slight.

The spiral ramp

Another idea is a spiral ramp, wrapping around the pyramid like a mountain road. It has the advantage of using the pyramid's own sides for support, and it is a familiar image from old illustrations. The problem is the corners. As the ramp passes around each corner, it would cover the edge of the pyramid, making it impossible for the surveyors to check the alignment of the corners with their instruments. Since the pyramid is so precisely aligned, it seems hard to imagine that the builders would risk blocking their sightlines. Some suggest the ramp could have been dismantled periodically to check the corners, but that adds work.

The internal ramp

In 2007, a French architect named **Jean-Pierre Houdin** proposed a different solution, with a book by Bob Brier and Houdin to follow in 2008. His idea was a combination: a straight external ramp to a height of about 43 meters, where the first third of the pyramid is complete, and from there, a spiral ramp built inside the pyramid itself. The internal ramp would run just inside the outer wall, making turns at the corners, and would be left in place as part of the structure. Houdin argued that the corners, where the ramp turns, could be seen as a small notch visible in the pyramid's northeast edge at a height of about 90 meters.

The theory is clever and has attracted a lot of attention. In 1986 and 1987, microgravity surveys by a French team found regions of different density inside the pyramid, which some have interpreted as consistent with an internal spiral, though other interpretations exist. More recent scans, including those of the ScanPyramids project, have found anomalies, but no confirmed spiral corridor has yet been identified. Many Egyptologists remain skeptical. Others find it plausible as part of the system. At present, the internal ramp theory is an interesting hypothesis, not an established fact.

The Hatnub evidence

The most important recent discovery bearing on this debate was made not at Giza but about 300 kilometers to the south, at the quarry of **Hatnub**, the source of Egyptian alabaster. Starting in 2012, a team led by **Yannis Gourdon** and **Roland Enmarch** of the French and British institutes excavated an extraordinary system there. It included a ramp with a steep slope of about 20 percent, flanked on each side by a staircase. On the ramp and next to it were numerous **post holes**, carefully cut into the rock in a regular pattern. The structure dates to the reign of Khufu, according to inscriptions at the quarry.

The excavators reasoned that the system was used to haul alabaster blocks up out of the quarry pit. Workers would have fastened ropes to wooden posts set in the holes and used them to pull a heavy sledge up the ramp. The posts acted as anchors, allowing the haulers on the stairs at each side to share the effort and to hold the load if it slipped. A sledge pulled by ropes tied to posts could climb a steeper slope than a simple team on a flat ramp, and the staircases gave the workers a stable footing.

> **KEY POINT**
>
> The Hatnub ramp was built at the same time as the Great Pyramid, by the same administration, and it used a steep slope with posts and ropes to lift heavy blocks. This is the first direct evidence of how Khufu's engineers moved heavy stone uphill. If the same system were used on the pyramid, a ramp with a slope of 20 percent would be only 730 meters long, and a series of smaller steep ramps would be even easier.

The Hatnub finds show that the Egyptians were capable of using steep ramps, anchored by posts and ropes, rather than depending on gentle slopes. This supports the idea of a system of shorter, steeper ramps used together with levers and other methods, though it does not tell us which of the ramp layouts was used at Giza.

Levers and rockers

Ramps are only part of the answer. At each stage, workers needed to move blocks sideways and to set them accurately in place. The tool for this was the **lever**, and it is widely accepted that wooden levers and wedges were used to nudge blocks into their final position. Wooden **cradle rockers**, curved timbers on which a block could be rocked from side to side and then lifted a little at a time, have been proposed as a means of raising blocks up a short step. A few objects interpreted as rockers have been found in Egyptian contexts, but whether they were used on the pyramids is disputed.

The Greek historian Herodotus, who visited Egypt around 450 BCE, said that the pyramid was built in steps, and that when the first level was complete, the stones were raised to the next by machines made of short timbers. He did not describe them. It is possible that he heard a garbled report about levers, rockers, or small ramps, and the tradition has endured.

There were no cranes or pulleys. The pulley does not appear in Egypt until much later, in the Greek and Roman periods. The Egyptians relied on ramps, ropes, levers, and muscle, and these were sufficient.

The granite beams

The hardest lifting job was the granite beams over the King's Chamber. The beams weigh between about 25 and 80 tonnes and had to be placed at a height of about 50 to 60 meters. How did the builders do it?

One answer is that a ramp of rubble, mud, and stone was built to the required height, and the beams were hauled up with sledges and rope teams, as at Hatnub, anchored by posts. Once the beams were in place, the chamber could be completed. Another is that the beams were hauled up the Grand Gallery, but the gallery is only about two meters wide at the base, and the beams are too large. A more likely answer is that they were placed during construction, before the upper courses of the pyramid were laid. The relieving chambers above them, which took the weight, were then built on top. The granite sarcophagus, which is too big to fit up the Ascending Passage, was almost certainly placed in the chamber at the same time, before the roof went on.

The placing of the capstone at the top was simple in comparison. It was the last block, a small one, and was lifted by the same methods used for the other blocks.

Combining ideas

Faced with a choice among ramp theories, many modern researchers suggest that more than one method was used. A sensible combination is:

- A **long, low ramp** (or ramps) from the quarry to the pyramid base, for the hauling of the bulk blocks.
- A **broad, straight ramp** against one face for the lower part of the pyramid.
- **Steeper, shorter ramps** or zigzag ramps for the upper levels, perhaps with post-and-rope assistance as at Hatnub.
- **Levers and rockers** for final positioning, and possibly to move blocks onto the next course at the top.

The advantage of this blended approach is that it matches the changing geometry of the pyramid. At the bottom, the course is wide and the blocks are large, so a broad ramp is used. At the top, the course is narrow, and the blocks are smaller, so a lighter system will do.

Theory	Strengths	Problems
Straight ramp	Simple. Remains of ramps at Lisht, Meidum. Tomb scenes of ramps.	Too long and bulky for upper levels. No remains at Giza.
Zigzag ramp	Uses less material. Flexible.	Sharp turns. Covers casing. Little evidence.
Spiral ramp	Uses the pyramid as support. Familiar.	Blocks corners and surveying. Needs dismantling.
Internal ramp (Houdin)	Explains the notch. Leaves no external trace.	No confirmed internal corridor. Complex.
Steep post-and-rope ramps (Hatnub)	Direct evidence from Khufu's reign.	Shown for a quarry, not yet at Giza.

What remains to be found

It may be a surprise that the question of how the Great Pyramid's blocks were raised is not fully settled. The reason is that ramps are temporary structures, made of rubble and brick, which were removed or left to erode once the work was done. The signs left behind are subtle: post holes, bedrock cuttings, small deposits of rubble, and marks on the pyramid's own surface. Archaeologists are still looking.

Despite this uncertainty, nobody has found a feature of the pyramid that could not have been made with ramps, ropes, sledges, and levers. The builders had the means, they had the evidence for their methods in other places, and they had the time. The exact form of the ramp is a puzzle of detail, not a gap in the case that humans built it.

KEY TAKEAWAYS

Most of the pyramid's volume is in its lower half, so the lifting work is concentrated near the base. A single straight ramp is too large for the full height. Zigzag, spiral, and internal ramps have each been proposed, and none is proven. The Hatnub quarry (dated to Khufu) shows steep, post-anchored ramps in use, a strong clue. Most researchers think a combination of methods was used. Levers placed the blocks. There were no cranes or pulleys.

CHAPTER 13

The Workforce: Who Really Built the Pyramids

Not slaves, not aliens: a rotating, fed, housed, and honored labor force, and the town they lived in.

Few questions about the pyramids have a more popular answer than "who built them?" For generations, the standard answer was slaves, driven by overseers with whips. That image was shaped by ancient writers, by Biblical stories, and by Hollywood. It has proven remarkably durable. It is also, as far as the archaeological evidence can tell, wrong. The people who built the pyramids were Egyptians, and the physical evidence of where they lived, what they ate, how they were buried, and how they were cared for shows a workforce that was organized, supplied, and in many respects respected.

Where the slave story came from

The notion of enslaved pyramid builders goes back to the Greek historian Herodotus, who wrote around 450 BCE, about two thousand years after the Great Pyramid was built. He reported that Cheops forced all the Egyptians to work for him, in gangs of 100,000 men who worked for three months at a time, for twenty years. He described them as suffering under a tyrant. His account is the oldest surviving description of the pyramid's construction by an outsider, and it was based on what Egyptian priests and guides told him. It has to be treated with care. The numbers are too large, the tone is that of a Greek moralist, and he was writing a long time after the fact.

A second source of the idea was the Bible. The Book of Exodus describes the Hebrews laboring under a Pharaoh, making bricks and building store cities. This is sometimes linked to the pyramids in popular memory. But the text mentions store cities, Pithom and Ramesses, and the setting, if it was historical, would belong to the New Kingdom, more than a thousand years after the Great Pyramid. The stories have no connection with Giza. Yet through centuries of retelling, and with the boost of twentieth-century films, they merged into a single mental picture.

MYTH AND FACT

Myth: "The pyramids were built by tens of thousands of slaves."
Fact: There is no archaeological evidence for slave labor on the pyramids. The evidence points to a force of Egyptian laborers, some permanent and many rotating in on a labor-service basis, who were fed, housed, buried with honor, and given medical care. Slavery existed in ancient Egypt, mostly in households and mostly involving foreign captives, but it is not documented as the basis of pyramid construction.

The discovery of the workers' cemetery

The first real challenge to the slave theory came in 1990, near the southern edge of the Giza plateau. According to a popular story, a tourist's horse stumbled on a mudbrick tomb. Whatever the truth of that anecdote, the Egyptian archaeologist **Zahi Hawass** and his team investigated and found a large cemetery that had been hidden under the sand. Over the years, the excavations uncovered several hundred tombs, in two main cemeteries, an upper one on the slope of the plateau and a lower one beside it. Many of them contained intact skeletons.

The tombs were modest, built of mud brick and local stone, but they were real tombs, with a burial chamber and often a small superstructure. Some of the upper tombs were built in the shape of small pyramids or stepped structures. Burials were placed in the traditional position, curled on one side, with the head to the west. Pottery jars, bread, and beer were left for the deceased. Some contained jewelry, such as simple beads. A few

of the larger tombs had inscriptions naming the owners, with titles connected to the work crews.

A person who was a slave, with no standing, would not have been given a tomb within sight of the royal tombs. These people were buried at Giza close to the king whose pyramid they built, in the manner of Egyptians of their rank, and equipped to take food and drink with them to the afterlife. This is how a respected laborer was buried.

What the bones show

The skeletons are a lasting record of working lives. Osteologists who examined them found clear evidence of hard physical labor: arthritis in the spine, knees, and hips, compressed vertebrae, and worn joints. Many of the men and women had stress injuries. These match the demands of heavy hauling and lifting.

But there is also evidence of care. Many skeletons show **healed fractures**, including broken legs and arms that had set well, which means the injured person received treatment and was allowed to recover. In some cases, bones show signs of surgery. A few individuals had survived **amputations**, one of a lower leg, with healing around the cut, which implies skilled surgical treatment and long recovery. A pattern of this kind is not what would be expected for slave labor, where an injured worker might simply be discarded. It reflects a system in which there was a medical service, and the workforce was valued.

The cemeteries also contain the graves of women and of children. This has been taken as evidence that the labor force included families, or that the settlements included women who

worked in bakeries, breweries, and workshops. The relative sparseness of the tombs of babies might suggest that families lived elsewhere, but the evidence is not conclusive.

The Lost City of the Pyramid Builders

If the dead are one half of the evidence, the other half is the living quarters. In 1988, the American Egyptologist **Mark Lehner** and Zahi Hawass began excavating an area south of the Sphinx, beyond a huge stone barrier known as the **Wall of the Crow**. Over the next several decades, Lehner's team, known as **Ancient Egypt Research Associates (AERA)**, uncovered what they called the **Lost City of the Pyramid Builders**, officially known as **Heit el-Ghurab**, "the Wall of the Crow."

The settlement dates to the reigns of Khafre and Menkaure, a generation or two after Khufu, but it is generally accepted as representative of the system. It was a planned community of about seven hectares, filled with streets, courtyards, houses, workshops, and storehouses.

The excavations revealed several distinct areas.

- **Bakeries.** Large installations, with rows of ovens and thousands of bell-shaped bread molds, indicate mass production of bread. The bakeries were on a scale that could feed thousands of people.
- **Breweries.** Vats and jars show that beer, a key part of the Egyptian diet and a source of calories, was brewed on site.
- **Fish and meat.** Excavators found large deposits of fish bones, and of cattle, sheep, and goat bones. A study of the

bones concluded that the workers ate a good deal of beef, which was a high-status food. It also found evidence that cattle and sheep were brought in on the hoof from farms in the Delta and elsewhere, a sign of a large supply network.

- **Barracks.** A series of long, narrow buildings, called the **Galleries**, each with a wide platform, seem to have been dormitories for the rotating laborers. They are estimated to have housed between about 1,600 and 2,000 workers at a time.
- **Houses.** Streets of smaller homes, probably for the permanent staff of overseers, craftsmen, and their families, and a district of larger houses for officials.
- **Workshops.** Remains of copper working, flint tool making, sculpture, wood working, and other crafts.

KEY POINT

The Lost City shows a settlement built for pyramid construction, with food production on an industrial scale, accommodation for thousands, a hierarchy of housing, and workshops. The workers were housed, fed, and supplied. Nothing about it suggests a population of captives.

How the labor was organized

The evidence of the Giza workmen's marks and of later documents shows that the workforce was divided into a hierarchy of units, a sort of military-style organization. Tentatively, the structure looked something like this.

- **The crew or gang**, the largest unit, with a name. Names found in the pyramid include "Friends of Khufu" and, at Menkaure's pyramid, "The Drunkards of Menkaure." A gang, according to Lehner's reconstruction, might have numbered in the range of 1,000 to 2,000.
- **The phyle**, a subdivision of a gang, of a couple of hundred men.
- **Divisions** of about 20 men, each with a leader. Merer's log-book shows a team of about 40 men led by an inspector.

Each unit had an overseer and a scribe who kept records. The names show a sense of pride and rivalry. A gang might compete to place stones faster than another.

Not all workers were equally skilled. There were different categories. A permanent core of craftsmen and overseers stayed on site year-round. These included stonemasons, carpenters, surveyors, scribes, copper workers, and rope makers. A much larger group of laborers did the heavy hauling and lifting, and rotated in for periods of weeks or months. The latter were drawn from villages and towns across Egypt and may have been part of a labor obligation to the state, known as a **corvée**. They worked for the king for part of the year, in return for rations and, in effect, as part of their tax. It was a system that was demanding, but not enslavement.

How big was the workforce?

Estimates vary. Herodotus's 100,000 is regarded as too high. Modern estimates, derived from the size of the workers' town, the food needed, the number of blocks to move, and the capacity of the quarries, settle on a total workforce of about 20,000 to 30,000 people at the peak. Lehner and others have suggested that a core of several thousand permanent workers was supplemented by a larger number of rotating laborers. The total is considerable but it is a small proportion of Egypt's population at the time, estimated at between 1 and 2 million. That would mean 1 to 2 percent of the population was engaged in building the pyramid at any given time.

The number needs to be put in perspective. Many more people were involved indirectly: farmers and herders who produced the food, boat builders, rope makers, potters, and the families who supported the workers. A pyramid project was a national economic undertaking.

Pay, food, and the beer ration

Pyramid workers were paid in kind, in the way most Egyptians were: with rations of bread, beer, and other goods, plus clothing and oil. Later records from the workers' village at Deir el-Medina, about 1,000 years after the Great Pyramid, show that rations were calculated in numbers of loaves and jars of beer, with the bread and beer an essential part of the diet. Egyptian beer was not much like a modern lager. It was a thick, nutritious, low-alcohol drink made from barley, drunk through a straw. It provided carbohydrates, vitamins, and safe hydration.

The food needs of the workforce were considerable. A modern estimate suggests that feeding 20,000 workers would require about 20 tonnes of grain a day. We return to this calculation in Chapter 14.

Why people worked

Why would tens of thousands of people leave their villages and spend months hauling stone? There are several possible explanations.

First, it was an obligation. Egyptian farmers owed labor service to the state, as part of their taxes. This was an old and deeply rooted institution.

Second, it was an opportunity. The work included regular meals, medical care, housing, and, for skilled workers, status. In a hard agricultural world, regular food could be an inducement. The inundation season, when the fields were flooded, meant farmers had no other work.

Third, it was a religious duty. Building a pyramid for the god-king was, for the Egyptians, an act of devotion and a contribution to the maintenance of the world order. A worker who helped to bring the king safely into the afterlife might hope to share some of his benefits. Many of the workers' tombs were built next to the king's tomb for just that reason.

Fourth, it was a source of pride. The gang names and the tombs show it. People took pride in their work and in the group to which they belonged.

None of this means that the work was voluntary or pleasant. A labor draft is a labor draft, and the work was hard and dangerous, and some people died. But it was part of an accepted social system, not a regime of chains.

Strikes and complaints

Direct evidence from the pyramid age on workers' complaints is scarce, but later records show that Egyptian workers were capable of organized protest. The earliest recorded strike in history took place around 1157 BCE, in the reign of Ramesses III, when the workers of the royal tomb village at Deir el-Medina laid down their tools and sat down outside the mortuary temples of earlier kings, because their grain rations were late. The strike worked, at least temporarily. The incident is recorded on a papyrus, the *Turin Strike Papyrus*, and it shows that workers on royal tombs had expectations and the power to enforce them. A state that depended on skilled labor could not afford to treat it as disposable.

KEY TAKEAWAYS

The pyramids were built by Egyptian workers, a mix of permanent skilled staff and rotating laborers, totaling perhaps 20,000–30,000 at the peak. The workers' town (Heit el-Ghurab) had bakeries, breweries, barracks, and workshops, and the workers' cemeteries show proper burials, healed injuries, and evidence of medical care. The slave story comes from Herodotus and later tradition, not from archaeology.

CHAPTER 14

Energy and Effort: The Numbers Behind the Pyramid

How much work did it take, where did the energy come from, and what do the numbers tell us?

When people ask about the energy behind the pyramids, they are often hoping for something mysterious. The real answer is both less exotic and more impressive. The energy was human: muscle, fueled by bread, beer, fish, and beef, delivered through a vast organization. When we put numbers on it, we find that the pyramid fits comfortably within what a large preindustrial society could do. It also becomes clear where the effort actually went, and the answer is not where most people expect.

This chapter works through the arithmetic step by step. Where a figure is an estimate, we say so. The aim is not to prove a single exact number but to show the scale, so that the claims in later chapters about "lost energy technologies" can be measured against reality.

Step 1: How much does it weigh?

The Great Pyramid has a volume of about 2.6 million cubic meters. Limestone has a density of about 2,200 to 2,700 kilograms per cubic meter, depending on type, and the pyramid also contains voids and mortar. A commonly used figure for the total mass is about 5.9 million tonnes, or 5.9 billion kilograms. If we divide by the 2.3 million blocks, we get an average of about 2.5 tonnes per block, which agrees with the typical block weight seen in the masonry.

Step 2: How much energy to lift it?

Raising a mass against gravity takes energy equal to the mass times the gravitational acceleration times the height. Because the pyramid is wider at the bottom, the average block did not have to be lifted all the way to the top. The **center of mass** of a pyramid is one quarter of the way up, so the average lift height is about one quarter of the pyramid's height: 146.6 divided by 4, or about 36.7 meters.

> **WORKED EXAMPLE**
>
> **Lifting energy.** $E = m \times g \times h = 5.9 \times 10^9$ kg $\times$ 9.81 m/s^2 $\times$ 36.7 m $\approx 2.1 \times 10^{12}$ joules. In more familiar units, that is about 590,000 kilowatt-hours, or 590 megawatt-hours.

To put this number in context: 590 megawatt-hours is what a single one-megawatt power station produces in about 24 days. It is the electrical energy used by about 55 average American

homes in a year. In terms of fuel, it is roughly the energy in 60,000 liters of gasoline, or a couple of tanker trucks.

That sounds small, and it is. A modern engineer would say that the useful work done in raising the stone was modest. This is the figure sometimes quoted to claim that the pyramid "needed" some special source of power. It does not.

Step 3: How many calories does that take?

Human muscle is not very efficient. The body converts food energy into mechanical work at about 20 to 25 percent efficiency; the rest becomes heat. If we use 20 percent, the food energy needed to supply the 2.1×10^{12} joules of lifting work is five times as much: about 1.06×10^{13} joules. A food calorie (kilocalorie) is 4,184 joules, so that equals about 2.5 billion kilocalories.

A working adult in a heavy job might need about 3,500 kilocalories per day. If a workforce of 20,000 people ate that much, the group would consume 70 million kilocalories per day. So the lifting work alone, in energy terms, corresponds to about 36 days of food for such a workforce.

KEY POINT

The energy needed simply to lift the stone is equal to about five weeks of food for a 20,000-person workforce. The pyramid was not hard because of the lifting energy. It was hard because of everything else.

Step 4: What was the rest of the effort?

The lifting energy is only a small part of the story. A human being can do about 75 to 100 watts of useful mechanical work over a long working day. Let us take 75 watts, a reasonable figure for sustained labor, and assume an 8-hour workday. A single worker delivers 600 watt-hours, or 0.6 kilowatt-hours, per day. A workforce of 20,000 delivers 12,000 kilowatt-hours, or 12 megawatt-hours, per day.

Over the 20 years of construction, assuming 300 working days a year, the total work output of the workforce would be about 6,000 days times 12 megawatt-hours, or 72,000 megawatt-hours. Compare that with the 590 megawatt-hours needed to lift the stone: the lifting was less than one percent of the total labor capacity of the workforce.

Where did the other 99 percent go? Several places.

- **Quarrying.** Cutting 2.3 million blocks out of rock by hand, with copper tools and pounders, is slow. Every cubic meter of stone removed takes many hours of labor.
- **Dressing.** Shaping blocks so that they fit, and dressing the outer casing to a polished face, was labor-intensive, particularly the casing.
- **Hauling on the level.** Pulling blocks across sand and up the ramps, against friction, takes far more energy than the lifting itself, as we calculate below.
- **Boats and rope.** Building, loading, rowing, and unloading ships, and making rope.

- **Building and removing ramps.** The ramps themselves were enormous heaps of material that had to be built, raised, and then taken away.
- **Support work.** Brewing, baking, farming, carrying water, making tools, keeping records, and managing the settlement.
- **Idle time and rest.** Even the best-organized workforce has downtime, sick days, and festivals.

Step 5: The cost of friction

To see how friction dominates, we need to calculate the work done in dragging blocks. Take the average block, 2.5 tonnes, dragged across level wet sand at a friction coefficient of about 0.25. The force is 0.25 × 2,500 × 9.81, or about 6,100 newtons. Work equals force times distance. If the average block was dragged 500 meters on the level between the quarry, the harbor, and the foot of the pyramid, the energy is 6,100 N × 500 m, or about 3 million joules per block.

For 2.3 million blocks, the total is about 7×10^{12} joules. That is more than three times the lifting energy. If we add the work of dragging blocks up ramps, where friction and lifting are combined and the sledge has to travel a longer distance than the vertical rise, the total frictional work grows again.

WORKED EXAMPLE

Friction versus lifting. Lifting: 2.1×10^{12} J. Hauling 500 meters on the level at a friction coefficient of 0.25: 2.3×10^{6} blocks × 3.1×10^{6} J ≈ 7.0×10^{12} J. Hauling up ramps at a 1-in-8 slope adds friction over the ramp length, perhaps another 3 to 5×10^{12} J. Total hauling energy is therefore several times the lifting energy. If the sand was dry, with a friction coefficient of about 0.5, the hauling energy would double. Wetting the sand saved about half of it.

This calculation shows why wetting the sand mattered so much. The saving of several trillion joules, achieved with a few jars of water, is one of the most cost-effective innovations in the history of construction. It also shows why the Egyptians built harbors near the pyramid and why they chose a quarry next door: every extra 100 meters of hauling added hundreds of billions of joules.

Step 6: Feeding the workforce

The work had to be paid for in food. At 3,500 kilocalories per person per day, a workforce of 20,000 needs about 70 million kilocalories per day. Barley, the main cereal, contains about 3,300 kilocalories per kilogram (as flour or grain). That means about 21 tonnes of grain per day, or about 7,700 tonnes a year. Of course, the Egyptian diet was not all barley. Some of the calories came from emmer wheat bread, beer, fish, meat, vegetables, and fruit. But the order of magnitude is what matters.

Over the 20 years of the project, the workforce would consume about 155,000 tonnes of grain, or the equivalent. This

sounds enormous, but it needs to be seen against the productivity of the Nile Valley. Egyptian farmland in a good year could produce on the order of a tonne of grain per hectare, and the cultivated area of the Nile Valley and Delta was very large. The pyramid's food needs were well within the capacity of the country. The workers' food also did not come from nowhere: much of it would have been eaten at home had the workers not been drafted. The state's added cost was the extra food for the specialist and permanent workers, and the logistics of moving it.

The Giza settlement shows the process: bakeries with bread molds by the thousand, breweries, fish processing, and animal pens. The economics worked because Egypt had a taxation system that collected grain and a transport network, the Nile, that moved it cheaply.

Step 7: Putting it all together

A rough budget for the work of building the Great Pyramid looks like this, using round figures and the assumptions stated above. The figures should be taken as orders of magnitude.

Item	Energy (joules)	Comment
Lifting the blocks (mechanical)	about 2×10^{12}	The lowest-cost part
Hauling on the level (wet sand)	about 7×10^{12}	Friction dominates
Hauling on ramps (extra friction)	about $3–5 \times 10^{12}$	Depends on ramp design
Quarrying, dressing, ropes, boats, ramps, support	the remainder	Most of the total effort

Item	Energy (joules)	Comment
Total workforce capacity (20,000 × 20 years)	about 2.6×10^{14}	At 75 W sustained

The totals do not match exactly, since most of the work in quarrying and dressing is hard to quantify. But the broad picture is clear. The labor force of 20,000 had enough capacity to do the work, with plenty of margin for every kind of inefficiency. A workforce of 5,000 would have been too small. A workforce of 100,000 would not have been needed. A figure in the range of 20,000 to 30,000, which is what archaeologists have estimated from the settlement and the logistics, matches the physics.

Comparisons

Sometimes it helps to compare the Great Pyramid with modern projects.

- The Hoover Dam used about 3.3 million cubic meters of concrete, comparable to the pyramid's volume, and took about five years, with a workforce of up to 5,000 and heavy machinery.
- The Great Pyramid took about 20 years with muscle, ropes, copper, and wood.
- A modern construction crane lifts a block of 2.5 tonnes to a height of 100 meters in about a minute, using a few tens of kilowatts. A work gang of Egyptians took several hours to bring the same block up the pyramid.

The comparison is not to belittle the Egyptians but to show that the task was big, and that the time and workforce the Egyptians used were reasonable for their technology.

What the energy numbers do not say

It is also worth being clear about the limits of these calculations.

- They do not prove how the pyramid was built. They only show that a human workforce with simple technology had enough power to do it. The method has to be established by other evidence, such as the quarries, the tools, the logbooks, and the workers' town.
- They are not precise. Changing the friction coefficient, the average haul distance, or the efficiency can shift the totals by a factor of two or more.
- They do not include the labor of the people who produced the food and materials. A fuller account would add the farmers, herders, and boat crews.

Nonetheless, the basic conclusion is robust. The pyramid does not require any energy source other than human muscle, animal power, the river, and the sun. There is no energy deficit to be explained.

MYTH AND FACT

Myth: "No ancient civilization had enough power to build the pyramids, so they must have used another energy source." **Fact:** The mechanical work of lifting the Great Pyramid is about 590 megawatt-hours, less than 1 percent of the work that a 20,000-person workforce could do in 20 years. The big costs were friction, quarrying, and logistics, all of which the Egyptians managed with muscle power and good organization.

KEY TAKEAWAYS

Lifting the Great Pyramid takes about 2×10^{12} joules, or 590 MWh, equal to about 5 weeks of food for a 20,000-person crew. The real costs were friction, quarrying, dressing, and logistics, together amounting to more than 99 percent of the labor capacity. Wetting sand halved hauling effort. Feeding the workers needed about 21 tonnes of grain a day, well within Egypt's means. The numbers match archaeological estimates for the workforce size.

CHAPTER 15

Inside the Great Pyramid: Shafts, Voids, and Hidden Doors

What is known about the interior's mysteries, from the star shafts to the cosmic-ray discoveries of the last decade.

Chapter 5 gave a tour of the Great Pyramid's main passages and chambers. This chapter takes up the parts that have generated the most speculation: the narrow shafts, the doors in the Queen's Chamber, the relieving chambers, and the new voids revealed by modern scanning. This is the frontier of pyramid research, and it is a place where careful science and enthusiastic speculation often meet.

The four shafts

The Great Pyramid has four narrow shafts, two leading out of the King's Chamber and two from the Queen's Chamber. They are roughly 20 centimeters across, square or nearly so, and they rise at angles through the masonry. They are unique among Old Kingdom pyramids, and nothing like them has been found elsewhere.

The King's Chamber shafts emerge at the pyramid's north and south faces. The northern one rises at about 31 degrees at first, then bends to avoid the Grand Gallery and continues at

about 32 degrees. The southern one rises at about 45 degrees. Both reach the outer surface of the pyramid, though the openings are now at a height that is not accessible without scaffolding. Their openings inside the chamber were noted by nineteenth-century visitors, and in the 1990s the shafts were explored with robots. A small fan was installed in the southern shaft in 1992 to circulate air in the chamber for visitors.

The Queen's Chamber shafts also run north and south, rising at about 39 degrees, but they never reach the surface. At their lower ends, the builders left the last few centimeters of stone in the chamber wall in place, so the shafts were invisible until Waynman Dixon and his team found them in 1872. Both were explored with robots, and each stops at a limestone slab or door.

The shafts are the subject of debate, and the debate has produced a number of ideas.

- **Ventilation.** An early assumption was that these were air shafts, to ventilate the chambers. But the Queen's Chamber shafts were sealed and the chamber was never meant to be used by living people. This idea is generally rejected.
- **Star alignment.** In the 1960s, the astronomer Virginia Trimble and the engineer Alexander Badawy noted that the shafts from the King's Chamber seem to have pointed at particular stars at the time of construction. Their analysis suggested that the southern King's Chamber shaft pointed at Orion's Belt (Al-Nitak), which was linked to Osiris, and the Queen's Chamber southern shaft pointed at Sirius, linked to Isis. The northern shaft of the King's Chamber was thought

to point to the region of the northern stars, including Thuban, the pole star of the time, or alternatively to the circumpolar stars. The northern Queen's Chamber shaft was thought to point to Kochab. This idea has been very popular.

- **Ritual function.** Another suggestion is that the shafts are symbolic "passages" for the king's spirit, or *ba*, to leave the pyramid and join the stars. This reading fits the Pyramid Texts, which speak of the king ascending to the sky. In this view, the shafts need not be aimed at any specific star, only at the sky in general.

> **KEY POINT**
> The star-alignment theory is attractive, but it has weaknesses. The shafts bend, the dating is uncertain, and the shafts at the Queen's Chamber were not open to the sky, so no one could see a star down them. It is possible that the builders aimed them toward general regions of the sky. The ritual interpretation does not require that stars be seen.

The Queen's Chamber doors

The Queen's Chamber shafts have been explored three times with robots, and each time produced a new surprise.

In **1993**, the German engineer **Rudolf Gantenbrink** sent a small tracked robot called *Upuaut 2* up the southern shaft. About 65 meters along, the robot found a small slab of limestone blocking the shaft, with two **copper fittings** that look like

handles. This was the first "door." The finding made headlines worldwide, and gave rise to expectations of a hidden chamber behind it.

In **2002**, a National Geographic team led by Zahi Hawass sent a more advanced robot, the *Pyramid Rover*, to the same spot. It drilled a small hole through the first slab and pushed a camera through. Behind the slab was a small empty space, and then a second slab, similar to the first. The second slab had no visible handles. The 2002 televised event was dramatic, but the result was inconclusive.

In **2011**, an international team that included the roboticist **Rob Richardson** sent a more refined robot called *Djedi* with a flexible fiber-optic camera that could be threaded through the hole. The camera was able to look around the corner and into the space between the slabs. It also looked at the back of the first door and saw **red painted marks**, written in what look like hieratic or masons' marks, on the back of the first slab. It also showed that the space beyond the first door is a small cavity. The second door was not breached.

Some interpretation has been attempted, but there is no scholarly consensus. The most common view is that the doors are **symbolic**: a representation of a gate through which the king's spirit would pass. Some propose that the slab is a stopper, set in place to close the shaft and perhaps hide a small chamber. In the absence of more evidence, the question stays open. Researchers have proposed new robot missions, but the permit system is slow.

MYTH AND FACT

Myth: "A hidden chamber full of treasure has been found behind the doors in the Queen's Chamber shafts." **Fact:** Behind the first door is a small empty cavity and a second door. No chamber has been entered, and no one has seen anything behind the second door.

The relieving chambers and Vyse

Above the King's Chamber there are five relieving chambers, between the ceiling beams and the apex of the structure. The lowest is called Davison's Chamber, after Nathaniel Davison, a British consul who found it in 1765 by crawling through a hole above the Grand Gallery. The other four were opened in 1837 by Colonel Howard Vyse and his engineer John Perring, who used gunpowder to blast his way upward through the ceiling stones. He was looking for hidden treasure and burial chambers. He found none.

What he did find were **graffiti** painted in red ocher on the stones of the topmost chambers, in places where no one ever expected them to be seen. They include the cartouches of Khufu, the names of work gangs, and various marks. Because these chambers had been sealed since construction, the graffiti are authentic marks from the time of building, and they are the best evidence for the pyramid's date.

MYTH AND FACT

Myth: "Vyse forged the Khufu cartouches inside the pyramid."
Fact: This claim was popularized by the author Zecharia Sitchin in 1980. But the paint inscriptions run across the joints between blocks and in places that Vyse could not have reached. Some are upside down or on the hidden faces of blocks that were placed before the chambers were formed. They match the style of Fourth Dynasty writing and gang names. Forgery would have required a level of Old Kingdom scholarship that did not exist in 1837, since hieroglyphs had been deciphered only in 1822. The marks are accepted as genuine by Egyptologists.

The subterranean chamber and the well

Beneath the pyramid, the unfinished Subterranean Chamber is a rough, irregular cavity cut in the bedrock, with a pit in the floor and a short dead-end passage. Its purpose is unclear. Some suggest that it was an early burial chamber, abandoned when the plan changed. Others suggest it was a symbolic tomb of Osiris. Its roughness suggests it was not completed.

Another feature is the **Well Shaft**, a narrow, irregular shaft connecting the lower end of the Grand Gallery to the Descending Passage. It is thought to have been an escape route for workmen who sealed the Ascending Passage's granite plugs. After lowering the plugs, the workers could have exited by the shaft, rather than being trapped inside. That is an engineering explanation, and is generally accepted.

ScanPyramids: using particles to see inside

For two centuries, explorers entered the Great Pyramid with ropes and candles. In the twenty-first century, they use particles from outer space. In 2015, an international project called **ScanPyramids**, involving Egyptian, French, and Japanese researchers, began using non-invasive technologies to look inside the pyramids without drilling. The methods included:

- **Muon tomography.** Muons are particles created when cosmic rays strike the upper atmosphere. They rain down in huge numbers and can pass through hundreds of meters of rock. Detectors placed inside or around the pyramid record how many muons arrive from each direction. Where there is a void, more muons arrive; where there is dense rock, fewer. Over weeks and months of exposure, a detector builds up an image of density, like a cosmic X-ray.
- **Infrared thermography.** Thermal cameras can detect small temperature differences on the surface, which may point to hidden structures.
- **Ground-penetrating radar and micro-gravimetry.** These detect differences in density and electrical properties.

The first results appeared quickly. In 2016, the team found a thermal anomaly on the east face of the Great Pyramid, in the stones at the base. In 2017, they announced a much bigger discovery, which they published in the journal *Nature*: a large void in the pyramid, above the Grand Gallery.

The Big Void

The **Big Void**, announced in November 2017, is a cavity at least 30 meters long, with a cross-section similar to that of the Grand Gallery. It lies above the Grand Gallery, at about the same slope, and it was detected by three independent muon detection techniques, using different detector types in different locations. It was the first major discovery inside the Great Pyramid since the nineteenth century.

What is it? No one knows. Scientists have made several suggestions.

- A second gallery, possibly a construction feature, built to relieve the weight on the Grand Gallery.
- A building-ramp remnant or an internal construction space.
- A chamber, empty or filled with something.

The void has not been entered. In 2023, the team announced a further step: they inserted a small endoscope through a joint in the pyramid's masonry, and imaged the interior of another structure, discussed below. The Big Void remains unexplored, and its meaning is still being debated.

> **KEY POINT**
> The Big Void shows that even after 4,500 years, the Great Pyramid still holds surprises. But the discovery of a void does not imply a hidden treasure room or burial chamber. Large pyramids contain many structural voids, such as relieving chambers and construction spaces.

The north-face corridor

The 2016 scans also detected a void behind the pyramid's north face, above the original entrance, at the place where large gabled stones form a chevron structure. In 2023, the team drilled a tiny hole, about six millimeters wide, through a joint between the stones, and inserted a flexible endoscope. The camera revealed a **corridor**, about nine meters long and about two meters wide, with a gabled ceiling, behind the chevron stones. The walls were roughly finished. It is a sealed passage that was not known to exist.

The team's interpretation was that the corridor was built to relieve the load on the entrance passage, which sits beneath the chevron. It runs along the same plane. It is a structural feature, not a burial chamber. It could also be an access route to something else. At the end of the corridor, a large block was seen, which may conceal a continuation.

The corridor is considered a clear example of the way new technology can add detail to the pyramid's design without damaging the monument. It also raises a question: if there is a corridor that has been invisible for 4,500 years, what else is there?

Reading the interior

The interior of the Great Pyramid is a record of a design in flux. The plans seem to have changed as the work went on. The Subterranean Chamber, the Queen's Chamber, and the King's Chamber, with their different degrees of finish, show the building plan being revised upward. The shafts, the relieving chambers, and the Grand Gallery are all adapted to weight management. The newly discovered voids are consistent with an elaborate set of structural and perhaps ritual features. None of them requires an explanation beyond ancient Egyptian engineering and belief.

KEY TAKEAWAYS

The Great Pyramid's shafts probably served a ritual purpose, possibly linked to stars. Robots have found small doors at the end of the Queen's Chamber shafts, but nothing beyond the second door has been seen. The relieving chambers contain authentic Fourth Dynasty graffiti, including Khufu's name. In 2017, ScanPyramids found the Big Void above the Grand Gallery, and in 2023, a corridor behind the north face. Neither has been entered, and neither implies a treasure room.

Interlude: A Day on the Plateau

An imagined reconstruction of one working day during the building of the Great Pyramid, with every detail tied to the evidence.

Before we turn from how the pyramids were built to what they meant, it is worth pausing to put the evidence together into a picture. This interlude is a work of reconstruction. The characters in it are invented, apart from one, Merer, the boat foreman whose logbook has survived. The day itself is imagined. But each detail is drawn from something archaeologists have found, and at the end of the chapter a table shows where each one comes from. Think of it as a way of testing the evidence: if the pieces fit together into a believable day, we are probably on the right track.

Before dawn

It is the season of the inundation. The Nile is high, and the fields of the valley are under water, glinting like a long, flat mirror in the faint light. On the desert edge, on the ridge above, the great pyramid is a pale mass against the stars, with its lowest courses sheathed in white and its upper steps still raw. Below the plateau, in the long low streets of the workers' town, the first smoke is rising.

Nakht, a baker, has been awake since the middle of the night. He tends a row of ovens at the edge of the settlement, where clay molds shaped like tall bells are stacked in the embers. He fills each with dough, a mixture of emmer wheat flour and water, and lets the bread bake in the heat. By the time the sky turns grey, there are hundreds of loaves, flat on one side and round on the other, with a dense, slightly sour crumb. Beside the bakery is the brewery, where Nakht's cousin stirs vats of mashed barley and bread dough, which will ferment into beer. It is a thick, nourishing, low-alcohol drink, closer to a liquid meal than to the beer of today. Neither product is a luxury. Together they make up the main part of a worker's pay.

Farther down the street, a butcher is already at work. A steer has been brought in from the royal estates in the Delta, and the meat will be divided among the work gangs. The town's refuse heaps, which archaeologists would one day dig through, are full of the bones of cattle, sheep, and goats, along with the remains of fish from the Nile. The workers of the pyramid eat better than the average farmer of the valley.

The gang assembles

As the light grows, men begin to gather in the lanes between the long galleries where they sleep. These are the work gangs. Each has a name, and the name is a point of pride: the men of one crew call themselves "Friends of Khufu," another "Followers of Khufu," or by a more playful name. The gang is divided into smaller parties, each with its own leader and its own mark, which will be painted on the blocks it moves. The marks are practical, so that the supervisors can tell which crew did what. They are also a way of claiming credit.

The foreman of this gang is called Senu. He is not a farmer but a permanent member of the project's staff, a skilled man who has been on the plateau for years. Most of the men in his gang are different: they are farmers from the villages of the Delta and the Nile valley, called up for a period of service during the weeks when their fields lie under water. They have come with their rations of bread and beer, and with bundles of clothing. They will work here for a stretch of months and then go home to plant their fields as the flood recedes. Others will take their places. The system gives the state a huge labor force and gives the farmers a task for the idle season, and in a society that has no money, labor is a form of tax.

Senu counts his men, and a scribe beside him enters the numbers on a sheet of papyrus. Each worker will receive a ration at the end of the day. The scribe's list is the foundation of the project: it tells the granaries how much grain to release, and it tells the overseers who is present.

At the quarry

The plateau itself is the quarry for the main body of the pyramid. A short walk from the base of the monument, a long gash has been cut into the rock, and there the stonecutters are at work. They are skilled men who have spent years learning to read the grain of the limestone. One of them, a veteran named Ipi, uses a copper chisel and a wooden mallet to cut a channel around a block. He works slowly and carefully, tapping, checking, tapping again. Then he drives wooden wedges into slots at the base of the block and wets them with water. As the wood swells, the stone cracks along the line he has prepared. A team of men, using wooden levers, edge it free.

The block weighs about two and a half tonnes. It is not a perfect cube. It is a rough, squarish mass, with edges that will be trimmed at the building site to fit its neighbors. Ipi paints a mark on its side with red ochre: the name of his gang and a number that will tell the masons where it belongs.

Copper chisels dull quickly, and a boy runs back and forth between the quarry and a nearby workshop, where a smith sharpens them, hammers them back into shape, and sometimes casts new ones. At the edge of the quarry lies a heap of limestone chips, a kind of waste that will be used to level the courses of the pyramid or to fill the spaces between blocks.

Dragging the stone

Once a block has been cut, it is levered onto a wooden sledge, a simple runner-based frame made from imported cedar or from local timber, and lashed in place with ropes. A team of a dozen or two men takes hold of the ropes. In front of the sledge, a man crouches with a jar of water and pours a thin stream onto the sand. The sand darkens. It will pack more firmly, and the sledge will glide more easily. Modern experiments have shown that sand wetted to the right degree can halve the force needed to pull a sledge. A scene in a tomb painting from a later period, in which a man pours water in front of a giant statue on a sledge, shows that the Egyptians knew the trick.

The men chant to keep time. A foreman walks beside them, calling out directions. Ahead of them runs a prepared track of packed gravel and mud, which has been laid with care, and in places lined with wooden beams. At the foot of the pyramid, the track turns into a ramp. The ramp is made of rubble, mud brick, and clay, and it climbs against the side of the monument. As the pyramid rises, the ramp is extended.

A short distance away, at a different quarry, a different gang is moving a different kind of block. These are the casing stones, which will form the smooth outer surface of the pyramid. They are cut from the fine white limestone of Tura, across the river, and they arrive by boat. Each is carefully dressed on its face and its sides, so that it will fit snugly against its neighbors, with a joint no wider than a fingernail.

The boat from Tura

By mid-morning, a fleet of boats approaches the harbor at the foot of the plateau. At the head of one of them stands a man with a roll of papyrus in his hand: Merer, an inspector in charge of a crew of about forty men. He has been traveling with his boat for several days, loading limestone at Tura and delivering it to the harbor at Giza, then returning for another load. He keeps a daily record of the journey, noting when his crew sets out, where they spend the night, and when they arrive. Some of the entries are dry and brief. Others mention the name of the high official who oversees the harbor, a royal prince, and the name of the monument for which the stone is intended: Akhet-Khufu, "the Horizon of Khufu."

Merer and his crew tie up the boats at the edge of the basin. The stones are unloaded onto sledges, and another scribe checks the delivery against a list. Merer receives his rations and an acknowledgment. His boats will head back across the river by afternoon, rowing against the current of the flood or sailing with the north wind behind them. It is a routine job, and Merer does not know that his log will survive for four and a half thousand years in a sealed gallery beside the Red Sea, or that his words will be the key to understanding how this monument was built.

The noon heat

By noon the sun is fierce, and the stone is hot to the touch. The work slows. In the shade of a long, low shed, men sit with their backs against the wall and share bread and beer. Some eat onions and a little dried fish. An overseer walks through, speaking to the foremen. The day's work is on schedule.

One of Ipi's apprentices has cut his hand on a flake of stone, and a medic dresses the wound with honey, linen, and a poultice. The skeletons from the workers' cemeteries show that injuries were common, including broken bones, crushed limbs, and the wear of heavy labor, and some of those injuries healed, a sign that the injured received treatment and time to recover. The project has a small medical staff, and a man who cannot work is taken care of. That is not charity: a worker lost is a skill lost.

At the edge of the shed, a priest of Khufu's cult offers a few words and a libation to the god of the workers. The pyramid is a sacred project as well as a practical one. Everyone who works on it takes part in the king's passage to eternity, and the king is a god. The labor is a service as well as a duty.

At the pyramid

In the afternoon, the masons are at the foot of the monument. Here the blocks are set. A gang of men on the lower courses guides each stone into position with levers and wooden rockers, nudging it a few fingers at a time. A mason checks the line with a taut cord, a plumb bob, and a wooden set-square. A thin layer of gypsum mortar, made from burnt and ground gypsum, is spread between the blocks to act as a lubricant while the stone is shifted, and as a filler once it is in place. Higher up, men work on the ramp, hauling the stones to the upper levels.

On the rising structure, the builders have marked the center lines and the corners with ink and chisel. The base was laid out in the earliest stage with great care, using the stars or the shadows of the sun to find true north, and a network of channels filled with water to establish a level. The accuracy they achieved is such that even today, nearly forty-six centuries later, the four sides of the pyramid still point nearly exactly to the compass points.

In one of the chambers inside, a team is setting the last of the stones that will form the roof of a room. Here and there, on the blocks that will soon be hidden forever, they paint the names of their gangs. The paint will survive undisturbed in the dark for thousands of years, until a nineteenth-century explorer, with a crowbar and gunpowder, breaks into the space and reads them.

The end of the day

As the sun sinks, the work stops. The gangs return to the town in the long evening light, tired and covered in dust. The ovens are lit again. At the doors of the galleries, the rations are handed out: loaves of bread, jugs of beer, meat, onions, perhaps a few figs or dates. The men eat and talk. Some play a board game, in a corner of a courtyard, with pieces of clay or stone. A few of them go to a small chapel, where they leave a loaf of bread for the spirit of a colleague who died of a fall some weeks before and is buried in the cemetery on the hill above the town.

That cemetery lies in the shadow of the pyramid itself. The tombs of the workers are small but carefully built, of mud brick and stone, some with a little pyramid shape on top, with a chamber below. Inside are jars of beer and loaves of bread, set beside the body for the journey to the next life. A few of the tombs carry inscriptions naming the owner and his title. The people who built the pyramid were permitted, in a modest way, to share in the same hope that the king enjoyed. They were buried within sight of his monument, as members of his household.

Night falls. Across the settlement, the lamps are lit, and the smell of baking bread and woodsmoke drifts over the roofs. On the plateau above, the half-built pyramid is a silhouette against the stars. It will take another decade of days like this one to finish it.

What is evidence and what is imagination

Detail in the story	Basis
Bread baked in conical clay molds; beer brewed on site	Bakeries and breweries excavated in the workers' town at Heit el-Ghurab, with large numbers of bread molds
Meat rations of beef, sheep, goat, and fish	Animal bones and fish remains in the town's refuse heaps
Gangs with names such as "Friends of Khufu"	Painted gang names and marks in the relieving chambers above the King's Chamber and on quarried blocks
Rotating labor, with farmers working in the inundation season	Inference from the scale of the labor force, the town's size, and the structure of Egyptian labor taxation; widely accepted, but details are debated
Quarrying with copper chisels, wooden wedges, and dolerite pounders	Tools found at quarries; wedge slots in quarry floors; experiments
Water poured in front of the sledge	A tomb painting of a statue's transport (Djehutihotep, c. 1900 BCE) and modern experiments showing that wet sand reduces friction
Casing stones from Tura, delivered by boat	The papyri of Wadi al-Jarf, including Merer's logbook
Merer's crew of about forty, journeys to Giza, Akhet-Khufu	The logbook of Merer, which records these details
Injuries and medical care	Skeletons in the workers' cemeteries with healed fractures and signs of surgery
Small tombs with bread and beer	Workers' cemeteries excavated at Giza in the 1990s
Names painted in hidden chambers	Work-gang graffiti discovered by Howard Vyse in 1837

Detail in the story	Basis
The characters Nakht, Senu, and Ipi, and the specific events	Invented for the story

Why reconstruct?

A reconstruction like this has risks. It can make uncertain things sound certain, and it can supply a human warmth that the surviving evidence does not directly provide. But it also does something that a list of facts cannot. It lets us check whether the facts fit together. And it reminds us that behind the numbers in the previous chapters, the tonnes and the kilojoules and the cubits, were people who got up in the dark, ate their bread, and went to work.

PART 4

Life, Death, and Belief

What the pyramids were for, and what happened to them

CHAPTER 16

The Pyramid Complex: Temples, Causeways, Boats, and Priests

The pyramid was the center of a machine for eternity, and a town that kept it running for centuries.

When people picture a pyramid, they usually picture a lone triangle in the sand. Egyptians would not have recognized the picture. For them, the pyramid was one part of a **complex**, a set of buildings with different functions that together formed the royal tomb. The complex reached from the desert plateau to the edge of the farmland, and it was supported by a community that lived and worked beside it. Understanding the complex is the best way to understand what the pyramid was for.

The standard plan

By the time of the Fourth Dynasty, a standard layout had developed. It is clearest at Giza, in the complexes of Khufu, Khafre, and Menkaure, and it was repeated, with variations, for the next several centuries. The parts, in order from the river to the burial, were as follows.

Part	What it was	Function
Harbor and valley temple	A building at the edge of the floodplain, linked to a canal or basin	The entry point. Boats arrived here. Ritual purification of the body and reception of offerings
Causeway	A long, covered or open walkway, sometimes decorated, running up to the plateau	The processional route. Connected the valley temple to the pyramid
Mortuary temple	A temple built against the east face of the pyramid	The main place for daily offerings and the royal cult
Pyramid	The monument itself, with a burial chamber	Protection of the body, ascent of the king
Cult (satellite) pyramid	A small pyramid, usually on the south side	Its function is debated. Perhaps for the king's *ka*, or symbolic of a southern tomb
Queens' pyramids	Smaller pyramids for royal wives	Burial of queens
Boat pits	Rock-cut trenches around the pyramid	Funerary boats for the king's journey
Enclosure wall	A wall of limestone around the pyramid	Separated the sacred area from the world

Not every complex had every feature, and some elements are missing from some sites. But the general logic was consistent, and the overall plan stayed fairly stable for a thousand years.

The valley temple

The **valley temple** stood at the bottom of the causeway, close to the edge of the cultivated land and the harbor. It was the formal entrance. When the king died, his body was brought by boat to the valley temple, where it underwent final rites of purification before the funeral procession. The building was also the start of the daily ritual cycle: offerings were presented there and carried up the causeway to the pyramid.

The best-preserved valley temple is Khafre's, built of massive limestone blocks cased in red granite, with a floor of alabaster. We saw in Chapter 6 how its pillared hall once held about twenty-three statues of the king. Other valley temples have been lost or are buried under modern buildings. Khufu's valley temple, for example, is thought to lie under the village of Nazlet el-Sissan, at the foot of the plateau, and has been only partly investigated.

The causeway

The **causeway** was a long ramp, running for hundreds of meters from the valley temple up the plateau to the pyramid. In Khufu's case, it was about 700 meters long, and in Unas's, 750. The causeways were often roofed and walled, giving them the appearance of a long corridor, with a narrow slit in the roof to let in light. The walls were covered in painted relief carvings.

The Fifth Dynasty causeway of **Unas** at Saqqara preserves some of the most informative reliefs. They show scenes of the transport of granite columns by boat, of craftsmen at work, of

foreign trade expeditions, and, famously, of people suffering from hunger. This last scene, which shows emaciated figures, has been interpreted as a reference to famine in the region. It is rare to find such a depiction of hardship in a royal monument, and it indicates that artists of this period could depict harsher realities.

The causeway also served the logistics of construction. While the pyramid was being built, the causeway may have been used as a haul road. Some researchers think that it was partly a ramp, serving first for construction and then, after the pyramid was complete, as the processional way.

The mortuary temple

On the east side of the pyramid, facing the rising sun, stood the **mortuary temple**. It was the place where, day by day, the king's spirit was fed. In the earliest pyramids of the Fourth Dynasty, the temple was a modest structure, but by the Fifth Dynasty, it had developed into an elaborate building, with an entrance hall, a pillared courtyard, five niches for statues of the king, storerooms, and an inner sanctuary with a false door facing the pyramid.

The false door was a carved panel that represented a door between the world of the living and the realm of the dead. The king's *ka* was thought to pass through it to receive offerings of food, drink, linen, and incense. Offerings were laid on a stone altar in front of the door. The reliefs on the temple walls depict processions of estates bringing produce, rituals, and scenes of the king with the gods.

Khufu's mortuary temple has almost entirely disappeared. What remains is a basalt pavement, 4,600 years old, one of the earliest uses of this stone in monumental architecture. The basalt came from a quarry at Widan el-Faras, and its transport, noted in the last chapter, was a considerable effort. The scale of the black pavement, still visible on the east side of the pyramid, gives a sense of how much expense was lavished on the temple.

The boat pits

Around the Great Pyramid, five rock-cut pits were found. Three were along the east side and two along the south. Some were originally covered with limestone slabs. Those on the south side were opened in 1954 and 1987, as described in Chapter 5, and they contained the disassembled boats of cedar wood. The pits on the east side were empty.

The purpose of the boats is debated. A traditional view is that these were "solar boats," intended for the king's voyage with the sun god through the heavens, an idea that fits the Pyramid Texts, which describe the dead king sailing with Ra. Another is that they were boats actually used in the funeral, carrying the body of the king from the capital to the valley temple, and then buried afterward, as the king's ships for eternity. Evidence from the first ship supports this second idea in part: examination showed signs of wear and traces of water, suggesting that it had been used, and the ropes found with it were worn. It may have been both. A boat used in the funeral procession was then buried as a vehicle for the afterlife. Boat pits have also been found at the pyramids of Khafre, Djedefre, and others.

The cult pyramid and queens' pyramids

Beside each Giza pyramid stands a small subsidiary pyramid. Egyptologists call it a **satellite pyramid** or **cult pyramid**. Khufu's stands at the southeastern corner. It has a short passage and a small chamber, but no burial. Its purpose is still debated. Theories include a tomb for the king's *ka*, a symbolic southern tomb that balanced the main northern tomb, or a place for ritual, such as for the reburial of the king's ritual regalia.

The pyramids of the queens, three in Khufu's case, stand in a row to the east. They housed the burial of royal women. It is thought that the three belong to important wives or mothers of Khufu, but we do not know who they were. In one of them, G1-a, excavators found a shaft containing the empty sarcophagus and funerary equipment of Queen Hetepheres I, which we met in Chapter 5, though it was in a shaft some distance away. Another small tomb is identified as the tomb of Meresankh III, a granddaughter of Khufu, whose tomb, decorated with colored reliefs and a row of statues of women, was found nearby in the 1920s.

Who served the pyramid?

A pyramid complex needed staff. Every day, offerings had to be prepared and presented, the buildings cleaned, supplies delivered, and records kept. The staff were organized into a **cult**, with priests and attendants, and supported by endowments.

Priests held titles such as *hem-netjer*, "servant of the god," and *wab*, "pure one." Most were part-time, working in rotating

groups, and most also held jobs outside the temple. The Abusir papyri give detailed rosters of duty, and show that priests were organized into five groups, each serving one month in turn. A priest might work one month in five, and spend the rest of the year in other occupations, such as farming, trade, or administration. They were paid in the goods that passed through the temple: bread, beer, meat, and linen.

The system was an elegant solution. Once the king died, the offerings placed before his statues were not wasted. After the gods and the dead king had received their share, in a ritual sense, the food was passed on to the priests and officials. This is called the **reversion of offerings**. It meant that the temple served as a distribution center, and the staff could live from the produce of the king's estates.

Pyramid towns

Around many pyramids grew small communities to support the cult. The best known examples come from different periods.

- **Sneferu's pyramid town at Dahshur.** A royal decree of King Pepi I, found at Dahshur and dated about 300 years after Sneferu's death, exempts the personnel of Sneferu's pyramid town from taxes and corvée labor, so that they could continue serving the cult. It is a remarkable document, which shows that the king's cult was still active 300 years after his death.

- **Khentkaus's town at Giza.** Beside the tomb of the queen Khentkaus I, excavators found a small town of mud-brick houses for her priests, laid out around the monument.
- **Kahun**, near the Middle Kingdom pyramid at Lahun, with its grid of streets, houses, and papyrus archive, which we saw in Chapter 8.
- **The settlement at Menkaure's valley temple.** After the king's death, his valley temple was rebuilt in mud brick, and a town of priests and officials grew up in and around it. Excavation revealed rooms and houses, with remains of seals and sealings, a testimony to continued activity for many generations.

These towns show that a pyramid complex was not a dead monument but an institution that lasted. In some cases, they lasted for centuries, and a king who had died hundreds of years before was still being served. The state had committed itself, in a legal sense, to supporting the royal cult, and it recorded the obligation in endowments of land and tax exemptions.

The cemetery of the court

The pyramid complex was also the center of a larger necropolis. Around the Great Pyramid, the tombs of princes, officials, and priests were laid out in orderly rows, called the **Eastern Field**, **Western Field**, and **Southern Field**. The mastabas were close to the king, in the hope of sharing in his afterlife. Many of the tombs have decorated chapels that tell us about the titles and duties of the people buried there: overseers of the king's works, scribes, priests, physicians, and craftsmen.

The idea of burying one's most important officials near the king was part of the way that the royal court operated. In life, they served him, and in death, they would continue to serve. At Giza, the plan of the cemetery reflects the hierarchy of the living court. The highest-ranking people were closest to the pyramid.

KEY POINT

The pyramid complex was a place of continuing ritual. The king was not simply buried and forgotten. Offerings were made every day, by priests who lived nearby, for as long as the cult lasted. In some cases, the cult of a king survived for more than a thousand years.

What the complex tells us about purpose

Taken together, the elements of the complex give strong evidence for the purpose of the pyramid. A monument that was only a landmark or an observatory would not need a harbor, a causeway, a temple with a false door, an altar, storerooms, and priests. A building that was a power plant or a machine would not need statues of the king, offering lists, and a cult that fed his spirit. The complex only makes sense as a system for the burial and eternal maintenance of a dead king.

The absence of a body in the Great Pyramid does not alter that conclusion. The Egyptians built the system as a whole, and the Great Pyramid, with its harbor, temples, boat pits, queens' pyramids, and cemetery, was built as part of it.

KEY TAKEAWAYS

A pyramid was one part of a complex that included a valley temple, causeway, mortuary temple, cult pyramid, and often boat pits. It was staffed by part-time priests and supported by endowments and pyramid towns. Royal decrees show that cults continued for centuries. The complex is strong evidence that the pyramid was a tomb, and that the king's spirit was supported by daily offerings.

CHAPTER 17

Burial and the Afterlife: Mummies, Spells, and the King's Journey

What the Egyptians did with a dead king, what they believed would happen next, and why no one has found Khufu.

If the pyramid was a machine for eternity, what did it do to the person inside? To answer that, we need to look at the Egyptians' own descriptions of the process. They are among the oldest texts about the afterlife in the world, written on the walls of the pyramids themselves. We can combine them with evidence from tombs, mummies, and later texts to reconstruct, with some caution, what happened to a king at his death.

From death to burial: the funeral

When a king died, his body was taken to the embalmers. The Egyptian word for the process, which we call **mummification**, has a rich vocabulary, and the practice changed over time. By the New Kingdom, the full process took seventy days. In the Old Kingdom, the practice was less developed, but already included removing the internal organs, drying the body with **natron**, a natural salt from desert lakes, and wrapping it in linen.

The evidence for early mummification is slim. The earliest known sign that organs were removed from a royal body comes

from the tomb of Queen Hetepheres I, Khufu's mother, where a chest held her internal organs, preserved in a solution of natron. This dates to about 2600 BCE. The chest is the oldest known evidence of evisceration for a royal burial. For the first pyramid-age kings, most of what we know about mummification comes from the later periods, which are better preserved.

The body of the king was then taken in a procession, probably by boat, from the royal residence to the valley temple at Giza. There it was purified, and the final rituals were performed in a special tent. A procession went up the causeway to the pyramid, with priests, mourners, and officials, carrying the body, burial goods, and statues. In the burial chamber, the body was placed in the stone sarcophagus, and the lid, if there was one, was closed.

One of the key rituals was the **Opening of the Mouth**, a ceremony performed on the mummy and on the statues of the king by priests using special tools, including an adze and a ritual implement shaped like a snake's tail. It was meant to restore the senses and abilities of the dead king, allowing him to breathe, eat, speak, and see in the afterlife. We know it from later periods in detail, and we know that its origins go back to the Old Kingdom. After the ritual, the burial chamber was sealed, the passages were closed by heavy stone plugs, and the funeral party left.

What went into the tomb

The grave goods of an Old Kingdom king are poorly known because every one of the royal burial chambers was robbed. But the tomb of Hetepheres gives a sample. When the American archaeologist George Reisner found it in 1925, in a deep shaft near the Great Pyramid, it contained gilded furniture: a bed with a headrest, a carrying chair, a canopy frame covered in gold leaf, boxes, and bracelets of silver inlaid with turquoise, carnelian, and lapis lazuli. The furniture had collapsed because the wooden parts had rotted, but the gold sheathing and inlays were intact. It took Reisner's team years to reconstruct the pieces. The set shows the quality of what a queen, and probably a king, took into the tomb.

A king would have been buried with similar objects on a larger scale: furniture, jewelry, statues, weapons, tools, ritual objects, and vast quantities of food and drink. Stone vessels, in the tens of thousands, filled the galleries of the Step Pyramid. The idea was to provide the dead king with everything he needed in the afterlife, and to provide him with the symbols of his status. In some cases, objects were also placed in boat pits, as we have seen.

Another feature was the **canopic equipment**: containers for the internal organs. The embalmers removed the stomach, liver, lungs, and intestines and stored them in four jars or compartments, each protected by a guardian deity. Hetepheres's chest has four compartments. In later periods, the jars had lids in the shape of the heads of the four sons of Horus.

The Pyramid Texts

The most important evidence for royal afterlife beliefs comes from the **Pyramid Texts**, mentioned in Chapter 7. They were carved on the walls of the burial chambers and corridors of the pyramids from Unas (about 2350 BCE) onward. The first complete set, in the pyramid of Unas, has about 283 spells. When all of the pyramids are counted, there are around 800 different spells, called **utterances**, though no single pyramid has them all.

The spells are not a continuous story. They are a collection of ritual speeches, recited by priests at the burial and presumably during the continuing cult. They have several kinds of content.

- **Offering ritual.** Spells that accompany the presentation of food and drink, with lists of items: bread, beer, meat, fowl, fruit, incense, linen.
- **Protection.** Charms against snakes, scorpions, and hostile spirits that might threaten the king.
- **Resurrection.** Spells that reassemble the king's body, identify each limb with a god, and call on him to rise.
- **Ascent.** Spells that describe the king's journey to the sky. He climbs a ladder or stairway, crosses the heavens by boat, or flies as a bird, a falcon, or a beetle. He is welcomed by the gods and takes his place among the stars.
- **Identification.** Spells that identify the king with a god: with Osiris, with Horus, or with Ra.

The language is archaic, compressed, and hard to translate. It is sometimes lyrical and sometimes violent. One of the most famous sections, called the **Cannibal Hymn**, found in the pyramid of Unas, describes the king as a hunter who captures and eats the gods, absorbing their powers into his own body. It is an unusual passage and has been interpreted as a metaphor for taking on divine energy. This and other spells show an older, more archaic layer of religious thought than the later, more familiar ideas of the afterlife.

> **KEY POINT**
> The Pyramid Texts do not describe the pyramid as a power plant, a machine, or an astronomical instrument. They describe it as the place where the king is resurrected and from which he ascends to join the gods. This is the Egyptians' own explanation, written in stone, two and a half thousand years before the Greeks first described the pyramids.

The journey

The picture that emerges is of a journey in several stages. First, the king's body is purified, reassembled, and animated. His *ka*, the life-force, is nourished by offerings, and his *ba*, the personality, is freed to move. Then the king rises. He may do this by ascending the sun's rays, by flying as a bird, or by climbing a stairway built for him. In the sky, he meets Ra and joins the sun god's boat, traveling across the sky by day and through the underworld by night. He also joins the **imperishable stars**, the circumpolar stars that never set, taking his place among the eternal.

At the same time, he is identified with Osiris, the dead god, and he rules the land of the dead. These overlapping paths were not seen as contradictory. For the Egyptians, the afterlife was multi-layered, and the king might exist in several forms.

In the New Kingdom, these ideas expanded to include the **Book of the Dead**, with its famous scene of the weighing of the heart against the feather of Maat. That judgment scene does not appear in the Pyramid Texts. In the Old Kingdom, the afterlife was primarily a royal matter. Over the course of the First Intermediate Period, the ideas spread to nobles and ordinary people, and a set of spells called the **Coffin Texts** was written on coffins. This process, sometimes called the democratization of the afterlife, made the promise of eternal life available to everyone who could afford a coffin. Later, it could be purchased on a papyrus roll.

Where is Khufu?

All of this brings us to the problem mentioned in the preface. No body has been found in any Fourth Dynasty pyramid. The sarcophagus in the Great Pyramid is empty, with no lid. The Red Pyramid had fragments of a body, and Menkaure's pyramid held bones of a later date. For Khufu, we have nothing. Has the king vanished?

There are several possibilities, and the most likely is the simplest: the burial was robbed. The Egyptians' own records show that tomb robbery was a problem from the Old Kingdom. In the chaos of the First Intermediate Period, around 2181 BCE, many royal tombs may have been plundered. A thief who found a

mummy of a king wrapped in linen and gold amulets would strip the body, destroy it, or discard it.

A second possibility is that the priests moved the body. We know the Egyptians did this in later periods. In the New Kingdom, when the tombs of the Valley of the Kings were robbed, priests collected the mummies of the kings and hid them in a few safe places. In 1881, a cache of more than 40 royal mummies, including Ramesses II, Seti I, and Thutmose III, was found in a hidden tomb at Deir el-Bahari. Perhaps something similar happened to Khufu, and his mummy was reburied elsewhere, and perhaps lost.

A third possibility is that the Great Pyramid was not the actual burial place of the king, and that his body was buried elsewhere, with the pyramid as a symbolic monument. This idea has long been popular, but there is no evidence for it. The sarcophagus, the sealed passages, the granite plugs, the burial-sized chamber, the funerary complex, and the fact that every other pyramid-age king was buried in his pyramid all point in the opposite direction.

KEY POINT

The absence of Khufu's body is an absence of evidence, not evidence of absence. Thieves, priests, and time all work against the survival of a 4,500-year-old burial. Most Egyptologists consider the Great Pyramid to have been Khufu's tomb, plundered long ago.

The afterlife and the pyramid's design

The ideas in the Pyramid Texts help to explain some of the design features of the Great Pyramid. The use of granite, a hard, dark stone, for the burial chamber, may have been linked with the king's eternal body and the imperishable stars. The orientation of the pyramid to the cardinal directions reflected the cosmos. The corbelled Grand Gallery may have represented the primeval hill or the heavenly path. The narrow shafts may have been paths for the king's *ba*, or soul, to leave the pyramid. And the sloping sides, as the Texts say, were a stairway to heaven. All the features are open to debate, but they are all consistent with the idea of a resurrection machine, not a machine in the physical sense but a symbolic and ritual one.

KEY TAKEAWAYS

Royal bodies were mummified, placed in stone sarcophagi, and buried with goods, though no Fourth Dynasty royal burial has survived intact. The Pyramid Texts (from Unas, c. 2350 BCE) contain about 800 spells for offerings, protection, resurrection, and ascent. They describe the pyramid as the king's route to the sky and the underworld. Khufu's body was likely lost through robbery or removal, as with other royal mummies.

CHAPTER 18

Robbers, Restorers, and Rediscovery

Four and a half thousand years of visitors: thieves, tourists, scholars, and the people who finally learned to read the stones.

A pyramid built to last forever has, in practice, spent most of its life being visited. Within a few centuries of its completion, thieves found the way in. Within a thousand years, Egyptians themselves were repairing and studying the monuments. Greeks and Romans came as tourists, medieval builders stripped them for stone, and European explorers measured them, blasted into them, and carried pieces home. This chapter follows that long history, because it explains both what has been lost and how we know what we do.

Why the pyramids were robbed

Tomb robbery was not a modern invention. As soon as a king was buried with gold, jewelry, and fine furniture, he became a target. For the Egyptians, such crimes were not only theft but sacrilege, an attack on the king's afterlife, and the penalties were severe. But the temptation was great, and the knowledge needed to commit the crime was widely available. Workers, priests, and officials all knew where the treasure was and how to get in.

Pyramids were designed with this in mind. The sealed entrances, granite plugs, false passages, and heavy sarcophagus

lids were all intended to stop thieves. But even the best security failed when people had time. Most of the pyramids of the Old Kingdom appear to have been entered and plundered by the end of the Old Kingdom or in the First Intermediate Period. The Egyptians themselves seem to have known it. Later texts speak of tombs being laid bare and the dead being cast out, though these passages are literary and hard to use as evidence.

The robbery trials

Our best evidence for how tomb robbery worked comes from a later period. In about 1110 BCE, during the reign of Ramesses IX, Thebes was shaken by accusations that royal tombs on the west bank had been plundered. A commission of officials was sent to inspect the tombs, and the findings were recorded on papyri, such as the **Abbott Papyrus** and the **Amherst Papyrus**, now in the British Museum.

The commission inspected the tombs of the kings of the Seventeenth Dynasty, who had been buried in small brick pyramids at Dra Abu el-Naga. Some were found intact. One, the pyramid of King Sobekemsaf, was found to have been broken into. The thieves had confessed, giving a detailed account of how they tunneled into the pyramid, found the burial chamber, took the gold from the mummies, and burned the coffins. The confessions were obtained under torture, and some of the accused were later executed.

The papyri show that robbery was often an inside job, involving people who worked in the necropolis and could easily gain access. They also show corruption among officials. They

are valuable evidence that the problem of security was permanent, and that no design could solve it. The solution adopted by the priests of the Twenty-First Dynasty was to take the royal mummies out of their tombs and re-hide them in secret caches.

The prince who was the first Egyptologist

The Egyptians were conscious of their own past. About 1,000 years after the Great Pyramid was built, a New Kingdom prince named **Khaemwaset**, a son of Ramesses II, took the title of High Priest of Ptah at Memphis. He took an interest in the old monuments of the Memphite necropolis. He inspected the pyramids of Giza, Abusir, and Saqqara, found them in ruins, and ordered repairs. He had inscriptions carved on the pyramids of Unas, Userkaf, and others, stating that he had restored them and recording the name of the king to whom each belonged.

Khaemwaset is sometimes called the first Egyptologist, and the title seems fair. He was motivated by piety and by dynastic pride, wanting to show his father's devotion to the ancestors. He did not know as much about the pyramids as we do, but he understood that they belonged to ancient kings, and he took pains to record their names. His inscription on the pyramid of Unas is one of the oldest known acts of monument preservation.

Later in Egyptian history, the Twenty-Sixth Dynasty (c. 664–525 BCE), known as the Saite Period, consciously revived Old Kingdom art and religion, restoring cults and monuments. At Giza, a chapel was built near the Sphinx, and the cult of Khufu and Khafre was revived. People of the Late Period saw

the Great Pyramid as the work of a great ancestor, and honored him.

The Greeks and Romans

By the time the Greeks arrived, the pyramids were ancient. **Herodotus** visited around 450 BCE. His account, written in Book II of his *Histories*, is the first detailed description by an outsider. He described the Great Pyramid, giving its dimensions, and said that it was built by "Cheops" using forced labor. He also gave details that cannot be verified, such as an inscription on the pyramid that recorded the sum paid for radishes, onions, and garlic for the workers. He was told things by local guides, some of which were garbled, and his account should be read as a record of what people said in his time.

The pyramids became part of the Greek list of the **Seven Wonders of the World**. The list appeared in the second century BCE in a poem by Antipater of Sidon, and it included the Great Pyramid of Giza first. Of those wonders, it is the only one that has survived.

Later Greek and Roman writers visited and wrote about them. **Diodorus Siculus**, who visited around 60 BCE, gave the names of the builders and the story that they were not buried in the pyramids. **Strabo**, around 25 BCE, wrote that the Great Pyramid had a stone that could be moved, a hinged door that opened to a passage inside the pyramid. His description fits the original north entrance. **Pliny the Elder** wrote in the first century CE that the pyramids were useless displays of royal wealth. The Roman emperor Hadrian visited in 130 CE. Visitors

scratched their names on the stones, and some of these graffiti survive, in Greek and Latin. The Great Pyramid has been a tourist attraction for more than two thousand years.

The caliph's tunnel

By the Middle Ages, the entrance to the Great Pyramid had been lost. Its location was hidden by debris and by the pyramid's own casing. According to a tradition recorded by later Arab historians, in about 820 CE, the caliph **Abdullah al-Ma'mun** decided to enter the pyramid in search of treasure and knowledge. He had his workmen tunnel into the north face by heating the stone with fire, pouring on vinegar to crack it, and hammering with battering rams. After days of work, they heard something fall inside. It was a stone that had been hidden in the wall of a passage, and it revealed the Ascending Passage. The workmen, blocked by the granite plugs, tunneled around them and reached the upper chambers.

Inside, according to the story, they found an empty sarcophagus. A famous legend claims that the caliph's men salted the pyramid with just enough gold to cover the cost of the excavation, so that the caliph would not be embarrassed. This is a later embellishment, and we cannot say how much of the story is true. What is certain is that someone cut the tunnel, called the **Robbers' Tunnel**, through the masonry at about this time. It is still the way visitors enter the pyramid today.

Stripped for stone

During the Middle Ages, Cairo grew into a major city, and it needed stone. The pyramids' fine Tura limestone casings were a convenient source. Beginning in the tenth and eleventh centuries, and especially after an earthquake in 1303 that loosened the stones, builders dismantled the casing and reused it in mosques, palaces, and city walls. This is why the pyramids are now stepped, with the rough core exposed, and why some of the stones in the old city of Cairo bear traces of Fourth Dynasty cutting.

The Giza complex also lost its smaller monuments. The queens' pyramids, mortuary temples, and causeways were quarried heavily. In 1196, as we saw, the son of Saladin tried to demolish Menkaure's pyramid and gave up. In the nineteenth century, the Egyptian ruler **Muhammad Ali** considered dismantling the pyramids for the stone to build the barrages on the Nile, but his engineers advised that it would cost more than quarrying new stone. The pyramids survived, partly because they were too big to destroy.

European travelers and early science

After the Renaissance, European scholars rediscovered the pyramids. In 1638–39, the English astronomer **John Greaves** traveled to Egypt, measured the Great Pyramid, and published *Pyramidographia* in 1646. He was one of the first to use scientific methods. A century later, in 1737, the Danish captain **Frederic Norden** made detailed drawings, including of the Sphinx without a nose, and in the 1760s, **Nathaniel Davison** found the chamber above the King's Chamber that bears his name.

In 1798, **Napoleon Bonaparte** invaded Egypt with an army and with a team of 167 scholars and artists, known as the *savants*. They surveyed the Giza plateau and other sites and produced the monumental *Description de l'Égypte*, published from 1809 to 1829. According to a report, before the Battle of the Pyramids, Napoleon told his soldiers that forty centuries looked down on them from the top of the pyramids. The line is probably an embellishment, but it captures the awe of the age. The *savants* also found the Rosetta Stone in 1799, which would allow **Jean-François Champollion** to decipher hieroglyphs in 1822, a key to everything that followed.

The age of explorers

In the early nineteenth century, a series of colorful figures explored the pyramids. The Genoese captain **Giovanni Battista Caviglia** cleared the Sphinx of sand in 1817 and explored the Great Pyramid's passages and the Well Shaft. The Italian **Giovanni Belzoni**, a strongman at a circus before he became an explorer, found the entrance to the pyramid of Khafre in 1818. The British officer **Howard Vyse** and engineer **John Perring** spent 1836–37 at Giza, using gunpowder and drills to find hidden chambers, as we have seen. They made a great deal of useful measurement but did much damage.

In the 1840s, the German scholar **Karl Richard Lepsius** led a Prussian expedition that mapped 67 pyramids along the Nile, creating a numbered list that scholars still use. The French archaeologist **Auguste Mariette**, who founded the Egyptian Antiquities Service in 1858, found Khafre's statues in 1860.

The age of scientific excavation

Modern archaeology of the pyramids begins with the British archaeologist **Sir William Matthew Flinders Petrie**, who surveyed Giza in 1880–82 and published *The Pyramids and Temples of Gizeh* in 1883. His measurements, made with great care, became the foundation for all later work. He also brought a principle that changed archaeology: pay attention to everything, including potsherds and everyday objects, not just treasures. He later excavated the pyramids of Meidum, Lahun, and Hawara.

In the early twentieth century, the American Egyptologist **George Reisner** led the Harvard–Boston expedition to Giza, from 1902 to 1942. He and his teams recorded and excavated the cemeteries, the queens' pyramids, and Menkaure's valley temple, and discovered the tomb of Hetepheres in 1925. The expedition's records, photographs, and field notes are still the basis of Giza research. The German scholar **Ludwig Borchardt** excavated the Fifth Dynasty pyramids at Abusir from 1902 to 1908, and the Egyptian archaeologists **Selim Hassan** and **Ahmed Fakhry** worked at Giza and Dahshur in the 1930s and 1950s. In 1954, **Kamal el-Mallakh** found Khufu's boat. The French architect **Jean-Philippe Lauer** worked for seven decades at Saqqara.

The modern era

Since the 1980s, research has changed in character. **Mark Lehner**, **Zahi Hawass**, and their teams have uncovered the workers' town and cemeteries, giving an unprecedented view of the people who built the pyramids. **Pierre Tallet**'s discovery of the Wadi al-Jarf papyri in 2013 gave us Merer's diary. **Yannis Gourdon** and **Roland Enmarch** found the ramp system at Hatnub. And since 2015, the ScanPyramids team has used cosmic rays, thermal imaging, and endoscopes to look inside the great monuments without harming them. The next chapters describe other sides of the modern study of the pyramids, including the myths that surround them.

KEY POINT

The history of pyramid research shows a steady shift from treasure-hunting to science, and from a few adventurers to teams of specialists. Each step, from the Rosetta Stone, to Petrie's careful measuring, to the muon detectors of today, has added evidence and eroded speculation.

KEY TAKEAWAYS

Pyramids were plundered from antiquity; Egyptian records of robbery trials survive. Prince Khaemwaset restored pyramids about 1250 BCE. Greek and Roman tourists wrote about them. Medieval Cairo stripped the casing. Explorers from Greaves to Vyse measured and sometimes damaged them. Petrie, Reisner, Lehner, Hawass, Tallet, and ScanPyramids transformed the study into a science.

PART 5

Myths, Mysteries, and Modern Science

Fair hearings for popular claims, and a look at the real frontier

CHAPTER 19

Machines, Chemicals, and the "Power Plant" Theory

Was the Great Pyramid a lost machine? What the claim says, where it came from, and how it holds up against the evidence.

Almost everyone who has looked at the Great Pyramid has felt the same pull. The thing is so large, so regular, and so precisely made that it does not feel like a tomb. It feels like equipment. A tomb is something you hide a body in. This looks like something that was supposed to *do* something.

That feeling is the seed of a family of ideas that treat the pyramids as machines: power plants, pumps, resonators, water-lifting devices, or generators of some unnamed force. This chapter takes those ideas seriously enough to describe them accurately, trace where they came from, and test them. Some are plain wrong. One, which looks like a machine theory at first glance, is an interesting engineering proposal about construction. And one piece of modern physics, which is often cited in support of the machine theories, turns out to say something much more modest than its popularizers claim.

Why machine theories feel natural

It helps to understand why these ideas are so persuasive before we try to answer them. Three things push people toward them.

The first is precision. As we saw in Chapter 9, the Great Pyramid's base is level to within a couple of centimeters over an area of more than five hectares, and its sides are aligned to the cardinal directions to within a small fraction of a degree. Precision on this scale is something modern readers associate with machine tools and lasers. It is natural to ask whether people with copper chisels could have achieved it, and if they could not, what else could have.

The second is the modern habit of reading old objects as technology. We live in a world where nearly every large structure has a function beyond itself: a dam holds back water, a tower broadcasts, a plant makes electricity. A giant structure that seems to do nothing invites a modern viewer to look for the hidden function.

The third is a quiet one. The people who built the pyramids left no engineering manuals. They left tomb inscriptions, offering lists, work-gang graffiti, and the logbook of a boat foreman, but no document saying "here is how we made the base level." The gap invites speculation. Silence in the record becomes a blank screen onto which each age projects its own technology.

None of this makes the machine theories correct. But it explains why they keep returning, and why a reader who has felt drawn to them is in good company.

The Giza Power Plant

The best-known machine theory was set out by the American engineer and machinist **Christopher Dunn**, who published a book in 1998 titled *The Giza Power Plant: Technologies of Ancient Egypt*. Dunn had spent his career working in precision manufacturing. He had been impressed by the accuracy of Egyptian stone-work, and he had read the measurements of the nineteenth-century archaeologist Flinders Petrie, who surveyed Giza in the early 1880s and reported extremely fine work on some granite objects. Dunn argued that such accuracy implied machine tools, and he went on to propose that the Great Pyramid as a whole was a machine.

Stated briefly, and as fairly as possible, the proposal goes like this.

1. The Great Pyramid was built to be a *power plant*, not a tomb. Its purpose was to generate and transmit energy of some kind.
2. Water flowed into the pyramid from underground channels. In the Queen's Chamber, Dunn proposed, two chemicals were combined: zinc, and hydrochloric acid (which he suggested might be made on site from other materials). The reaction between them produces hydrogen gas.
3. The hydrogen rose through the narrow shafts that run from the Queen's Chamber. The shafts were not for stars or souls; they were pipes.
4. In the King's Chamber, which is lined with granite, the hydrogen was ignited or otherwise interacted with a vibrating

system. Sound waves, produced in part by the flow of gas and in part by the natural vibrations of the Earth, made the granite resonate. Granite contains quartz, and quartz is *piezoelectric*, meaning it generates electricity when squeezed or vibrated.

5. The result, Dunn said, was a resonant system that converted the Earth's vibrations into microwave energy, which was then beamed out through the shafts or conducted away.

There are elements of real science in this chain. Hydrogen *can* be made from zinc and hydrochloric acid, which is a standard classroom demonstration. Quartz *is* piezoelectric. Rooms *do* resonate. A reader with a technical background can follow each link and say, yes, that is something that exists.

The trouble is that a chain is only as strong as its weakest link, and nearly every link in this one breaks when tested against the record.

Testing the claims, one by one

Let us go through the main points in order.

There is no physical trace of the apparatus. If a chemical plant had operated in the Great Pyramid, there would be pipes, tanks, vessels, pumps, valves, or at least the places where they were fixed. There would be corrosion, residue, scorched surfaces, and a great deal of chemical contamination. Hydrochloric acid is highly corrosive. Hydrogen is explosive. Over a long operating life, the chambers would show it. Instead, the interior shows plain, dry limestone and granite, carved graffiti from the

work gangs, and the scars of nineteenth-century explorers' crowbars and gunpowder. No archaeologist, chemist, or geologist who has studied the interior has reported any trace of reaction products, metallic residues, or the mounting points of any equipment.

The salt deposits do not point to a reaction. Dunn and others have cited the crystalline salt crust on the walls of the Queen's Chamber as evidence that something chemical went on there. Salt crusts are real, and visitors can see them. But they are common in limestone chambers in the Egyptian desert, and the explanation is ordinary. Rock in contact with moisture slowly dissolves trace minerals, and when the water evaporates at the surface the salts are left behind. This is the same process that attacks the stones of many tombs and temples in Egypt. Salt deposits of this kind are found in rooms of other pyramids, and in tombs that were never anything but tombs.

The shafts do not behave like pipes. Chapter 15 described the narrow shafts that rise from the King's Chamber and the Queen's Chamber. The King's Chamber shafts reach the outer faces of the pyramid, though the openings are not where one might expect for ventilation. The Queen's Chamber shafts, by contrast, do not begin at the chamber floor; they start some distance above it inside the masonry, and their inner ends were never opened until modern times. Robots sent up the southern one in 1993 and 2002, and again in 2011, found small limestone slabs, with copper fittings, blocking the way. A pipe that is sealed with a stone slab at one end and closed off in the masonry

at the other is not carrying gas. Whatever the purpose of these shafts, it was ritual or symbolic, not plumbing.

No source of zinc or hydrochloric acid has been found in Old Kingdom Egypt. The metal that Dunn's reaction required, zinc in a pure form, does not seem to have been produced anywhere in the ancient world until a very late period. Brass, an alloy of copper and zinc, is known in later antiquity, but pure metallic zinc is difficult to produce because it boils at a temperature close to the one needed to smelt it, and it was a centuries-later achievement. Hydrochloric acid, for its part, is generally credited to alchemists of the early medieval Islamic world. There is no evidence for either substance in the Egyptian Old Kingdom, not as finished products, not as raw materials, not in residues in any tool or vessel.

The energy does not add up. This is the deepest problem, and it does not even require any archaeology. A chemical reaction can only release the energy that was stored in its ingredients. When zinc dissolves in acid and makes hydrogen, the energy that eventually comes out as heat, or as a flame, came from the zinc and the acid. But zinc, in its metallic form, was itself produced by smelting ore, which takes at least as much energy as you can get back by reacting it. In other words, a plant that runs on zinc and acid is a way of storing and releasing energy that had to be put in first. It is a battery, not a generator. It cannot be a net source of power.

WORKED EXAMPLE

Suppose you wanted to get one megajoule of heat from burning hydrogen. Burning one mole of hydrogen (two grams) releases about 286 kilojoules, so you would need about 3.5 moles, or about 7 grams of hydrogen. Making that hydrogen from zinc would consume about 3.5 moles of zinc, or roughly 230 grams. To refine that much zinc from ore takes more energy than you get back from the hydrogen, and the acid has to be made as well. So you would be spending far more energy to produce the zinc than you would ever recover. This is why nobody today makes power by dissolving zinc in acid.

Resonance is not generation. Dunn's central idea is that vibrations in the granite generate electricity. But resonance, the tendency of a structure to vibrate strongly at certain frequencies, does not create energy; it can only amplify energy that is already being supplied. A bell rings loudly if you strike it, but the ring is made from the energy of the blow. For the King's Chamber to act as a generator, there would have to be a strong, constant source of vibration feeding it. The Earth's ambient seismic noise is tiny. The proposed gas flow would have to supply the rest, and we have just seen that there is no gas.

The piezoelectric argument fails for a specific reason. Quartz is indeed piezoelectric, and granite contains quartz. But the grains of quartz in granite are oriented randomly. When the rock is squeezed, each grain generates a small voltage, but the voltages point in all directions and largely cancel each other out. This is the reason why a block of granite is not a battery, and

why industrial piezoelectric devices are made from single crystals or carefully aligned ceramics, not from rocks.

There is no consumer. Finally, a power plant exists to supply something. The Great Pyramid is not connected to anything. There are no cables, no transmission towers, no remains of any apparatus anywhere on the Giza plateau that could have used the power. There is no wire, no lamp, no motor, no sign of any electrical or mechanical load. All the evidence points the other way: around the pyramid there are temples, boat pits, cemeteries, and a workers' town.

What the Egyptians themselves said and did

Set against all this is a body of evidence that points, consistently, in a different direction. The pyramid stands at the head of a complex whose parts make sense as a royal tomb and a cult center. The causeway leads down to a valley temple on the edge of the old harbor. The mortuary temple at the pyramid's foot held statues of the king and a place for offerings. Boat pits flanking the pyramid held full-size cedar boats, one of which, the **Khufu ship**, has been reassembled and conserved. Around the pyramid are the cemeteries of Khufu's relatives and officials, with their tomb chapels carrying inscriptions that name him. A priesthood served his cult for centuries. Chapter 16 described all of this.

Moreover, the *builders themselves* wrote their messages in the stone. In the sealed relieving chambers above the King's Chamber, work gangs painted names such as "Friends of Khufu" and the king's own name in red ochre on the blocks, with quarry marks. Nobody going into a machine room scribbles

gang names on the walls before it is sealed for good. But a crew finishing a building, who want their work to be credited, does exactly that.

MYTH AND FACT

Myth: The Great Pyramid has no connection to burial because no mummy was found. **Fact:** The absence of a body in the chamber is real, but it is not unusual. Almost every large pyramid was robbed in antiquity, and the Great Pyramid was found open and empty when it was entered in the ninth century CE. Absence of the king does not make a power plant. What the Great Pyramid lacks is a body, not a funerary purpose; what it does *not* lack is everything else, including a sarcophagus, a complex of temples and boats, and a surrounding cemetery.

The physics paper everyone cites

Supporters of machine theories sometimes cite a scientific study published in 2018 in the *Journal of Applied Physics*. A team of physicists from Russia and Germany, with ties to ITMO University in Saint Petersburg and the Laser Zentrum in Hannover, modeled how the Great Pyramid, treated as a block of limestone of that shape, would respond to radio waves. Their calculations suggested that at certain wavelengths, between about 200 and 600 meters, the pyramid would resonate, and that under those conditions energy would concentrate in the internal chambers and beneath the base.

Headlines at the time said that the Great Pyramid "can focus electromagnetic energy." That is accurate as far as it goes, but it

needs context. The study was a computer model of a *shape*. The authors used it as a way to design tiny particles that could produce similar effects at optical wavelengths, with the aim of building better nanosensors and solar cells. They did not claim that the Egyptians knew about resonance, that the pyramid was designed for it, or that any of it had anything to do with ancient purpose. Their model also rested on assumptions, for example that the pyramid is a uniform block of limestone with no unknown cavities. Any large object of a given size and shape has resonances. A football stadium does. A church does. A mountain does. Finding that a pyramid-shaped body interacts with long radio waves says nothing about why anyone built it.

> **KEY POINT**
> A resonance is a property of a shape. It tells you what a structure *would* do if you drove it with the right wave. It does not tell you that anyone ever did, or that anyone built the structure in order to do it.

A different kind of machine: the hydraulic lift

Not every machine idea in the pyramid literature is mystical. In August 2024, a research team led by the French engineer **Xavier Landreau**, of the private research institute Paleotechnic in Paris, published a paper in the open-access journal *PLOS ONE* proposing that the builders of Egypt's first pyramid, the Step Pyramid of Djoser at Saqqara, raised stones with a **hydraulic lift**.

The proposal is specific and ingenious. Landreau's team pointed to the large vertical shaft on the north side of the pyramid's complex and to a system of channels and basins, including a huge walled enclosure called the Gisr el-Mudir and a dry moat around the complex. They suggested that seasonal floodwater was collected and fed into the shaft, where it raised a floating platform carrying building stones; when the water was drained, the platform dropped to the bottom, and the cycle repeated. A wooden float and ropes over pulleys balanced the platform. The idea is that water, not muscle, did the heavy lifting.

The proposal drew much attention, and it also drew sharp criticism from specialists. Archaeologist **Oren Siegel** of the University of Toronto argued that the enclosure could not have held enough rainwater to run such a system, and suggested it was an early experiment in stone enclosures. Egyptologist **Kamil Kuraszkiewicz** of the University of Warsaw pointed out that no ancient Egyptian text mentions the lake the model requires, and that Djoser's stones, which average about 300 kilograms, are light enough that human labor would have been simpler than building a hydraulic device. Other researchers said they were unconvinced.

The hydraulic proposal is useful to compare with the power-plant theory because it shows what a *testable* machine theory looks like. It was published in a peer-reviewed journal, it makes specific claims about specific places, and it can be checked against the excavated landscape and against the texts. Its critics were able to answer it with specific objections. Whether it

stands or falls, it sits squarely inside the mainstream argument about how Egyptians built. It also does not make the pyramid a power plant; it treats the pyramid as a monument and the lift, if it existed, as a *construction tool*.

The valid core: the Egyptians were good with water and chemistry

There is a grain of truth in the machine theories that is worth rescuing. The Egyptians were extremely capable practical chemists and hydraulic engineers, but they used these skills for other purposes. They made faience, a glazed ceramic, and Egyptian blue, the world's first synthetic pigment. They refined gold and worked copper. They made plaster and gypsum mortar by burning and grinding gypsum. They used natron, a natural salt mixture from desert lakes, to dry bodies for mummification. They dug canals, built harbors, and managed floods. As we saw in Chapter 11, they probably used water to wet the sand ahead of heavy sledges. They had hydraulic skill. They did not have, and did not need, a chemical power plant.

When we say that the Egyptians built the Great Pyramid with copper tools and rope, we are not saying that they were primitive. We are saying that they were very good at the things they actually did, and that they applied their cleverness to organization, geometry, and material handling, not to the kind of energy technology we associate with the industrial age.

How to judge a machine theory

The power-plant theory gives us a useful checklist that applies to any claim that an ancient monument was a lost machine. You can ask:

1. **What is the machine supposed to do?** A real machine has an input and an output. Where does the energy come in, and where does it go?
2. **Where are the parts?** Machines leave hardware, wear, residues, and repair marks.
3. **Where are the raw materials?** If the theory needs zinc, acid, copper wire, or lenses, where were they found and in what quantities?
4. **Does the energy balance work?** No process creates energy from nothing.
5. **What does the simplest explanation say?** If the building was demonstrably a tomb complex, with a tomb chamber, a sarcophagus, a funerary temple, boat pits, and a cemetery, a machine theory has to explain why all that is there.
6. **What would count against the theory?** A good theory states in advance what evidence would refute it. If no discovery could ever count against a claim, it is not a scientific claim.

The Giza Power Plant fails the first five checks. It also fails the sixth, because each time a missing piece is pointed out, the answer is that the evidence must have been removed. A theory that explains away every gap with "it was taken away" cannot be

tested, and is no longer in competition with an explanation that fits the evidence we do have.

Chapter summary

The Great Pyramid's precision and size make it tempting to see it as a machine. The best-known version of that idea, the Giza Power Plant, rests on a chain of real scientific facts (hydrogen from zinc and acid, piezoelectric quartz, resonant rooms) that do not connect when set against the evidence: no apparatus, no residue, no zinc or hydrochloric acid in the Old Kingdom, shafts that were sealed, no way to get net energy from chemistry, and no consumer for the output. A 2018 physics study of the pyramid's radio resonance is a model of a shape, not a claim about ancient purpose. A 2024 proposal of a hydraulic lift at the Step Pyramid is a serious and testable engineering hypothesis about construction, which many specialists doubt. What the Egyptians did have was extraordinary organizational skill, practical chemistry for pigments, plaster and mummification, and a mastery of water and stone.

The next chapter turns to a different kind of claim: that the pyramids, and the art around them, show evidence of alien technology.

CHAPTER 20

Ancient Astronauts and "Impossible" Imagery

Where the alien-technology idea came from, what its supporters point to, and what Egyptologists say about each piece of evidence.

If you search the internet for the pyramids, a large share of what you find says that they were built or assisted by beings from elsewhere in the universe. The claim is common enough that it deserves a fair, careful chapter. Many readers of this book will have met it first on television, in a documentary, or in a viral video, and will want to know what is behind it.

This chapter does four things. It describes where the "ancient astronaut" idea came from. It lays out the specific pieces of imagery that are usually cited from Egypt. It examines each one. And it considers what the idea says, often without meaning to, about the people who actually built the monuments.

A short history of an idea

The notion that ancient wonders were made by visitors from space is far younger than the wonders themselves. Earlier writers of the nineteenth and early twentieth centuries sometimes speculated about contact with other worlds, but the idea in its modern form took shape in the 1960s. In 1960, two French writers, Louis Pauwels and Jacques Bergier, published *The Morning of the Magicians*, a bestselling mix of history, science, and occult speculation that floated the notion of vanished civilizations and cosmic visitors. Then, in 1968, a Swiss hotel manager named **Erich von Däniken** published *Chariots of the Gods?*, which presented the idea to a mass audience. The book sold in the millions and was followed by many others and by a television franchise.

Von Däniken's argument rested on a pattern that is easy to recognize. He would point to a monument, a carving, or a myth, say that it was too advanced, too strange, or too precise for the people credited with it, and offer extraterrestrial contact as the simplest explanation. The Great Pyramid was an obvious candidate. Over the following decades, other writers expanded the theme, and in the twenty-first century a long-running cable television series brought a version of it to a wide audience.

It is worth noting that critics who checked von Däniken's sources, including a well-known television investigation in the late 1970s, documented numerous errors and misrepresentations in his books. That does not settle the matter by itself, since a claim can be wrong or right regardless of the track record of its

proponent, but it helps explain why professional historians and archaeologists have not treated the ancient-astronaut idea as a serious competitor to ordinary explanation.

The argument in outline

The case for alien involvement in Egypt, as its supporters make it, usually has four parts.

1. **The pyramids are too big and too precise.** The Great Pyramid's size, its alignment, and its smooth finish are said to exceed what ancient people could have done.
2. **Egyptian stonework includes "impossible" objects.** Fine granite vessels, huge stone boxes, and precisely cut blocks are cited as evidence of advanced tools.
3. **Egyptian art shows technology.** A handful of carvings and objects are said to depict aircraft, light bulbs, or other machines.
4. **Egyptian art shows non-human beings.** Elongated heads, animal-headed gods, and star-linked deities are read as depictions of visitors.

The first of these was dealt with in Parts Three and Four of this book. We have the quarries, the tools, the ramps, the workers' town, the bakeries, the cemeteries, and the logbook of Merer. It was not necessary to assume anything but human beings. In what follows we will concentrate on the other three.

Case one: the Abydos "helicopter"

The most famous piece of "evidence" is a set of carved hieroglyphs on a beam in the temple of **Seti I** at Abydos, in southern Egypt. Seti I was a king of the Nineteenth Dynasty, who reigned around 1290 to 1279 BCE, more than twelve hundred years after the Great Pyramid. In photographs, one section of the carving looks strikingly like a helicopter, with what could be a submarine, a jet, and other shapes beside it. Variations of the image have spread across the internet since the late twentieth century, and many people have seen it.

The explanation is mundane, and it is something that anyone who has studied Egyptian inscriptions will recognize. The carving was made twice. A first inscription, giving a set of the king's titles, was cut into the stone. After the death of Seti I, his son **Ramesses II** completed and usurped parts of the temple. The older text was filled in with plaster, and a new inscription with different titles was cut into the plaster. Over the centuries, the plaster eroded and fell away, and the older carving came back to light underneath, overlapping with the newer one. The "helicopter" is what you get when two different sets of hieroglyphs occupy the same space, with some of the signs of one running into the signs of the other. When the two layers are traced separately, each resolves into ordinary hieroglyphic signs, reading as phrases about the king.

This type of overlap is called a **palimpsest**, a word for a surface that has been written on more than once. Egyptians re-used and re-carved inscriptions constantly, because kings liked to

claim the work of their predecessors. Egyptologists see such overlaps all the time. The Abydos example only looks mysterious because it has been photographed at a flattering angle, in good light, and cropped.

> **MYTH AND FACT**
>
> **Myth:** The Abydos carvings show a helicopter, a submarine, and a jet. **Fact:** They are overlapping hieroglyphs from two different inscriptions, one carved over a plaster repair of the other. When the layers are separated, each reads as a normal royal title.

Case two: the Dendera "light bulbs"

In the great temple of **Hathor** at Dendera, about sixty kilometers north of Luxor, are underground chambers called the crypts. On the walls are reliefs that show a large oval object shaped a little like a long bulb, in which a wavy line rises like a filament. A figure with raised arms stands behind it, and a pillar supports it. To some viewers it looks like a light bulb connected to a cable. The comparison has been made for more than a century, and it appears in books and videos that claim Egypt had electric light.

Egyptologists read the same scene differently, and their reading rests on the inscriptions around it. The wavy line is a **snake**, rising from a **lotus flower**. The oval is the lotus, or a container called a *hen*, which may stand for the womb of the sky goddess Nut. The snake is a manifestation of a god, and the whole scene is connected to Egyptian creation myths, in which a god

emerged from the primeval waters on a lotus. The pillar is a *djed*, a symbol of stability that is common in Egyptian art. The captions around the scenes describe the sacred statues of the gods that were kept in the crypts, and the rituals through which they were revived in the light of the sun on the temple roof.

There are two further points. First, the Dendera temple as it stands today is a late building, constructed in the Ptolemaic and early Roman periods, about two thousand years ago. That is roughly twenty-five centuries after the Great Pyramid. If these reliefs were evidence of lost pyramid-age technology, the technology would have to have survived the whole long gap without leaving anything else. Second, there is no hardware. No bulb, no wire, no filament, no battery, no generator, and no lamp socket has ever been found in any Egyptian site. The ancient Egyptians have left us thousands of oil lamps, which are small bowls of oil with a wick, and many tomb scenes showing people using them.

The "no soot" argument

A related claim is that Egyptian tomb ceilings, decorated with fine paintings in deep passages far from daylight, show no soot from torches or lamps. How, the argument goes, did artists work underground without leaving soot unless they had electric light?

The question has a number of plausible answers, and it is worth noting that a claim about what is absent is only as good as the survey behind it. Egyptians used oil lamps, and the lamps could be made to burn cleanly by using good oil and trimmed wicks, and some researchers have proposed that salt was added to reduce smoke. Mirrors of polished metal could bounce sun-

light into passages in relays. And soot marks have in fact been noted in a number of tombs. Researchers have also pointed out that artists often worked in tombs that were open to the outside and decorated the burial chambers last, when the work was small and the lamps could be few. The absence of soot, where it is absent, is far from demonstrating an alternative technology, and the absence of any physical evidence for the technology (no bulbs, no wires) is a much stronger fact.

Case three: the Saqqara bird

In 1898, in the tomb of a man named Pa-di-Imen at Saqqara, excavators found a small wooden figure that is now in the Egyptian Museum in Cairo. It is a bird, carved from sycamore wood, with outstretched wings and a pointed beak. It has a wingspan of about eighteen centimeters and weighs less than forty grams. It dates to around 200 BCE, the Ptolemaic period.

In 1969, an Egyptian physician and researcher named **Khalil Messiha** proposed that the figure is a model of a glider, and that the ancient Egyptians must therefore have understood aerodynamics. The idea was taken up in books and documentaries. It looks, to a viewer, rather like a model airplane: the body is slim, the wings are long, and the shape is simple.

The objections are straightforward. The bird, as it survives, has no vertical tail fin, which a real glider needs for stability. Enthusiasts who build copies add one. A 2023 computational study of its aerodynamics by researchers at the Institute of Aerospace Technology in Bremen found that the object has a poor glide performance, is unstable in pitch, and would roll uncon-

trollably in flight. An aviation historian had earlier called it far too heavy and unstable to fly. The object seems to have been a decorative or ritual piece. Egyptians carved many bird figures, and birds were closely associated with gods, especially the falcon. Other suggestions include a child's toy, a weather vane, or an ornament for the prow of a ceremonial boat. In any case, an object that dates to two thousand years after the pyramids and cannot actually fly tells us nothing about how pyramids were built.

Case four: elongated heads and "strange" beings

Some of the images used in the alien argument are not machines but people. The most common example is the art of the **Amarna period**, around 1350 BCE, under the pharaoh **Akhenaten**. In the reliefs and statues of this reign, the royal family is shown with long faces, narrow, stretched skulls, thin limbs, swollen bellies, and broad hips. Popular writers have suggested that this reflects an alien lineage.

Scholars of Egyptian art have long recognized that Amarna style was a deliberate artistic revolution. Akhenaten broke with Egyptian tradition by promoting the worship of the sun disk, the Aten, over the old gods, and his artists developed a new, expressive style that exaggerated the human form. The style is found not only on the king and queen but on ordinary officials, courtiers, and even the sculptors' own workshop pieces, and it softens in the later years of the reign. Some modern writers have suggested the king had a medical condition. A 2010 study led by

Egyptian researchers, which examined the mummies of the royal family using DNA and imaging, concluded that there was no evidence of the genetic disorder some had proposed, though the study itself has been debated. The best-supported view is that the elongated forms were a stylistic choice with religious meaning, an artistic way of expressing the king's special role as the intermediary of the sun.

Akhenaten, we should note, lived well over a thousand years after the Great Pyramid. His art tells us about Amarna theology. It tells us nothing about who built Giza.

The same reasoning applies to Egyptian gods with animal heads, such as Anubis, Thoth, Horus, and Sekhmet. The Egyptians combined human bodies with animal heads to express the qualities of a deity: the keen sight of the falcon, the watchfulness of the jackal, the fierceness of the lioness. These were symbolic images, and the Egyptians themselves wrote at length about what they meant.

Case five: "impossible" stonework

A fifth line of argument holds that some Egyptian stone objects are so finely made that modern machine tools would be required. The cited examples include thin-walled stone vases, which are sometimes described as lathe-turned, and the huge granite boxes in the underground galleries of the Serapeum at Saqqara, which weigh many tonnes and have very flat surfaces.

Egyptologists have answered these claims in some detail, using three kinds of evidence.

- **Tools and workshops.** Excavations have turned up stoneworking tools: copper chisels, dolerite pounders, flint blades, bow drills, and the abrasive materials, such as quartz sand, used in cutting and polishing. They have found workshops where vessels were made, and tomb paintings that show craftsmen working on stone vases.
- **Unfinished pieces.** Abandoned and half-finished objects show the stages of the work: roughing out, hollowing, grinding, polishing. The **unfinished obelisk** at Aswan, abandoned in its quarry after it cracked, preserves the marks of the pounding stones used to cut it.
- **Replication.** Modern stonemasons and archaeologists have reproduced Egyptian techniques, drilling and grinding hard stone with copper tools and sand, and have made vessels of a similar quality. The work is slow, but it can be done.

Precision work in hard stone is a mark of skill and patience, not of machinery. The Egyptians had centuries to refine their methods, and a craft tradition in which fathers taught sons.

> **KEY POINT**
> Almost every "impossible" object in Egyptian archaeology turns out to be made of a hard material by a long, laborious, but entirely traceable process. What looks impossible is usually just *slow*.

A summary of the cases

Claim	Where it is from	What it actually is
Helicopter, submarine, jet	Temple of Seti I, Abydos, c. 1280 BCE	Two overlapping inscriptions, one carved over plaster that later fell away
Electric light bulbs	Hathor temple, Dendera, c. 2,000 years ago	A lotus and snake in a creation scene, with a djed pillar
Glider	Saqqara bird, c. 200 BCE	A small wooden bird figure, unable to fly, of unknown ritual or decorative use
Alien heads	Amarna art, c. 1350 BCE	A deliberate religious art style used for the royal family and courtiers
Impossible stonework	Vases, granite boxes, Aswan	Tool marks, workshops, unfinished pieces, and modern replications show ordinary methods

The deeper problem: what the idea leaves out

Beyond the cases, there is a further, more serious objection to the ancient-astronaut argument. It has to do with what the idea does to the people it claims to explain.

Think about what the argument is really saying. It says: *the achievement is so great that the people credited with it must not have done it.* That is a claim about the people, and it deserves to be examined. It assumes that a society of farmers, craftsmen, and scribes, working for a few generations, could not have de-

veloped the geometry, the organization, and the stonecraft to build monuments, even though we have their tools, their villages, their written records, their failed attempts, and their own graffiti. It treats the evidence of their work as irrelevant.

Archaeologists and historians have pointed out that ancient-astronaut claims tend to be applied far more to ancient cultures outside Europe, such as the Egyptians, the Maya, the Inca, and the builders of Easter Island, than to the Greeks and Romans, whose monuments are not generally said to need extraterrestrial help. Critics see in this pattern a quiet assumption that some peoples were less capable than others. Whether or not any individual writer intends it, the effect is to take credit from the actual builders.

There is also a logical problem. The ancient-astronaut explanation is an **argument from ignorance**: we cannot immediately see how this was done, therefore it must have been done in a way we cannot see. But the more we learn, the less room that leaves. In the 1960s, it was truly unclear how the pyramids were organized. Since then, archaeologists have found the workers' town, their cemeteries, the quarries, the harbor, the papyri, and the tools. Each discovery has made the human explanation stronger and the extraterrestrial one more redundant.

Finally, the idea fails a basic test of explanation. If aliens had built the pyramids, we would expect to see something that the ancient Egyptians *could not* have made: an alloy beyond their reach, an electronic component, a tool that does not fit into any known tradition. Instead, we find things that fit squarely into what we know of the Bronze Age: copper, stone, wood, rope,

and mud brick. We find mistakes, repairs, and changes of plan. We find the collapse at Meidum, the changed angle at the Bent Pyramid, and the successful third attempt at Dahshur. Superior beings, with advanced technology, would not need a century of trial and error.

> **MYTH AND FACT**
>
> **Myth:** If we cannot explain exactly how something was done, aliens must have done it. **Fact:** "We do not know exactly how" is a very common situation in history, and it has always been resolved, when it was resolved, by finding more human evidence. Gaps in knowledge are gaps in knowledge. They are not evidence for a particular alternative.

Respecting the question

It would be a mistake to treat everyone who is drawn to these ideas as foolish. The longing behind them is a natural one. People want the past to be full of mystery. They want to feel the strangeness of the pyramids, and the sense that those monuments are older and bigger than ordinary life. That feeling is real, and it is a good one.

The honest answer is that the strangeness is real, but the explanation is human. A group of people, with no machines, took on the biggest construction project in history and finished it, and they did it so well that it is still standing. They organized food for tens of thousands, trained crews, ran quarries, ran boats, and kept records. The evidence for this is richer and more specific

than the evidence for any alternative. The truth, once you know it, is not a smaller story. It is a bigger one, because it is about us.

Chapter summary

The ancient-astronaut idea took its modern form in the 1960s and has been applied to the Great Pyramid and to a handful of Egyptian images. Examined individually, each claimed piece of evidence has a straightforward explanation: the Abydos "helicopter" is an overlapping inscription, the Dendera "bulbs" are a lotus and snake in a creation scene, the Saqqara bird is a small bird figure that cannot fly, Amarna art is a religious style, and "impossible" stonework is the product of known tools and techniques. The idea also fails on logic and evidence, since it rests on arguments from ignorance and finds none of the hardware that outside technology would leave behind. And it takes credit away from the real builders.

The next chapter looks at a final cluster of ideas: star maps, secret numbers, "pyramid power," and a lost civilization.

CHAPTER 21

Star Maps, Secret Numbers, and "Pyramid Power"

The Orion correlation, the pi and golden-ratio claims, the pyramid inch, the razor-blade sharpener, and a lost civilization: what each says and how it holds up.

The previous two chapters dealt with machines and with visitors from space. This one deals with a third family of ideas, which is in some ways the most human. These are the claims that the Great Pyramid *encodes* something: a map of the sky, a table of cosmic constants, a prophecy, or a mysterious energy that works on objects placed inside a pyramid shape. Some of these ideas contain real observations, and one of them, the possible alignment of the narrow shafts with particular stars, remains an open question among scholars. Others have been tested many times and found empty.

We will take them in order, beginning with the one that has the best claim to be taken seriously.

The Orion Correlation

In 1994, a construction engineer and writer named **Robert Bauval**, who was born in Alexandria and raised in Egypt, published a book called *The Orion Mystery*. His central idea, now known as the **Orion Correlation Theory**, is that the layout of the three great pyramids at Giza reproduces on the ground the pattern of the three stars of Orion's Belt in the sky.

The argument runs like this. The three pyramids of Giza, those of Khufu, Khafre, and Menkaure, are not in a straight line. The third, Menkaure's, which is the smallest, is offset slightly to the east from the diagonal line formed by the first two. The three stars of Orion's Belt, called Alnitak, Alnilam, and Mintaka, show a similar pattern: the smallest and faintest, Mintaka, is offset slightly from the line through the other two. Bauval proposed that the pyramids were placed so as to mirror those stars, with the Nile standing in for the Milky Way.

There is a real basis for connecting Orion with Egyptian royal burial. The Egyptians identified the constellation of Orion with **Sah**, a god associated with **Osiris**, the god of the dead and the afterlife. The Pyramid Texts, which we met in Chapter 17, say that the dead king ascends to the sky and becomes a star in the company of Orion. In addition, as Chapter 15 mentioned, in the 1960s the astronomer Virginia Trimble and the engineer Alexander Badawy noticed that the southern shaft of the King's Chamber seems to have pointed toward the region of Orion's Belt at the time of construction, while a shaft in the Queen's

Chamber may have pointed toward Sirius, which the Egyptians associated with Isis. Bauval built on those suggestions.

So the Orion Correlation has a few legs of real material to stand on. The trouble is that each of them is weaker than it first looks.

The match between the pyramids and the stars is loose. To make the pyramids match the Belt, you have to turn the map upside down, or otherwise rotate or flip it, because, in the natural orientation, north on the ground does not correspond to north on the star map. The stars of Orion's Belt run diagonally across the sky, with a slight bend. The pyramids also run diagonally, with a slight bend. But a bend is a bend: with three points and a few degrees of freedom, there are very few ways to place three dots that *cannot* be matched to some three-star group in the sky. When one tries the match carefully, the angles and relative distances are only approximately similar, and the match depends on which way you orient the sky and the ground.

The other pyramids do not fit. Egypt has dozens of pyramids. At Abu Rawash, a few kilometers north of Giza, stand the ruins of the pyramid of Djedefre, Khufu's son. At Zawyet elAryan, Abusir, Saqqara, Dahshur, and Meidum there are many more. The Orion Correlation concentrates on three pyramids and ignores the others. Bauval suggested that a larger map of the Nile Valley might fit the entire constellation, but this extension requires choosing which pyramids to include and which to leave out. A pattern that holds only for the points that are picked is not a strong pattern.

The three pyramids were not built as a single design. As we saw in Chapters 5 and 6, the Giza pyramids were built by three different kings over about eighty years. Khufu's pyramid was placed first, on a good site, and the later pyramids were placed with reference to it and to the terrain. Khafre's pyramid was placed on higher ground to the southwest, and Menkaure's farther southwest still. The arrangement is easily explained by available space and the slope of the plateau. The idea that three different kings, in three different reigns, followed a single predetermined star map requires a continuity of plan across generations for which there is no evidence.

The Egyptians did not leave any such map. We have a large body of Egyptian astronomical material, including star clocks, ceiling paintings of the sky, and lists of constellations and decans. None of them refers to a pyramid layout as a star map. The Egyptians who built pyramids in the Fourth Dynasty were most concerned with the circumpolar stars, the "imperishable ones" that never set below the horizon, because those were the stars that symbolized eternal life. The orientation of the Great Pyramid to the cardinal directions, to north in particular, fits that concern. The Orion Belt idea is much less central to the evidence than it first appears.

The dating does not line up easily. Part of Bauval's theory depended on a particular date in the sky: he argued that Orion's position in about 10,500 BCE matched the layout of the pyramids. In later versions of the argument, the pyramids were said to have been planned much earlier than the traditional date. But the evidence of dating, including the papyri of Merer, which are

dated to Khufu's reign, the workers' town, and radiocarbon work on short-lived organic material, puts the Great Pyramid squarely in the reign of Khufu, in the twenty-sixth century BCE. (Early radiocarbon tests on mortar gave dates a few centuries too old, an effect usually blamed on old wood burned to make the mortar, and not on an older pyramid.)

> **KEY POINT**
>
> It is true that the Egyptians linked Orion with the dead king's afterlife. It is *not* shown that the Giza pyramids were laid out as a map of Orion's Belt. The first claim is in Egyptian texts. The second is a modern proposal that has not been supported by the evidence.

The question of the shafts is a separate one, and it remains open. It is entirely possible that the builders aimed the King's Chamber shafts at stars that were important to them, in order to guide the king's soul. Several scholars have suggested alignments of this sort, and they are consistent with Egyptian religion. The difficulty is that the shafts are not perfectly straight, the stars' positions are only roughly known across the centuries, and the Queen's Chamber shafts have never been explored beyond the small stone doors. The question will be settled, if at all, by more exploration, not by a theory of the layout of the pyramids.

The Sphinx, the Younger Dryas, and a "lost civilization"

A related idea has to do with the claim that the monuments of Giza are *much older* than they appear to be, perhaps by thousands of years, and were built or begun by a civilization that has vanished from the record.

The best-known proponent of this idea is the British writer **Graham Hancock**, whose 1995 book *Fingerprints of the Gods* became an international bestseller, and who has continued to write and present on the subject, including in a streaming television series in 2022. Hancock has argued that a lost advanced civilization existed during the last ice age and was destroyed in a catastrophe around 12,000 years ago, a period geologists call the **Younger Dryas**, and that its surviving members passed on knowledge to later cultures, including Egypt.

A related piece of the argument concerns the Great **Sphinx**. In the early 1990s, the geologist **Robert Schoch** of Boston University examined the Sphinx and its enclosure and argued that the pattern of weathering on the walls, with deep, rounded, vertical fissures, looked like the result of long-term rainfall rather than wind and sand. Since the climate of Giza has been very dry for the whole of the historic period, he suggested, the Sphinx must date from a much earlier, wetter time, thousands of years before the traditional date.

It would be unfair to dismiss the Sphinx question as stupid. It was posed by a trained geologist, and it received serious atten-

tion. Still, the great majority of Egyptologists and geologists who have examined it have not accepted it, for several reasons.

- **The weathering has other explanations.** Limestone weathers differently depending on the quality of the layers. The soft, shaley layers of the Sphinx's body erode quickly, while the harder layers resist. Salt crystallization, dew and humidity, and wind erosion all produce the type of rounded profile seen at the Sphinx. Similar weathering is seen on other monuments of known age at Giza.
- **The Sphinx is tied to the surrounding monuments.** The enclosure around the Sphinx is the quarry from which the blocks of the nearby Sphinx Temple and Valley Temple were cut. Those temples are in the architectural style of the Fourth Dynasty and are connected to the causeway of Khafre's pyramid. The Sphinx sits on the axis of that complex. Archaeologists therefore date it to the reign of Khafre, around 2500 BCE.
- **There is no sign of a people who could have done it.** If a society capable of carving and moving enormous blocks existed in Egypt in the tenth millennium BCE, it would have left evidence in the form of settlements, tools, food remains, burials, and rubbish. The archaeology of Egypt in that period shows hunter-gatherers and early herders, with no evidence of a society with the necessary resources and organization. In contrast, for the pyramid age we have all of these things in abundance.

- **The hypothesis is not needed.** The chain of development, from Djoser's step pyramid to Sneferu's experiments to Khufu's masterpiece, shows the builders learning. A civilization that had inherited the knowledge of a lost master culture would not have needed that.

In 2022 a streaming series presenting Hancock's ideas drew public criticism from archaeologists and from professional organizations, including the Society for American Archaeology, who said that it misrepresented the evidence and downplayed the accomplishments of known peoples. Hancock has rejected that criticism and says that his work calls for a more open search of the evidence. Readers are encouraged to read both sides. But it is fair to say that the proposal has not persuaded the specialists in the field, and that no physical evidence of the lost civilization it describes has been presented that has survived independent examination.

> **MYTH AND FACT**
>
> **Myth:** Giza was built by a civilization that vanished 12,000 years ago. **Fact:** There is no archaeological evidence for such a civilization in Egypt, and the pyramids, temples, and workers' town at Giza have been dated to the twenty-sixth century BCE by several independent methods.

Is the Great Pyramid a table of cosmic constants?

The second family of ideas concerns numbers. The Great Pyramid has famous proportions, and for nearly two centuries writers have found in them mathematical "secrets."

Pi

In 1859, a London publisher named **John Taylor** published a book, *The Great Pyramid: Why Was It Built, and Who Built It?*, in which he noticed that the perimeter of the Great Pyramid's base, divided by twice its height, comes out very close to the number pi, the ratio of a circle's circumference to its diameter. Taylor proposed that the builders had known pi and had deliberately designed the pyramid to represent a circle and a square at once. The idea has been repeated ever since.

The observation is correct, as far as the arithmetic goes. With a base of 440 cubits on each side and a height of 280 cubits, the perimeter is 1,760 cubits, which divided by twice the height, 560, gives 3.1428..., a value that matches the fraction 22/7. That is about 0.04 percent above pi. It is a striking coincidence.

But there is a simple reason for it, and the reason has to do with the **seked**, the Egyptian measure of slope that we met in Chapter 5. The builders chose a slope of 5½ palms per cubit, which works out to a rise of 14 for a run of 11. It is easy to measure out with ropes and to repeat in practice. A pyramid with a 14-to-11 slope necessarily has a perimeter-to-height ratio of 44 to 7, or 6.2857, and 22/7 is just the half of that. In other words, the builders were not trying to represent pi; they chose a *practi-*

cal, repeatable slope, and the pi-like ratio is a consequence of that choice. As the Egyptians had no way to write a symbol for pi, and as their own mathematical documents use quite a different approximation, it is hard to see how they would have designed in a value they did not have.

That last point is worth emphasizing. The one Egyptian text that gives us their method for finding the area of a circle, the Rhind Mathematical Papyrus (Problem 50), tells the scribe to subtract one-ninth of the diameter and square the result. That gives an effective value of pi of about 3.16, or 256/81, which is not the same as 22/7 (3.1428) and further from the true value of 3.1416. If the Egyptians had wished to build in an exact pi, they would have used the value they actually knew.

The golden ratio

A parallel claim is that the Great Pyramid embodies the **golden ratio**, phi, which is about 1.618. The claim goes back to the nineteenth century and was widely repeated in the twentieth. A form of it was even attributed to Herodotus, who is said to have reported that the square of the pyramid's height equals the area of each face.

Again, the arithmetic is close. If the slope is 14 to 11, the ratio of the slant height of a face to half the base is about 1.6186, which is very close to phi, which is 1.6180. This is another coincidence that follows directly from the chosen slope. And here is the telling point: a pyramid that matches pi *and* the golden ratio at the same time is evidence that neither was the goal. The two numbers are not exactly related (the closest link is that four di-

vided by pi happens to come out within about a tenth of a percent of the square root of phi), so a design could not aim exactly at both at once. The slope of 14 to 11 simply lands close to both. Both appear because of the same practical measurement.

WORKED EXAMPLE

How many coincidences should we expect by chance? Suppose we take 15 measurements of the Great Pyramid (the base, the height, the slant height, the diagonals, the dimensions of the chambers, and so on). From these we can form about 105 different ratios. Suppose we compare each ratio with 10 well-known constants such as pi, e, phi, and the square root of two, and we allow a match if the ratio comes within one half of one percent of the constant, after multiplying or dividing by any power of ten. A random number has about a 0.4 percent chance of falling in such a window for any one constant. With 105 ratios and 10 constants, that is 1,050 chances, and 1,050 times 0.4 percent is about four expected matches by pure chance. So finding a few striking "coincidences" in a large structure measured in many ways is not surprising. It is what chance predicts. Real evidence of design requires a prediction made *before* looking, not a pattern found afterwards.

The pyramid inch and pyramidology

In 1864, **Charles Piazzi Smyth**, who was Astronomer Royal for Scotland, published *Our Inheritance in the Great Pyramid*. Smyth had measured the pyramid himself and became convinced that its builders had used a special unit, which he called the **pyramid inch**, a length very slightly greater than the British inch. He argued that the pyramid's dimensions, expressed in these units, encoded the circumference of the earth, the distance to the sun, the length of the year, and even the history and future of the human race, an idea that Smyth connected to biblical prophecy. This approach was called **pyramidology**, and it attracted many followers.

It had one weakness: it did not survive more careful measurement. In 1880, a young surveyor named **Flinders Petrie**, who had been inspired by Smyth's work, went to Giza and measured the pyramid with unprecedented care. His results did not support the pyramid-inch theory. The precision Smyth's numbers required was not in the monument, and Petrie came to reject the idea. His survey became the foundation of modern pyramid measurement, and it showed that the pyramid had been built with a degree of care that was high, but not mathematically perfect: the sides differ in length by only a few centimeters, a tiny fraction of their 230 meters, and the corner angles are not exactly right angles.

The real unit in use was the **royal cubit**, about 52.4 centimeters, which is used in other Egyptian buildings and is attested by surviving measuring rods. The Great Pyramid's base of about 230.3 meters corresponds to 440 royal cubits, and its original

height to 280. These are round numbers in the Egyptian system, which is exactly what one would expect from engineers working in cubits.

> **KEY POINT**
> When a monument's dimensions turn out to be round numbers in the builders' own units, that is evidence of design. When they turn out to be round numbers only in a unit invented centuries later, adjusted until it fits, that is evidence of the investigator's imagination.

"Pyramid power"

The last set of ideas is the most popular in daily life. It holds that the *shape* of a pyramid has special properties. Objects placed inside a pyramid, it is said, are preserved, sharpened, healed, or energized.

The modern version has a few well-known origins. In the 1930s, a Frenchman named **Antoine Bovis** claimed that when he visited the Great Pyramid, he noticed that small animals that had wandered into the King's Chamber and died there had not rotted, but had dried up, like mummies. He reasoned that the shape of the pyramid had preserved them. Back home, he built a small wooden model of the pyramid and reported that a dead cat placed inside also mummified rather than rotted.

In the 1940s and 1950s, a Czech radio engineer named **Karel Drbal** claimed that a used razor blade placed beneath a pyramid-shaped cardboard model, aligned with the earth's north-south axis, would regain its sharpness. He reportedly obtained a

Czechoslovak patent for a "Cheops pyramid razor blade sharpener." In the 1970s, a bestselling book on Soviet-bloc parapsychology brought the idea to a wide public, and "pyramid power" spread across the world. People built pyramid-shaped tents, hats, and bed frames, and sold pyramid-shaped devices to preserve food, improve plant growth, and promote health.

When tested under controlled conditions, none of these effects has held up. Dead animals in the dry, still air of an Egyptian chamber do dry out, and they do so for the same reason that mummies do: low humidity prevents decay. The same result is achieved in any dry, sealed container, whatever its shape. Razor blades last longer or shorter depending on the metal, the shaving technique, and the care with which they are dried and used, and a blind test that compares blades under a pyramid with blades on a shelf finds no difference. Experiments on plants, fruit, and milk yield the same: no effect that depends on the pyramid shape.

Pyramid power persists, in part, because it is hard to disprove casually. A person who sets a razor blade under a pyramid and feels it is sharper has no easy way to know if the difference is real, and the expectation of a result can influence the judgment. Controlled testing, with blinded comparisons, is the reliable way to answer such questions, and it gives the same answer every time.

Why these ideas persist

It is worth asking why these claims keep coming back. Writers on the psychology of belief have given a number of reasons that apply well to the pyramids.

- **Pattern-seeking.** Human minds are tuned to see patterns, even in random data. A monument with many measurements will always yield some striking numbers.
- **The appeal of hidden knowledge.** There is a deep satisfaction in feeling that one knows a secret that others do not.
- **Awe and scale.** The pyramids are so impressive that an equally impressive explanation seems called for.
- **Distrust of authority.** If one suspects that experts conceal things, a "forbidden" explanation is attractive.
- **Media incentives.** Mysteries sell books and attract viewers. A television program that says "the pyramids were built with ramps, ropes, and a lot of well-fed workers" is not likely to draw the same audience as one that hints at secret chambers.

None of these motives is shameful, and many of them are part of what makes people curious about the past in the first place. The point of understanding them is not to ridicule anyone but to recognize the pull in ourselves, and to build the habit of asking for evidence.

Chapter summary

The Orion Correlation has a basis in Egyptian religion, but the match between the pyramids and the stars is loose, selective, and not supported by Egyptian texts; the possible stellar targeting of the shafts is a separate and still-open question. The notion that Giza was built by a lost ice-age civilization has not been supported by evidence, and the Sphinx is tied by its architecture to the reign of Khafre. The pi and golden-ratio "secrets" are by-products of a practical slope of 14 to 11, and the pyramid inch collapsed when Petrie measured the monument carefully. Pyramid power has failed every controlled test. Taken together, these ideas show how readily pattern-seeking and a love of mystery can run ahead of evidence.

The final chapters of the book turn from claims to discoveries: what researchers are actually finding, right now, about the pyramids.

CHAPTER 22

The Frontier: What Science Is Finding Now

Cosmic rays, radar, buried rivers, papyrus logbooks, and a new museum: the discoveries of the last decade, and how to read the headlines about them.

The most exciting thing about the pyramids is that the story is not finished. For most of the twentieth century, research at Giza meant digging, measuring, and reading inscriptions. Those methods are still essential. But since about 2010, a flood of new technologies has joined them: particle detectors, ground-penetrating radar, satellite imagery, drone mapping, radiocarbon labs of unprecedented precision, and flexible cameras no thicker than a pencil. The result has been a run of discoveries that has changed some old debates and opened new ones.

This chapter surveys the main lines of current research, as of the autumn of 2026. It also tries to do something that may be even more useful: to show how to read a news story that says "hidden chamber discovered in pyramid." Such headlines appear regularly, and they are often more dramatic than the research behind them.

A logbook from the Red Sea

The most important single discovery for understanding how the Great Pyramid was built came, oddly, from a cave far from Giza. In 2013, the French archaeologist **Pierre Tallet** and his team, working at **Wadi al-Jarf** on the Red Sea coast, about 180 kilometers southeast of Cairo, uncovered among the oldest inscribed papyri ever found. They had been stored in a man-made gallery cut into the rock beside what turned out to be an ancient harbor, probably the oldest port yet found in Egypt, used for expeditions to the copper mines of Sinai.

The papyri are administrative records from the reign of Khufu. The best-known is the logbook of an official named **Merer**, who led a crew of about forty men. Over several months, his team transported blocks of fine white limestone from the quarries at Tura to Giza, using boats on the Nile and its canals, and they delivered them to the construction site that the document calls "the Horizon of Khufu." The diary gives dates, destinations, and rations. It also mentions a high official, Ankhhaf, a half-brother of the king, who was in charge of the harbor at Giza. For the first time, we have a primary source from the very people who were moving the stone.

The papyri have been studied and published by Tallet and Egyptologist **Mark Lehner**, who has spent decades excavating at Giza. Their joint account is told in a book, *The Red Sea Scrolls*, which appeared in 2021. Tallet's work continues. Each season brings new fragments, and each one adds to what we know of the administrative machine behind the pyramid.

KEY POINT

Merer's logbook is the oldest known papyrus archive, and the only first-hand account of work on the Great Pyramid. It proves that stone from Tura was sent by boat to Giza under Khufu's direct administration, and it shows an organized state bureaucracy, with dated records, named officials, and standard rations.

The lost branch of the Nile

In May 2024, a team led by **Eman Ghoneim**, a geomorphologist at the University of North Carolina Wilmington, published in the journal *Communications Earth & Environment* the discovery of a long-lost branch of the Nile that once ran along the edge of the desert plateau where many of the pyramids were built. The team named it the **Ahramat Branch**, from the Arabic word for pyramids.

The researchers used a combination of satellite radar imagery, which can penetrate dry sand to detect buried features, ground-penetrating geophysics, and deep soil cores taken from the fields. They traced a buried channel roughly 64 kilometers long, running parallel to the plateau, and found that thirty-one pyramids from the Old and Middle Kingdoms, including the pyramids at Giza, the Step Pyramid of Djoser, and the Bent and Red Pyramids at Dahshur, lie along its former course. Many of the pyramids' causeways, the ceremonial roads leading from the pyramid to the valley temple, seem to end at the proposed riverbank.

The implication is that the pyramid builders did not have to move stone across miles of dry land. A branch of the river ran right at the foot of the plateau, close to the construction sites, and so boats could bring in materials, from quarries and from the south, almost to the building site. The study also suggests the branch shifted eastward over the centuries and eventually dried up, perhaps as a result of drought and the build-up of desert sand.

The study fits well with older evidence, such as Merer's log-book and the harbor found at Giza, but it also has limits. It was published without an exact date for when the branch dried up, and the details of how the branch worked in different seasons are still being worked out. Outside specialists praised the study as showing how different the floodplain was in the pyramid age. Some, including the Egyptologist Zahi Hawass, noted that it does not change the main picture, since the papyrus logbook had already shown that materials were delivered to the Great Pyramid through a nearby harbor.

Cosmic rays, voids, and the Great Pyramid's hidden spaces

Chapter 15 described the **ScanPyramids** project, which began in 2015 and has used muon imaging, thermography, and other methods to look inside the Great Pyramid. Its main results have been the **Big Void**, a space at least thirty meters long above the Grand Gallery, announced in 2017, and a smaller corridor-like void behind the north face, above the entrance, which was confirmed in 2023 when the team threaded an endoscope through a joint in the stonework and saw the interior of a gabled space. The researchers have suggested that the north-face space is most likely a structural device to relieve pressure on the entrance passage, and have been cautious about any further interpretation.

It is worth being careful about what these results show. They show that there are empty spaces inside the pyramid that were not known before. They do not show that there is a burial chamber, treasure, or any other particular thing. The Great Pyramid already contains several known voids and relief spaces, and the new ones are consistent with the same architectural logic, though the Big Void's purpose remains unexplained. A press story that says "secret chamber found" is true in a narrow sense, and misleading in the broad one.

The Menkaure scans

In November 2025, researchers from Cairo University and the Technical University of Munich, working as part of the Scan-Pyramids project, published results of a different kind of study, this time of the **pyramid of Menkaure**, the smallest of the three great pyramids at Giza. Rather than muons, they used a combination of ground-penetrating radar, ultrasound, and electrical resistivity tomography, and combined the data using a technique they call image fusion.

The team detected two air-filled voids behind the smooth granite facing on the eastern side of the pyramid. One is about 1.4 meters deep, about a meter high, and about 1.5 meters wide. The other is about 1.1 meters deep and smaller. They lie behind an area of polished granite blocks, about four meters high and six meters wide, which is unusual because similar granite facing elsewhere on the pyramid is found only around the known entrance on the north side.

That similarity led the team to suggest that the voids might be associated with a second entrance on the east face, an idea that had been proposed earlier, in 2019, by the researcher Stijn van den Hoven. The scientists were careful to describe their finding as "initial evidence" in support of that hypothesis and not as proof. One of the authors, Christian Grosse of the Technical University of Munich, called the entrance hypothesis "very plausible," but nothing has been opened, and no entrance has been found. The results were published in the journal *NDT & E International*.

This is a good example of how to read such a story. The result is a real measurement of a real anomaly. The interpretation, a hidden entrance, is a reasoned hypothesis. What is still needed is a way to look inside, perhaps with a small borehole and an endoscope, and the permission of Egypt's antiquities authorities to carry it out. Until then, the answer is "we do not know yet."

Radiocarbon and the calendar of kings

Another quiet revolution has been in dating. For a long time, the dates of the Egyptian kings depended on king lists, records of astronomical events, and the reconstruction of how long each king ruled. These methods are good but produce a margin of error of a few decades or more in the Old Kingdom.

In 2010, a team led by **Christopher Bronk Ramsey** of the University of Oxford published a major study in the journal *Science*. They dated short-lived plant material, such as seeds and reeds, from the sealed contexts of many Egyptian archaeological sites, and used statistical methods to link the radiocarbon dates to the historical chronology. Their results largely supported the framework of the traditional chronology, with the radiocarbon and historical dates agreeing to within a few decades for most reigns, though some dates shifted. Earlier radiocarbon work on pyramid mortar had given dates that seemed too old, but those samples included old wood used as fuel, which skews the result.

This matters for the claims in the last chapter. Radiocarbon dating gives an independent check on the age of the Great Pyramid, and it places the monument in the twenty-sixth century BCE, as the historical records say.

The workers' town and the human story

Archaeology at Giza has not been only about technology. Since the 1980s, **Mark Lehner** and his colleagues in the Giza Plateau Mapping Project, later AERA, have been excavating the settlement south of the pyramids, known as **Heit el-Ghurab**, "Wall of the Crow," and often called the "Lost City of the Pyramid Builders." They found bakeries, breweries, workshops, long galleries that may have housed work gangs, and a great deal of refuse showing what people ate. Chapter 13 described what they found.

The animal bone studies show that the workers ate substantial amounts of beef, as well as sheep and goat. This implies a state-supplied diet that was richer than the diet of the average Egyptian farmer, and it contradicts the picture of starving, whipped slaves. Their cemeteries, excavated by **Zahi Hawass** and his team in the 1990s, contain simple but carefully built tombs, some with bread and beer placed beside the body. A number of skeletons show healed injuries, which suggests medical care.

Excavation continues. Each season of work adds details about the layout of the town, how the work gangs were housed, and how supplies reached the plateau.

Quarries and ramps

Another area of active research is the study of how stone was moved up the pyramid. In earlier chapters, we met the steep ramp discovered at the alabaster quarry of **Hatnub** in the Egyptian desert. A joint French and British team has re-examined the site in recent years, and it has revealed a system of a central ramp flanked by staircases and post holes, which seems to have allowed heavy blocks to be hauled up on sledges at a steep angle using ropes anchored to the posts. It dates to about the time of Khufu and shows a plausible technique, though not necessarily the method used on the Great Pyramid itself.

Modern engineers have also tested ideas about ramps with computer models and with small-scale experiments. The debate remains open, but the range of possibilities has narrowed: whatever the builders used, it probably involved a combination of ramps of different types, levers, and a lot of well-organized rope work.

The new museum

In November 2025, after many years of construction, Egypt formally opened the **Grand Egyptian Museum**, on the desert plateau only a few hundred meters from the pyramids of Giza. It is the largest museum in the world devoted to a single civilization. Among its exhibits are the full collection of treasures from the tomb of Tutankhamun, shown together for the first time, and the conserved **solar boat of Khufu**, the ancient cedar ship that was buried beside the Great Pyramid.

The museum's significance for pyramid research is practical as well as symbolic. It houses conservation laboratories and displays objects that had been scattered across storerooms. For visitors, it offers a way to connect the monuments on the plateau with the objects and people who used them.

Technologies on the horizon

Looking ahead, several tools may change the picture again.

- **Better muon detectors.** New detectors, more sensitive and easier to deploy, can image smaller structures and cover more of a pyramid's volume.
- **Robotic and fiber-optic exploration.** Small robots and flexible cameras can enter spaces that people cannot, and their cameras keep getting smaller.
- **Satellite and airborne radar and lidar.** These can map buried structures, ancient channels, and quarries across a wide area.
- **Artificial intelligence.** Machine learning can help find patterns in satellite imagery, sort thousands of archaeological fragments, and read damaged papyri.
- **Ancient DNA and isotopes.** Analysis of the remains of workers and of animals can show where people came from, what they ate, and what diseases they suffered.
- **Digital twins.** Detailed 3D models of the pyramids, combining scans, drone images, and survey data, let researchers test ideas virtually.

Each of these promises to answer a question that was out of reach a generation ago, and each will probably raise a new one.

How to read a pyramid headline

Because pyramid discoveries attract so much attention, it is worth having a short guide for reading them. When you see a headline that promises a hidden chamber, a lost civilization, or a secret, you can ask a few questions.

1. **Where was it published?** A peer-reviewed journal is a better sign than a press release, a social-media post, or a documentary. Peer review is imperfect, but it means that other specialists have examined the methods.
2. **What exactly was detected?** "An anomaly," "a void," and "a chamber" are different statements. A void is a measurement. A chamber is an interpretation.
3. **How certain do the authors say they are?** Careful researchers use words like "initial evidence," "consistent with," or "suggests." Be cautious of headlines that turn those into "proves" or "reveals."
4. **Has anyone else confirmed it?** An independent team using a different method is the strongest check.
5. **Who benefits from the story?** A television program, a tour operator, or a person selling a book has a reason to make a story sound bigger.
6. **Does it change the main picture?** Most new discoveries refine the story without overturning it. Be wary of any claim that "everything we know is wrong."

MYTH AND FACT

Myth: Each new scan finds a secret chamber that the authorities are hiding. **Fact:** Most scans find structural voids and construction spaces, and the scientists who find them publish the results openly. Whether an anomaly is a room, a gap in the masonry, or a pressure-relieving space is determined by further investigation, and that takes time and permission, not a cover-up.

What we still do not know

For all the progress, many things are still unknown, and a good ending for this chapter is a list of open questions.

- **The exact ramp system** used to raise the stone, especially to the upper levels of the Great Pyramid.
- **The purpose of the Big Void** above the Grand Gallery, and whether it can be explored.
- **What lies beyond the small stone doors** in the Queen's Chamber shafts.
- **Whether Menkaure's pyramid has a second entrance** and what it leads to.
- **Where the kings were actually buried.** No intact Fourth Dynasty royal burial has been found.
- **How the workforce was recruited and rotated,** in detail, and how many people were involved at any one time.
- **The precise methods of alignment** used to orient the base with such accuracy.

- **How the Sphinx's builders** organized the work and what, exactly, it was meant to represent.

None of these is a "mystery" in the sense of something beyond explanation. Each is a question with a possible answer, waiting for evidence.

Chapter summary

New technologies have opened a second great age of pyramid research. Merer's logbook gives a first-hand account of moving stone to the Great Pyramid. The discovery of the Ahramat Branch shows that the Nile once ran close to the pyramid fields. Muon imaging has revealed voids in the Great Pyramid, and radar and ultrasound have found two anomalies in Menkaure's pyramid, which may point to a second entrance but have not been explored. Radiocarbon dating confirms the traditional chronology, and continuing excavation of the workers' town fills in the human story. Every discovery needs to be read with care, because the leap from "an anomaly" to "a secret chamber" is a long one.

CHAPTER 23

Questions People Ask

Short, direct answers to the questions that come up most often, with pointers to the chapters that explain them.

Over the course of this book, a number of questions have come up repeatedly. This chapter gathers them in one place, in the form of short answers. If you want the reasoning and the evidence behind any answer, the chapter references will take you there.

About the pyramids themselves

How old is the Great Pyramid?

It was built around 2600 to 2560 BCE, during the reign of Khufu of the Fourth Dynasty, which makes it about 4,600 years old. It was already more than two thousand years old when Cleopatra VII was born. The dating rests on several independent lines of evidence: the work-gang marks inside the pyramid that name Khufu, the Wadi al-Jarf papyri from Khufu's reign, the style of the surrounding cemeteries, and radiocarbon dating of short-lived organic materials. See Chapters 5, 11, and 22.

How many pyramids are there in Egypt?

Counts vary depending on what is called a pyramid, since many are now only heaps of rubble. A commonly cited figure is around a hundred and twenty known pyramids in Egypt, built mostly between about 2670 and 1500 BCE. Sudan, in ancient Nubia, has more than Egypt, because the kings of Kush built pyramids for centuries. The best known Egyptian ones are at Giza, Saqqara, Dahshur, Meidum, Abusir, and Lisht. See Chapters 4 through 8.

Why isn't the Great Pyramid smooth and white anymore?

When new, the Great Pyramid was covered with a casing of fine white limestone from Tura, polished smooth, which would have shone in the sun. Most of the casing stones were removed in the Middle Ages, after earthquakes loosened them in the fourteenth century, and they were reused for buildings in Cairo. A few casing stones remain at the base. What we see today is the stepped core masonry. See Chapters 5 and 18.

Was the capstone made of gold?

We do not know for certain. The top of a pyramid was crowned by a small pyramid-shaped stone called a **pyramidion**. Pyramidia from other pyramids survive, made of limestone or granite, and some bear inscriptions. Egyptian texts speak of the sun's first rays striking the pyramid's tip, and some scholars have suggested that the pyramidion of the Great Pyramid may have been covered in electrum, an alloy of gold and silver, so that it would flash in the light. No such covering has survived, and the original pyramidion of the Great Pyramid has never been found. See Chapter 5.

Who designed the Great Pyramid?

We do not have a certain answer, but the best candidate is **Hemiunu**, a vizier and nephew of Khufu, who is called "overseer of all the king's works" in his tomb at Giza. A famous statue of him, found in his tomb chapel, shows an older, heavyset man, and is in a museum in Hildesheim, Germany. Many Egyptologists think he directed the project, though no text says in so many words that he designed it. See Chapter 5.

How tall is the Great Pyramid, and was it the tallest building in the world?

Its original height was about 146.6 meters, or 481 feet. Today it stands about 138.5 meters, because the top and the casing are missing. It was the tallest human-made structure in the world for roughly 3,800 years, until the central spire of Lincoln Cathedral in England was completed in the early fourteenth century. See Chapter 5.

About how it was built

How long did it take?

Ancient and modern estimates agree on about twenty years, matching the reign of Khufu, which was around twenty-three years by one reckoning. To finish in that time, the builders had to place, on average, many blocks per day, a rate that was achievable with several work crews operating at the same time on different parts of the monument. See Chapter 14.

How many people built it?

Estimates range from about ten thousand to thirty thousand workers at a time, with the most common range being twenty to thirty thousand, including quarrymen, haulers, masons, boat crews, bakers, brewers, and administrators. Many of them rotated in and out. The figure of a hundred thousand reported by Herodotus is almost certainly an exaggeration. See Chapter 13.

Were the builders slaves?

The evidence says no. They were housed in a town, fed bread, beer, meat, and fish, buried in tombs beside the pyramids with offerings, and given medical care. The tombs of the workers are decorated with the names of their gangs. The labor force seems to have combined permanent skilled specialists with farmers who were conscripted for a period of service, as part of the state's labor taxation. The conscripts were not free in the modern sense, but they were not slaves either. See Chapter 13.

How did they move the stones?

Most blocks were cut from limestone quarried within a few hundred meters of the pyramid, and dragged on wooden sledges. Wetting the sand in front of the sledge reduces friction considerably, as an Egyptian tomb painting seems to show and as modern physics experiments have confirmed. Finer stone, such as the limestone for the casing from Tura, and granite from Aswan, traveled by boat on the Nile. The logbook of Merer shows exactly how one boat crew did it. See Chapter 11.

How did they lift them to the top?

Nobody knows for sure, and this is one of the real open questions. The main proposals involve ramps: a straight ramp, a zigzagging ramp on one face, a spiral ramp around the pyramid, or an internal ramp, or a combination. Levers and rocking devices were probably used for final placement. A steep ramp with post holes found at the Hatnub quarry shows that steep hauling was practiced. See Chapter 12.

Could the pyramid be built today?

Yes. Modern engineers could build a copy with cranes and trucks, and the task would be a matter of cost, not possibility. The harder question is whether it could be done with only the tools of the Old Kingdom. Experiments, including televised attempts to build small pyramids with replica tools, have shown that small crews can quarry limestone with copper and stone tools and can drag multi-ton blocks on sledges, but that lifting blocks to a height is the difficult part, and that experiments at small scale do not tell us everything about work at large scale. See Chapter 12.

Is it true that the stones fit so tightly that you cannot slip a knife between them?

For the original outer casing stones, the joints were very fine, a fraction of a millimeter wide in places, and the claim is roughly true for those stones. For the inner core blocks, which were not meant to be seen, the joints are much wider, and the gaps were filled with mortar and rubble. The tight joints were a mark of care, not of magic. See Chapter 9.

How could they get the base so level and square?

By using simple tools with great care: a water-filled network of trenches cut into the bedrock as a level, plumb bobs, set squares, and knotted ropes. The north-south alignment was probably set by sighting stars or the sun. The base is level to within a few centimeters and aligned to true north within a small fraction of a degree. See Chapter 9.

About what they were for

Were the pyramids tombs?

Yes. The weight of the evidence supports that they were royal tombs and the centerpieces of cult complexes where the dead king was honored. The pyramid shape probably symbolized the sun's rays, or the mound of creation, or a staircase to the sky, or all three. No intact royal burial has been found in a Fourth Dynasty pyramid, since they were robbed in antiquity, and that remains a gap in the evidence. See Chapters 16 and 17.

Why is there no mummy in the Great Pyramid?

It was robbed long before the modern era. By the Middle Kingdom, if not earlier, the pyramid's contents were gone. When Caliph al-Ma'mun's men broke in during the ninth century CE, they found the chambers empty, according to later accounts. The sarcophagus in the King's Chamber is lidless and cracked, and the chamber has been open to visitors for more than a thousand years. See Chapters 17 and 18.

What about the "curse of the pharaohs"?

The story took off after the discovery of Tutankhamun's tomb in 1922, when its sponsor, Lord Carnarvon, died the following spring of an infection that followed a mosquito bite. Newspapers drew a connection. In fact, many of the people who worked on the tomb lived for decades afterward, and studies comparing the lifespans of those present at the opening with those who were not have found no difference. Inscriptions with curses do exist on some tombs, but they were mostly warnings to deter vandals, and had no more effect than a "no trespassing" sign. The curse has no connection to the pyramids as such.

Did Napoleon really say that forty centuries looked down on his troops?

He is reported to have said, before the Battle of the Pyramids in July 1798, "Soldiers, from the summit of these pyramids, forty centuries look down upon you." The line comes from accounts written after the fact, and its exact wording and even whether he said it are uncertain, but it was part of the legend that he and his supporters built around the Egyptian campaign. See Chapter 18.

About the controversies

Are the pyramids aligned with Orion's Belt?

Not in any way that the evidence supports. Egyptians associated Orion with the god Osiris and the dead king's afterlife, and the southern shaft of the King's Chamber may have been aimed at that region of the sky. But the claim that the three Giza pyramids are a map of Orion's Belt rests on a loose and selective match, and the pyramids were built by three different kings over about eighty years. See Chapter 21.

Do the pyramids hold secret mathematical codes?

The Great Pyramid's proportions are impressive and are based on a slope of 14 to 11, a simple practical slope that happens to produce ratios close to pi and the golden ratio. They are by-products of that slope, not evidence of hidden knowledge. See Chapter 21.

Was there a lost civilization before the Egyptians?

There is no evidence of one in the archaeology of Egypt. The idea has been argued by some popular writers, but it has not been supported by physical evidence that has survived independent examination. See Chapter 21.

Did aliens help?

There is no evidence of that, and a great deal of evidence of human work. See Chapter 20.

Could the Great Pyramid generate energy?

No. The power-plant theory has no physical remains to support it, requires materials the Egyptians did not have, and cannot work as a source of net energy. See Chapter 19.

Why do so many people believe these theories?

Because the pyramids are awe-inspiring, because the human mind seeks patterns, because mystery is exciting, and because stories about hidden knowledge are more fun to tell than an account of ramps and rations. Most people who are drawn to these ideas are curious and sincere. Checking claims against evidence is the way to turn that curiosity into knowledge. See Chapters 19 through 22.

Practical questions

Can I go inside the Great Pyramid?

Visitors can usually enter the Great Pyramid, though access depends on the day, the ticket, and the conservation work in progress, and rules change over time. Entry is through a tunnel cut by tomb robbers in the Middle Ages, which joins the original passages. The ascent through the Grand Gallery to the King's Chamber is steep and cramped, and it is not suitable for anyone who has trouble with confined spaces. Check the current rules before you travel, as they change.

What is the best time to visit Giza?

Early morning, when it is cooler and the crowds are thinner, and the light on the limestone is soft. Winter and early spring are more comfortable than summer, when temperatures are very high. The nearby Grand Egyptian Museum is worth a separate visit.

Where can I read more?

The Further Reading appendix at the end of this book lists the books and sources on which the book is based, including the works of the specialists who have done the digging.

Epilogue: What the Pyramids Teach

Four lessons from a mountain made by hand.

We began this book at the foot of the Great Pyramid, with a feeling that is familiar to almost everyone who sees it: that something so large and so exact cannot have been made by ordinary people. We end by saying that it *was* made by ordinary people, and that this is the most remarkable thing about it.

Let us gather the main lessons.

Lesson one: great things come from organization

The Great Pyramid is a triumph of logistics before it is a triumph of engineering. The stone was cut at the quarries, floated on the Nile, dragged on sledges, and set in place by tens of thousands of people who had to be fed, housed, and supervised. Bread and beer were the currency of the project. A boat foreman named Merer kept a log, a vizier named Hemiunu directed the work, and an army of bakers, brewers, carpenters, butchers, and scribes kept it running. The achievement was a collective one. It tells us that what we call technology is often a matter of coordinating people.

Lesson two: progress is made by trial and error

The pyramid did not appear fully formed. Djoser's builders learned to work in stone by stacking small blocks like bricks. Sneferu's builders tried a stepped design, saw it collapse, changed the angle halfway up a second pyramid when it began to crack, and got it right on the third. Khufu's builders applied what they had learned and perfected it. The failures are as instructive as the successes, because they show human minds at work: observing, adjusting, and improving. An advanced outside intelligence would not have needed to learn. The Egyptians did, and that is why we can respect what they achieved.

Lesson three: evidence beats mystery

It is thrilling to imagine that the pyramids hold secrets that no one has discovered. But the real secrets turned out to be better. A logbook found in a cave beside the Red Sea tells us how stone traveled to Giza. A buried town tells us who baked the bread. A series of failed pyramids tells us how the builders learned. Cosmic rays tell us that there are hollow spaces inside the monument that no one has entered. Each of these discoveries came from careful, patient work, and each is more satisfying than any story about lost machines or visitors from the stars. When people say "we don't know how it was done," the right answer is not to give up and imagine something exotic. It is to go and look.

Lesson four: the past deserves respect

Perhaps the most important lesson is the simplest. The people who built the pyramids were as intelligent as we are. They lacked iron, wheels for heavy loads, and engines, but they did not lack insight or will. They solved problems that would challenge a modern engineering firm, with materials that were at hand and skills passed from one generation to the next. To say that they could not have done it is to underestimate them. To learn how they did it is to honor them.

A closing image

Imagine a morning in the twenty-sixth century BCE, on the Giza plateau. Before dawn, the bakers have been up for hours. Smoke rises from the ovens in the workers' town. Gangs of men, with their names painted on their walls, gather their tools. At the quarry, a stonecutter strikes a copper chisel with a stone mallet. On the harbor, a boat arrives from Tura loaded with white limestone, and the crew, who have been traveling for days, hand over their load and receive their rations. A scribe records the delivery. The sun comes up over the ridge. On the plateau, a pyramid stands half-finished, its steps catching the light.

Everyone in that scene is doing something ordinary. The cutter is cutting. The baker is baking. The scribe is writing. The boatman is rowing. And the sum of all those ordinary acts is the Great Pyramid.

That is what this book has tried to show: that the greatest monument of the ancient world was not made by a miracle, but

by millions of small, skillful, patient acts, and that it has been standing for forty-six centuries to remind us what people can do together.

If you ever have the chance to stand at the foot of that great wall of stone, take a moment to look at one block, a single block, at about the height of your chest. Somebody cut it. Somebody dragged it. Somebody set it. Put your hand on it. That is as close as you will ever come to touching the hand of the person who did it.

Appendix A — A Timeline of the Pyramids and Their Study

Key dates, from the first kings to the most recent discoveries. Ancient dates are approximate and vary somewhat between chronologies.

The pyramid age

Date	Event
c. 3100 BCE	Egypt is unified under a single king; the first dynasty begins. Royal tombs at Abydos are brick-lined pits.
c. 2670 BCE	King Djoser of the Third Dynasty builds the Step Pyramid at Saqqara, designed by Imhotep, the first large stone building.
c. 2613–2589 BCE	Sneferu, first king of the Fourth Dynasty, builds or completes pyramids at Meidum and Dahshur (the Bent Pyramid and the Red Pyramid).
c. 2589–2566 BCE	Khufu builds the Great Pyramid at Giza, 146.6 meters high. Merer's logbook is written during this reign.
c. 2558–2532 BCE	Khafre builds the second pyramid at Giza and, most likely, the Great Sphinx.

Date	Event
c. 2532–2503 BCE	Menkaure builds the third and smallest of the Giza pyramids, with a lower casing of granite.
c. 2494–2345 BCE	Fifth Dynasty. Smaller pyramids and sun temples; the Pyramid Texts first appear in the pyramid of Unas (c. 2375–2345 BCE).
c. 2345–2181 BCE	Sixth Dynasty. Pyramids of Teti, Pepi I, Merenre, and Pepi II, all with Pyramid Texts.
c. 2181–2055 BCE	First Intermediate Period. Central authority weakens; pyramid-building largely stops.
c. 2055–1650 BCE	Middle Kingdom. Kings return to pyramid building, now with mud-brick cores, at Lisht, Dahshur, and Hawara.
c. 1550 BCE	The Rhind Mathematical Papyrus is copied; it preserves problems on the slope (seked) of pyramids.
c. 1250 BCE	Prince Khaemwaset, a son of Ramesses II, inspects and restores older pyramids.
c. 750–650 BCE	The Kushite kings of the Twenty-fifth Dynasty rule Egypt and revive pyramid building in Nubia.

Visitors, plunderers, and scholars

Date	Event
c. 450 BCE	Herodotus visits Egypt and records stories about the building of the Great Pyramid.

Date	Event
c. 820 CE	Caliph al-Ma'mun's men force a tunnel into the Great Pyramid, still used as the visitors' entrance.
1303 CE	A major earthquake loosens the casing stones of the Great Pyramid; over the following decades, many are carried off to build in Cairo.
1798	Napoleon's expedition brings scholars to Egypt; the resulting *Description de l'Égypte* is a landmark publication.
1837	Howard Vyse explores the Great Pyramid and discovers the upper relieving chambers with the work-gang marks naming Khufu.
1859–1864	John Taylor and Charles Piazzi Smyth publish the first major works of "pyramidology."
1880–1882	Flinders Petrie surveys Giza with unprecedented accuracy and refutes the pyramid-inch theory.
1954	The sealed boat pit beside the Great Pyramid is opened; the Khufu ship is found.
1993	Rudolf Gantenbrink's robot finds the small stone door in the Queen's Chamber southern shaft.
2002	A National Geographic team's robot drills through the first door and finds a second.
2010	A radiocarbon study published in *Science* largely supports the traditional chronology of Egyptian kings.

Date	Event
2011	The Djedi robot sees red ochre marks behind the first door in the Queen's Chamber shaft.
2013	Pierre Tallet's team discovers the Wadi al-Jarf papyri, including the logbook of Merer.

The era of non-invasive science

Date	Event
2015	The ScanPyramids project begins, using muon imaging, thermography, and other non-invasive methods.
2017	ScanPyramids announces the Big Void above the Grand Gallery in the journal *Nature*.
2023	ScanPyramids publishes its detailed study of the corridor-shaped void behind the north face, and views its interior with an endoscope.
2024	A team led by Eman Ghoneim reports the buried Ahramat Branch of the Nile along the pyramid fields. A separate team proposes a hydraulic lift at the Step Pyramid; specialists remain skeptical.
2025	Cairo University and the Technical University of Munich report two air-filled voids behind the eastern facade of Menkaure's pyramid. The Grand Egyptian Museum opens near the Giza plateau.

Appendix B — The Pyramids at a Glance

Quick-reference tables. Figures are standard published estimates and vary a little by source.

Major Old Kingdom pyramids

Pyramid	King	Site	Height (approx.)	Notes
Step Pyramid	Djoser (3rd Dyn.)	Saqqara	62 m	First large stone building; six steps; designed by Imhotep
Meidum	Huni / Sneferu (3rd–4th Dyn.)	Meidum	about 92 m originally	Stepped core converted to smooth sides; outer casing collapsed
Bent Pyramid	Sneferu (4th Dyn.)	Dahshur	about 105 m	Angle changes from about 54 to 43 degrees partway up
Red Pyramid	Sneferu (4th Dyn.)	Dahshur	about 105 m	First successful true smooth-sided pyramid; angle about 43 degrees
Great Pyramid	Khufu (4th Dyn.)	Giza	146.6 m originally (about 138.5 m now)	Largest; 230 m base; about 2.3 million blocks

Pyramid	King	Site	Height (approx.)	Notes
Pyramid of Khafre	Khafre (4th Dyn.)	Giza	143.5 m originally	Appears tallest because it stands on higher ground; top casing survives
Pyramid of Menkaure	Menkaure (4th Dyn.)	Giza	about 65 m	Lower courses faced in granite; 2025 scans found voids on the east side
Pyramid of Unas	Unas (5th Dyn.)	Saqqara	about 43 m	First pyramid with the Pyramid Texts carved inside

The Great Pyramid in numbers

Quantity	Value
Original height	about 146.6 m (280 royal cubits)
Present height	about 138.5 m
Base side	about 230.3 m (440 royal cubits)
Base area	about 5.3 hectares (13 acres)
Slope	about 51.8 degrees (a seked of 5½ palms; 14 up for 11 across)
Volume	about 2.6 million cubic meters
Number of blocks	about 2.3 million (estimates range higher)
Mass	about 5.9 million tonnes
Typical block	about 2.5 tonnes
Largest blocks	granite beams above the King's Chamber, some 50 tonnes or more
Time to build	about 20 years

Quantity	Value
Workforce	estimated 20,000 to 30,000 at a time (estimates vary)
Energy to lift all the stone	about 2 trillion joules
Food energy used (about 20 percent efficiency)	about 10 trillion joules
North-south alignment error	a few minutes of arc, a fraction of a degree

Units of measure used by the builders

Unit	Size
Royal cubit	about 52.4 cm (7 palms, or 28 fingers)
Palm	about 7.5 cm (4 fingers)
Finger	about 1.9 cm
Seked	the horizontal run, in palms, for each cubit (7 palms) of rise; a measure of slope
Khet (rope)	100 cubits, about 52.4 m

Appendix C — Glossary

Terms used in this book, in alphabetical order.

- **Akh.** In Egyptian belief, the transfigured, effective spirit of a dead person who has successfully entered the afterlife.
- **Ba.** An aspect of a person's spirit, often shown as a human-headed bird, that could travel between the tomb and the world of the living.
- **Benben.** The sacred mound or stone at Heliopolis associated with the sun god; the pyramid shape may recall it.
- **Canopic jars.** Containers for the preserved internal organs of a mummy.
- **Cartouche.** An oval ring around a king's name in hieroglyphs.
- **Casing stones.** The finely cut outer limestone blocks that gave a pyramid its smooth surface.
- **Causeway.** The covered or walled ceremonial road linking a pyramid's valley temple to its mortuary temple.
- **Corbelled vault.** A roof or ceiling made by stepping successive courses of stone inward until they meet.
- **Cubit (royal).** The standard Egyptian unit of length, about 52.4 centimeters.
- **Djed.** A hieroglyphic and religious symbol of stability, often drawn as a pillar with horizontal bars.

- **Dolerite.** A very hard dark stone used for pounding balls that shaped granite and other hard rock.
- **Dynasty.** A line of kings; Egyptologists, following the ancient historian Manetho, number the kings in about thirty dynasties.
- **False door.** A carved stone panel in a tomb chapel through which the spirit of the dead was believed to receive offerings.
- **Hieratic.** The cursive handwriting form of Egyptian script, used for everyday records on papyrus.
- **Hieroglyphs.** The picture-signs of formal Egyptian writing, used on monuments.
- **Inundation.** The yearly Nile flood, called Akhet; it drove the farming year and freed labor for state projects.
- **Ka.** The life-force or spiritual double of a person, which required offerings in order to survive.
- **Kemet.** "The Black Land": the fertile floodplain of Egypt, as opposed to the red desert.
- **King's Chamber.** The granite-lined chamber high inside the Great Pyramid that holds an empty sarcophagus.
- **Maat.** The Egyptian principle of truth, order, and balance, which the king was bound to uphold.
- **Mastaba.** A flat-roofed, rectangular tomb structure with sloping sides, from the Arabic word for "bench."
- **Mortuary temple.** The temple attached to the east face of a pyramid, where offerings were made to the dead king.

- **Muon.** A subatomic particle created when cosmic rays strike the upper atmosphere, used to image the inside of large structures.
- **Natron.** A natural salt mixture used to dry bodies for mummification and for many other purposes.
- **Necropolis.** A "city of the dead," a large cemetery complex.
- **Nome.** An administrative province of ancient Egypt, from the Greek word.
- **Nubia.** The region south of Egypt along the Nile, in present-day Sudan.
- **Old Kingdom.** The period of about 2700 to 2200 BCE, the age of the great pyramids.
- **Palimpsest.** A surface that has been written or carved on more than once, with traces of the earlier text still visible.
- **Papyrus.** A paper-like writing material made from the pith of the papyrus plant.
- **Pharaoh.** From the Egyptian *per-aa*, "great house"; the title of the Egyptian king.
- **Pyramidion.** The small pyramid-shaped capstone at the top of a pyramid.
- **Pyramid Texts.** The oldest known religious writings, spells carved inside pyramids from the end of the Fifth Dynasty.
- **Queen's Chamber.** The middle chamber of the Great Pyramid; it was probably never used for a queen.
- **Relieving chambers.** Five empty spaces above the King's Chamber, built to spread the weight of the stone above.

- **Sarcophagus.** A stone coffin.
- **Sed festival.** A royal jubilee in which the king renewed his powers, usually after thirty years of rule.
- **Seked.** The Egyptian measure of the slope of a wall or pyramid face.
- **Serdab.** A small closed chamber in a tomb holding a statue of the dead; from the Arabic word for "cellar."
- **Serekh.** A rectangular frame representing a palace facade in which early kings' Horus names were written.
- **Shabti.** A small figure placed in a tomb to do work for the dead in the afterlife.
- **Sledge.** A wooden runner-based sled used to drag heavy loads, such as stone blocks and statues.
- **Stela.** An upright stone slab with an inscription or image.
- **Tura.** The quarry area across the Nile from Memphis that supplied fine white limestone for casings.
- **Valley temple.** The temple at the lower end of a pyramid causeway, near the harbor or river.
- **Vizier.** The chief minister of the king, head of the administration and the judiciary.
- **Wadi.** A dry valley or watercourse in the desert.
- **Wadi al-Jarf.** A site on the Red Sea coast, with an ancient harbor and the cache of papyri that includes Merer's logbook.

Appendix D — Further Reading and Sources

Books and papers for readers who want to go deeper, and the sources behind the recent findings described in Part Five.

Books on the pyramids

- Mark Lehner, *The Complete Pyramids* (Thames & Hudson, 1997). The standard illustrated survey of every Egyptian pyramid, by the archaeologist who has led excavations at Giza for decades.
- Miroslav Verner, *The Pyramids: The Mystery, Culture, and Science of Egypt's Great Monuments* (Grove Press, 2001). A thorough account by the leading Czech Egyptologist, covering history, construction, and religion.
- Zahi Hawass and Mark Lehner, *Giza and the Pyramids: The Definitive History* (University of Chicago Press, 2017). A large, authoritative history of the plateau.
- Pierre Tallet and Mark Lehner, *The Red Sea Scrolls: How Ancient Papyri Reveal the Secrets of the Pyramids* (Thames & Hudson, 2021). The discovery and meaning of Merer's logbook.
- I. E. S. Edwards, *The Pyramids of Egypt* (Penguin, revised edition 1985). A classic one-volume introduction.

- Bob Brier and Jean-Pierre Houdin, *The Secret of the Great Pyramid* (Smithsonian Books, 2008). The case for an internal ramp.
- Dieter Arnold, *Building in Egypt: Pharaonic Stone Masonry* (Oxford University Press, 1991). A technical study of quarrying, transport, and construction.
- W. M. Flinders Petrie, *The Pyramids and Temples of Gizeh* (1883). The foundational survey, still worth reading for its detail and clarity.
- Herodotus, *Histories*, Book 2. The ancient Greek account of Egypt and the pyramids; read it with care, as much of it is hearsay.

Books on Egypt more broadly

- Toby Wilkinson, *The Rise and Fall of Ancient Egypt* (Random House, 2010). A readable narrative history.
- Barry Kemp, *Ancient Egypt: Anatomy of a Civilization* (Routledge, 3rd edition 2018). A study of how Egyptian society worked.

On myths, pseudoarchaeology, and how to evaluate claims

- Kenneth L. Feder, *Frauds, Myths, and Mysteries: Science and Pseudoscience in Archaeology* (McGraw-Hill, several editions). A clear guide to testing extraordinary claims about the past.

- Garrett G. Fagan (editor), *Archaeological Fantasies: How Pseudoarchaeology Misrepresents the Past and Misleads the Public* (Routledge, 2006). Essays by professional archaeologists.
- Jason Colavito, *The Cult of Alien Gods: H. P. Lovecraft and Extraterrestrial Pop Culture* (Prometheus Books, 2005). On the literary and cultural roots of the ancient-astronaut idea.
- Martin Gardner, *Fads and Fallacies in the Name of Science* (Dover, 1957). Includes a chapter on pyramidology.

Works cited for the claims discussed in Part Five

- Christopher Dunn, *The Giza Power Plant: Technologies of Ancient Egypt* (Bear & Company, 1998). The source of the power-plant theory discussed in Chapter 19.
- Robert Bauval and Adrian Gilbert, *The Orion Mystery* (Heinemann, 1994). The source of the Orion Correlation Theory.
- Graham Hancock, *Fingerprints of the Gods* (Heinemann, 1995).
- Erich von Däniken, *Chariots of the Gods?* (1968).
- Charles Piazzi Smyth, *Our Inheritance in the Great Pyramid* (1864); John Taylor, *The Great Pyramid: Why Was It Built, and Who Built It?* (1859).

Scientific papers and reports on recent findings

- K. Morishima and colleagues, "Discovery of a big void in Khufu's Pyramid by observation of cosmic-ray muons," *Nature* 552 (2017). The Big Void.
- S. Procureur and colleagues, "Precise characterization of a corridor-shaped structure in Khufu's Pyramid by observation of cosmic-ray muons," *Nature Communications* 14 (2023). The north-face corridor.
- C. Bronk Ramsey and colleagues, "Radiocarbon-based chronology for dynastic Egypt," *Science* 328 (2010).
- Kate Spence, "Ancient Egyptian chronology and the astronomical orientation of pyramids," *Nature* 408 (2000). A proposal on how the pyramids were aligned by stars.
- E. Ghoneim and colleagues, study of the Ahramat Branch of the Nile, *Communications Earth & Environment* (2024).
- X. Landreau and colleagues, study proposing a hydraulic lift at the Step Pyramid, *PLOS ONE* (2024).
- Cairo University and Technical University of Munich researchers, study of voids in Menkaure's pyramid, *NDT & E International* (2025).
- M. Balezin and colleagues, study of the electromagnetic response of the Great Pyramid, *Journal of Applied Physics* (2018).

Web sources consulted for recent findings

These pages were consulted in preparing the discussion of recent discoveries. Web pages change, so check the original papers where possible.

- Technical University of Munich press release on the Menkaure pyramid scans: https://www.tum.de/en/news-and-events/all-news/press-releases/details/detection-of-air-filled-anomalies-in-menkaure-pyramid-could-indicate-new-entrance
- Archaeology magazine, "Scientists Scan Egypt's Menkaure Pyramid with High-Tech Tools" (13 November 2025): https://archaeology.org/news/2025/11/13/scientists-scan-egypts-menkaure-pyramid-with-high-tech-tools/
- University of North Carolina Wilmington, "Researchers uncover buried branch of Nile River" (May 2024): https://uncw.edu/news/2024/05/researchers-uncover-buried-branch-of-nile-river
- Live Science, "Long-lost branch of the Nile was 'indispensable for building the pyramids,' research shows": https://livescience.com/archaeology/ancient-egyptians/long-lost-branch-of-the-nile-was-indispensable-for-building-the-pyramids-research-shows
- Science News, "Was Egypt's first pyramid built with hydraulics? The theory may hold water": https://www.sciencenews.org/article/hydraulic-lift-egypts-first-pyramid

- Phys.org, "Study reveals the Great Pyramid of Giza can focus electromagnetic energy" (July 2018): https://phys.org/news/2018-07-reveals-great-pyramid-giza-focus.html
- The Past, "Exploring the Great Pyramid void": https://the-past.com/news/exploring-the-great-pyramid-void
- Euronews, report on the opening of the Grand Egyptian Museum (30 October 2025): https://euronews.com/culture/2025/10/30/great-expectations-the-worlds-biggest-museum-dedicated-to-ancient-egypt-opens-this-week
- Wikipedia entries on the Saqqara Bird and the Dendera light, used as summaries of the objects and the published literature about them.

A note on this edition

This book was written as a general introduction and prepared with the assistance of artificial intelligence. It draws on the published scholarship listed above, and the claims in it should be checked against those sources before being cited in academic work. Archaeology moves quickly, and discoveries reported after the autumn of 2026 are not included.